

Data Analytics with Power BI

Creating a Dashboard from Scratch Learner's Guide



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Use Case Overview:

Use Case: Technology Training Utilization Dashboard

Note: This Use Case is taken from Real Project but data is fabricated for training purposes.

Digital Training Unit under Aramco Corporate Academy is responsible for delivering Technology courses and programs on various IT technologies including General IT Training, SAP Training, Information Security, Digital Technologies etc. These trainings are delivered as virtual sessions, Physical Classrooms, Seminars, workshops etc. both in the kingdom & out of kingdom. The Objective of the unit is to upskill Saudi Aramco workforce on various IT technologies related to their work and business needs.

The training unit management and the planning group requested to develop a dashboard which provide them a holistic overview of training utilization by various perspectives as well as to simplify the analysis of training cost & budget consumption.

Working as an analyst within the group, you are assigned to develop a dashboard using **Microsoft Power BI** application & deploy into the server in order to fulfill the management & planning team requirements.

Data Sources:

Training completions data is captured in **SAP SuccessFactors**. The data is pushed to **SAP PRW** (SAP Business Warehouse) every night through **SAP Jobs**. We created **BEx Queries** in PRW to fetch the required data into Power BI through **Web** option. We also created **Power BI scheduler** on a Power BI Server to update the dashboard from SAP PRW on daily basis. Training data contains completion of all training courses along with the required details such as employee information, course details, completion status etc.

Training Cost is maintained manually by planning group and provided on monthly basis. That's include Plan budget (initial planning cost), actual cost (actual cost of training) etc. we update it manually in our dashboard.

Note: For training purposes, the training completions data is directly exported from SuccessFactors as **CSV** file.

To Summarize, Data is coming from **two** different sources required for the dashboard.

- Training Data

- Training Cost **KPI's**:

Following KPI's & measures are required to present in dashboard from various perspectives.

- **Total Hours.** *(Participant Contact Hours (PCH). The Hours credited to participants on course completion. Its calculated as course duration * Hours per day. For example, if course duration is 2 full days then Total hours will be $2 * 8 = 16$).*
- **Total Participants.** *(No. of Participants successfully complete the training. We can calculate it by count the employee badge numbers from training data based on completion status column).*
- **Total Courses.** *(No. of distinct courses delivered. Again, its calculated from training data).*
- **Total Sessions.** *(Same course can be delivered multiple times that's called **Session**. To calculate number of sessions, we need to group the data at **course level** & then do the required calculation based on **End Date** column).*
- **Plan Budget.** *(Initial Budgeted allocated for training. This is coming from Training Cost file).*
- **Actual Cost.** *(Actual cost of training).*
- **Budget Variance** *(Difference between Plan vs. Actual. This is very important KPI as management wants to minimize the gap between what was plan vs. actual cost).*
- **Budget Consumption %.** *(How much budget is consumed in %).*
- **Avg. Cost Per Participant.** *(Actual Cost divided by Total Participants. This is another very important KPI).*

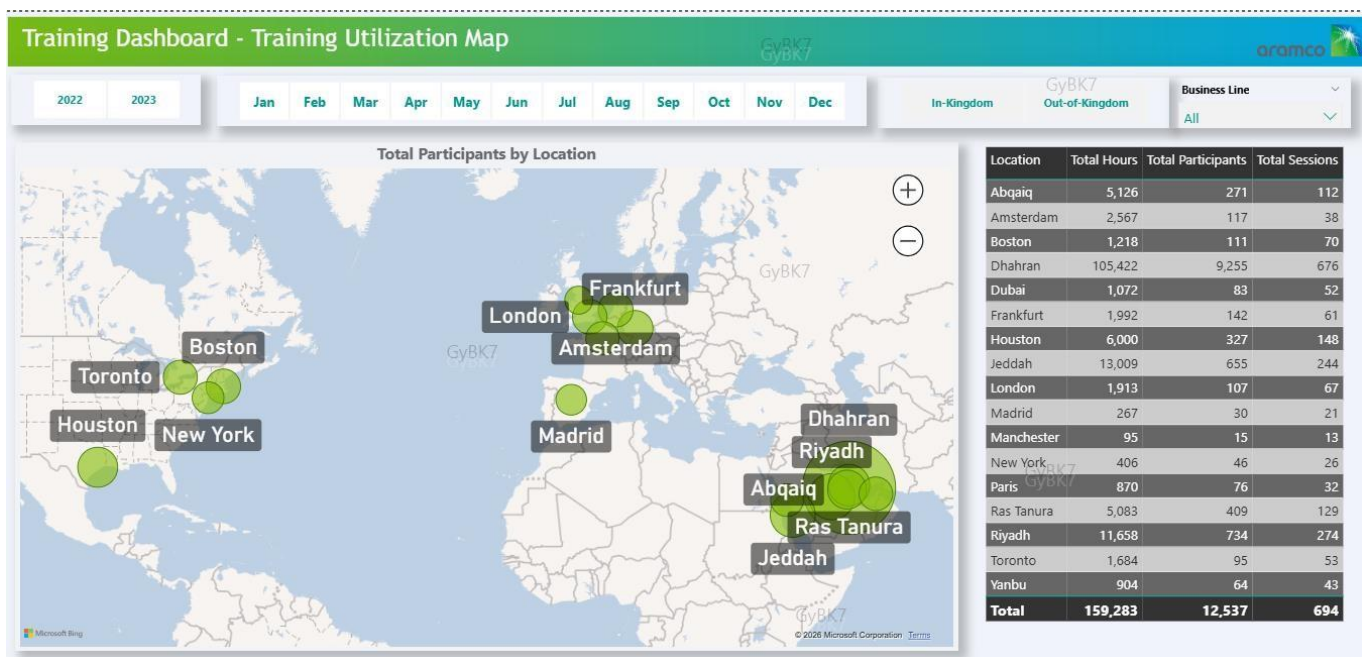
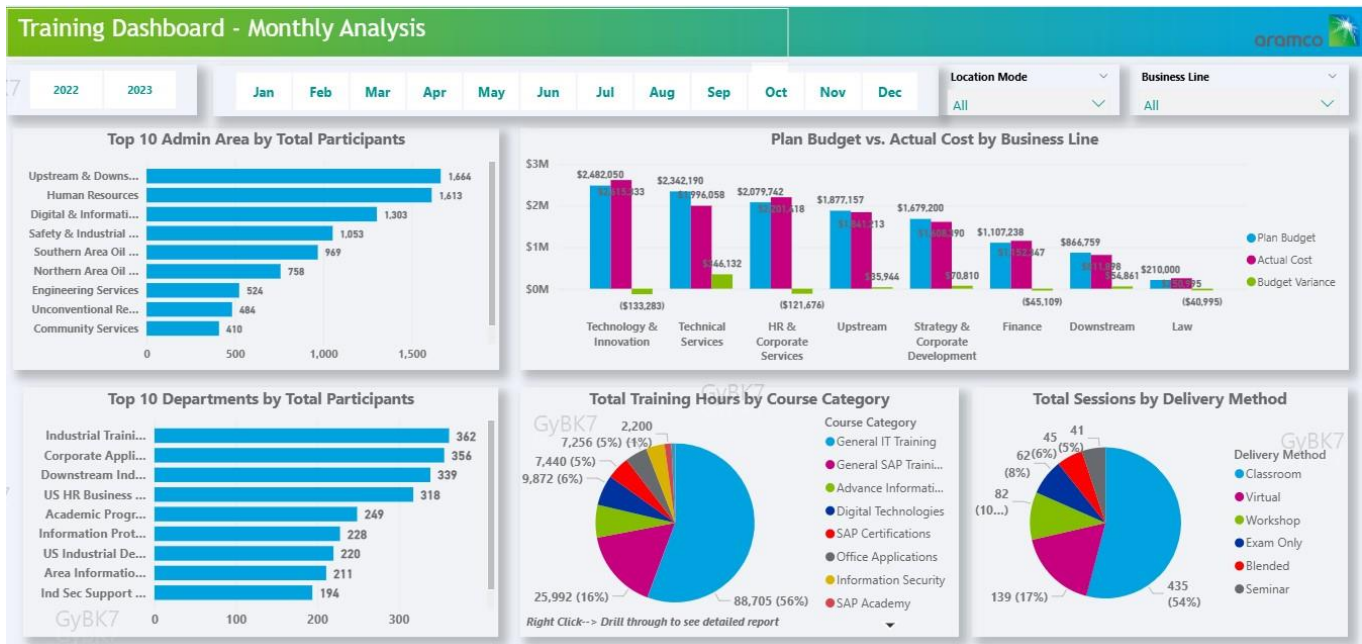
Your Tasks:

1. Create the dashboard from scratch using **Microsoft Power BI**.
2. Transform & Clean the Data using **Power Query Editor**.
3. Create **Visuals** to present the Data.
4. Build **Data Model** as the data comes from multiple sources.
5. Create KPI's & Measures using **DAX**.
6. Apply **Slicer or filters** so the users can able to filter the data as needed.
7. Publish the dashboard in **Power BI Report Server**.
8. Create the **scheduler** so the data in Dashboard will refresh automatically.
9. Work with **Advance Features**.

All these tasks are divided into **8** different exercises.

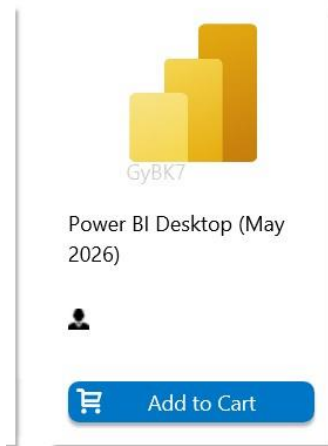
At the end of the course, you will be able to design a complete dashboard from scratch which looks similar to below. *(It contains three pages **QBR Summary**, **Monthly Analysis** & **Training Utilization Map**).*





Note: Exercise steps are not documented in detail. The purpose of these steps is to guide you on the flow of developing dashboard. Some of the details you need to [explore by your own](#).

Also note that these exercises are based on Microsoft Power BI Desktop Version **May 2026**. You can download Power BI Desktop from Aramco App Center. [App Center \(aramco.com.sa\)](https://aramco.com.sa)



Exercise 1: Understanding the User Interface

In this exercise, we will focus on understanding the user interface of Power BI.

What is Power BI:

Microsoft Power BI is a collection of software services, apps, and connectors that work together to turn your unrelated sources of data into coherent, visually immersive, and interactive insights. Your data may be an Excel spreadsheet, or a collection of cloud-based and on-premises hybrid data warehouses.

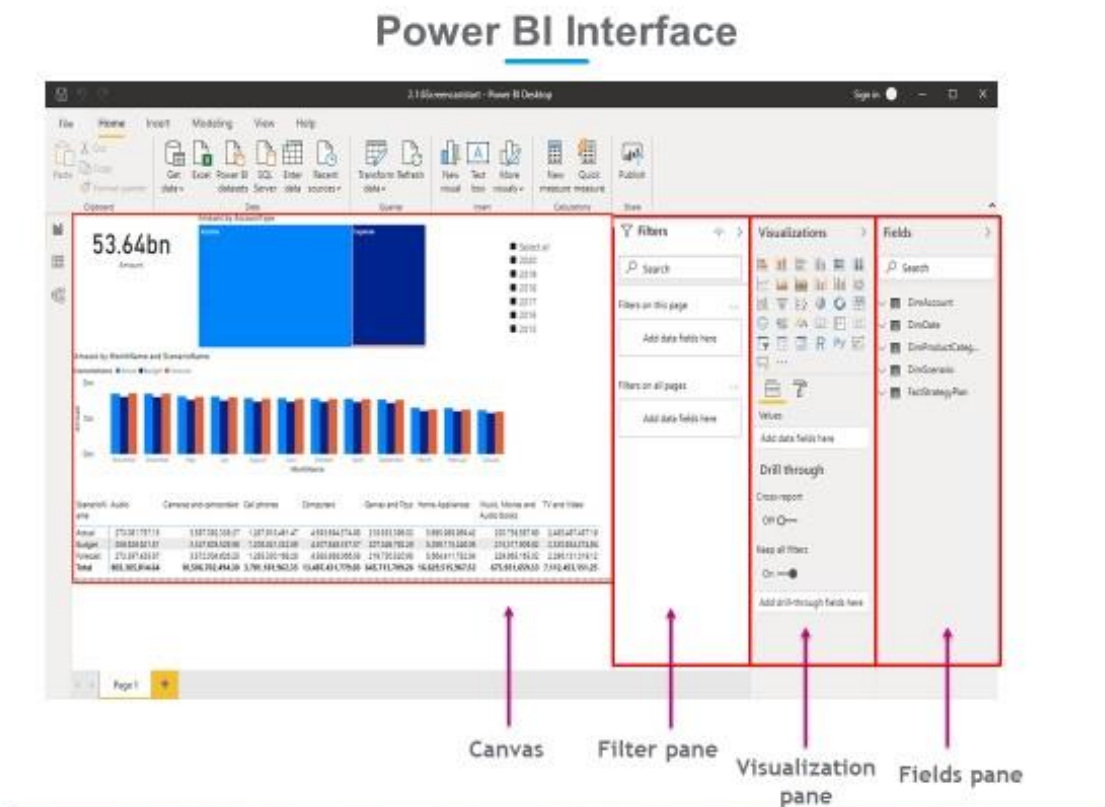
Power BI lets you easily connect to your data sources, visualize and discover what's important, and share that with anyone or everyone you want.

To learn more about Power Query, visit the link below:

[What is Power BI? - Power BI |](#)

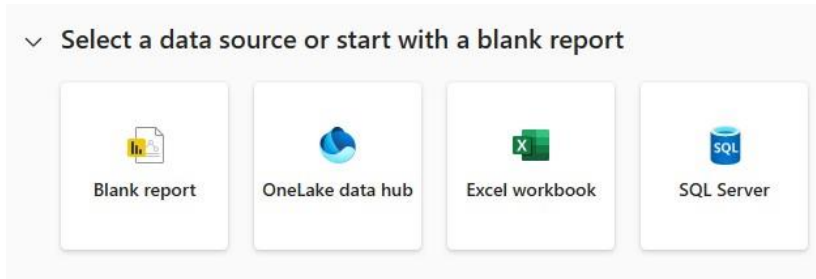
[Microsoft Docs](#) **Power BI User**

Interface:



Steps:

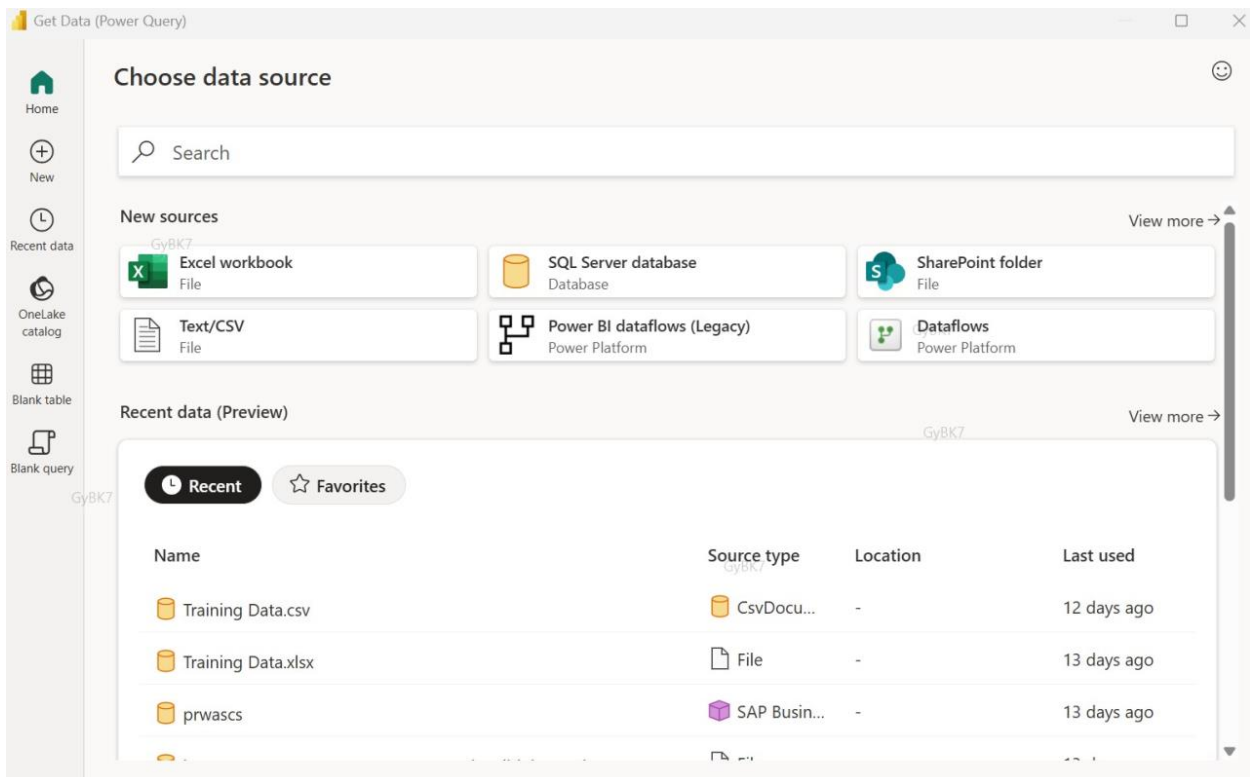
1. Open the Power BI Desktop Application.
2. Click on **Blank report**.



3. **Explore yourself:** Spend some time to explore the user interface of Power BI particularly the menu bar.



4. From **Get data** more, see the list of available data sources from where you can fetch the data in Power BI.



5. Select **Text/CSV**, click **Browse** & select the file named **Training Data** from the Dataset folder.

Keep the default selection for **Authentication kind**. (*Anonymous*).

Get Data (Power Query)

Connect to data source

Text/CSV File [Learn more](#)

Connection settings

File path *
Example: https://contoso-my.sharepoint.com/personal/ana_c ↑ Browse...

Connection credentials

Authentication kind
Anonymous

Back Next Cancel

6. Click Next.

Cancel Next

7. In the next screen, click **Load** to load the data into Power BI Desktop.

Preview file data

File path: C:\Users\siddiqf\Desktop\Trainings\Courses\Data Analytics with Power BI\Training Utilization Dashboard\For Participants\Data Analytics with Power BI\Dataset\T...

File origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data type detection: Based on first 200 rows

Column1	Column2	Column3	Column4	Column5	Column6
Employee Item Completion Detail					
Date	10/22/2024				
Employee Badge No	Employee Name	Gender	Business Line	Admin Area	Department
105134	XXXXXXXXXX	M	Technical Services	Upstream & Downstream Project Manag...	
105134	XXXXXXXXXX	M	Technical Services	Upstream & Downstream Project Manag...	
105423	XXXXXXXXXX	M	Technical Services	Upstream & Downstream Project Manag...	Oil Facilities Projects Dept
105437	XXXXXXXXXX	M	Technical Services	Upstream & Downstream Project Manag...	Tanajib Gas Plant Projects Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept
105510	XXXXXXXXXX	M	Technical Services	Infrastructure & JV Project Management	RAK Onshore Infrastructure Projects De

Back | Cancel | Add table using examples | Load

8. From the right side, click on **Training Data** to see the columns.

Data

Search

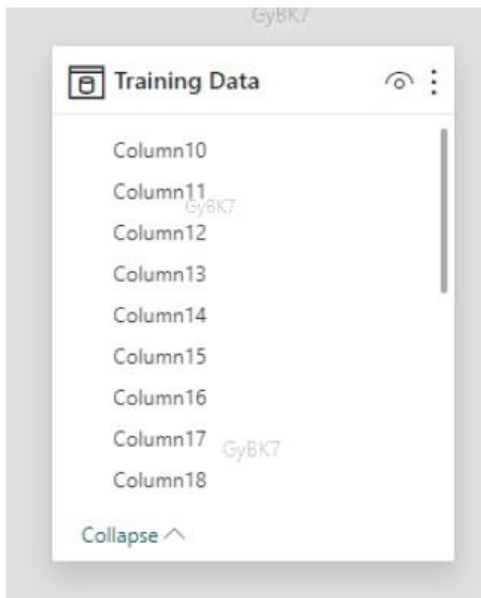
Training Data

- ☐ Column10
- ☐ Column11
- ☐ Column12
- ☐ Column13
- ☐ Column14
- ☐ Column15
- ☐ Column16
- ☐ Column17
- ☐ Column18
- ☐ Column19
- ☐ Column2
- ☐ Column20
- ☐ Column21
- ☐ Column3
- ☐ Column4

9. From the left side, click on **Table view** to view the data itself.

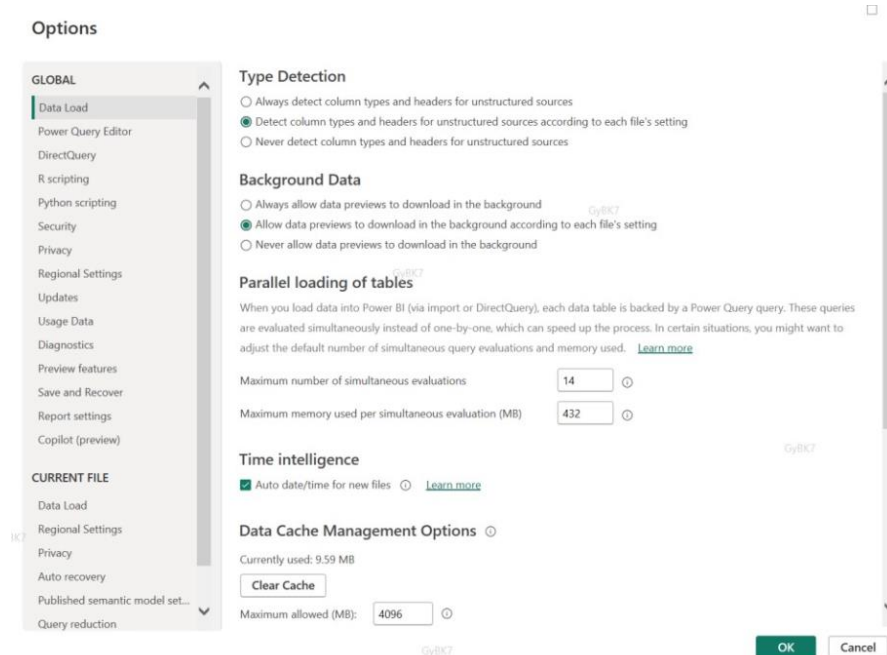
Employee Item Completion Detail	Column2	Column3	Column4	Column5	Column6	Column7	
105463	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept	30025343	Appren
106013	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Organization Consulting Dept	30024052	Change
106148	XXXXXXXXXX	M	HR & Corporate Services	Human Resources		30001419	Training
106859	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Academic Programs & Partnerships Dept	30022002	Profess
106895	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Mgmt & Professional Development Dept	30015036	Talent &
106231	XXXXXXXXXX	F	HR & Corporate Services	Human Resources		30039703	HSE Cu
109266	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	Personnel Dept	0	Dummy
109266	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	Personnel Dept	0	Dummy
109266	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	Personnel Dept	0	Dummy
109271	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	HR Policy & Planning Dept	30027754	Affiliate
109320	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	HR Systems Support Dept	30002073	Human
111547	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	Academic Programs & Partnerships Dept	30022002	Profess
111700	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	HR Policy & Planning Dept	30027754	Affiliate
111823	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Local Workforce Development Dept	30035332	Stakeho
112091	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Industrial Training Dept	30001530	Process
112178	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	Academic Programs & Partnerships Dept	30021527	Univers
112496	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	HR Policy & Planning Dept	30003020	Benefit
112617	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Learning Solutions & Services Dept	30001430	Technic
112617	XXXXXXXXXX	M	HR & Corporate Services	Human Resources	Learning Solutions & Services Dept	30001430	Technic
112830	XXXXXXXXXX	F	HR & Corporate Services	Human Resources	Academic Programs & Partnerships Dept	30022002	Profess

10. Click on **Model view** to view the dataset as table.



Note: Power BI stores everything as a table either data comes from Excel, CSV, databases, web services, SAP BEX Queries etc.

11. From the file menu, click on **Options & Settings**  **Options**.



12. You can set different options from here if needed. For example, under the security, click on the checkbox **Use Map and Filled Map visuals**. This is required to enable map visualizations in Power BI.

Map and Filled Map visuals

☒ Use Map and Filled Map visuals

13. Click OK.
14. From Help menu, click Examples  Sample Reports. Explore some sample Power BI dashboards provided by Microsoft.
15. From File, click Save. Call it **Technology Training Utilization Dashboard – Your Badge Number**.

Note: Save in your local machine. later, we will publish it into the server. To avoid conflict, append your badge number at the end of the file.

Exercise 2: Transforming & Cleaning the Data with Power Query

In order to present the data as needed, we need to perform some transformations. Also, we need to clean the data as well. For that purpose, we can utilize the **Power Query Editor**.

What is Power Query:

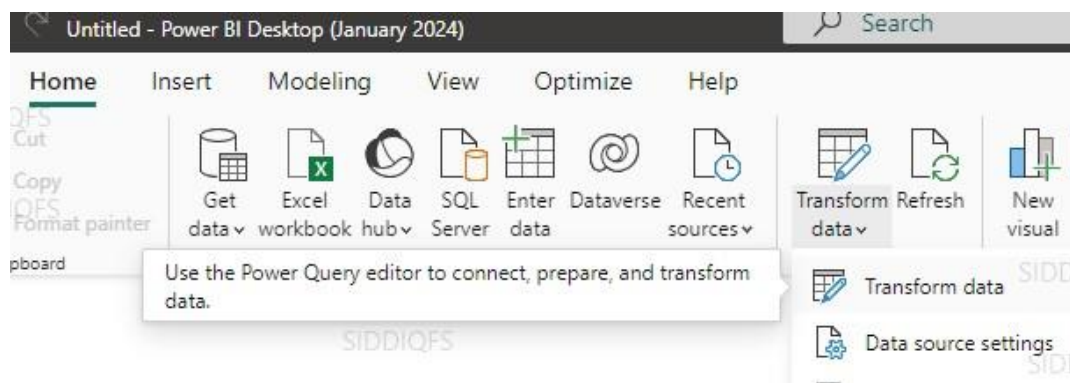
Power Query is a data transformation and data preparation engine. Power Query comes with a graphical interface for getting data from sources and a Power Query Editor for applying transformations. Because the engine is available in many products and services, the destination where the data will be stored depends on where Power Query was used. Using Power Query, you can perform the extract, transform, and load (ETL) processing of data.

To learn more about Power Query, visit the link below:

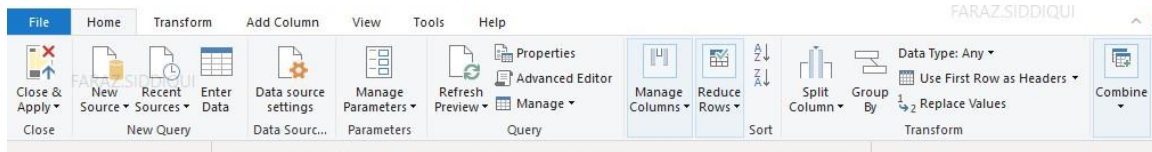
[What is Power Query? |](#)

[Microsoft Docs](#) Steps:

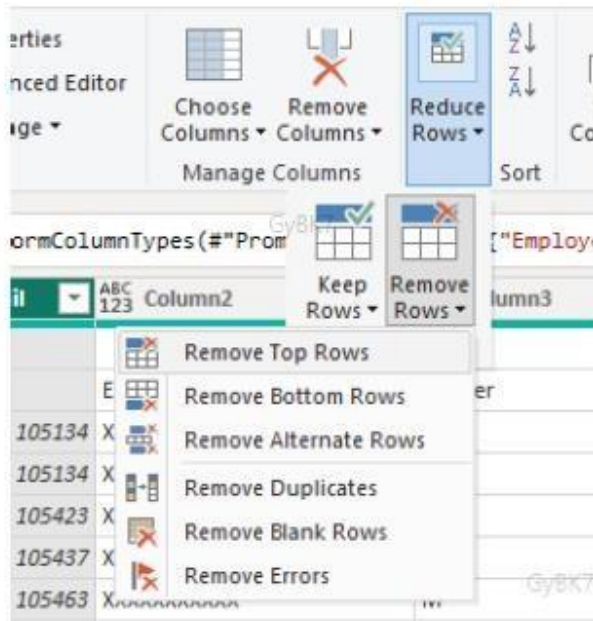
1. Continue with the same Power BI project file created in last exercise i.e. **Technology Training Utilization Dashboard – Your Badge Number**
2. From Home ➡ Transform data ➡ Transform data to access the **Power Query Editor**.



3. Power Query Editor open in a separate screen. It provides rich functionality to transform and clean the raw data.



4. For example, when we extract the data from the system (SuccessFactors), there are blank rows in a file. We have to remove them in order to process the data further. From Home, click on **Remove Rows→Remove Top Rows**.



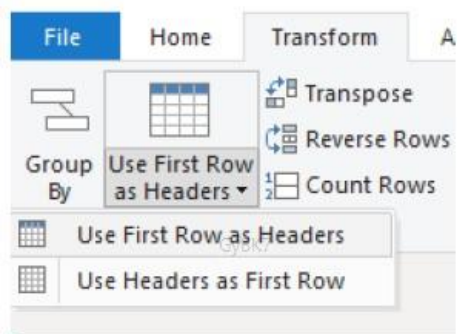
5. In Remove Top Row screen, type **2** and click OK.



6. Now we want to promote top row as column headers.

ABC 123	Employee Item Completion Detail	ABC 123	Column2	A ^B C	Column3	A ^B C	Column4
1	Employee Badge No		Employee Name		Gender		Business Line
2	105134		XXXXXXXXXX		M		Technical Services

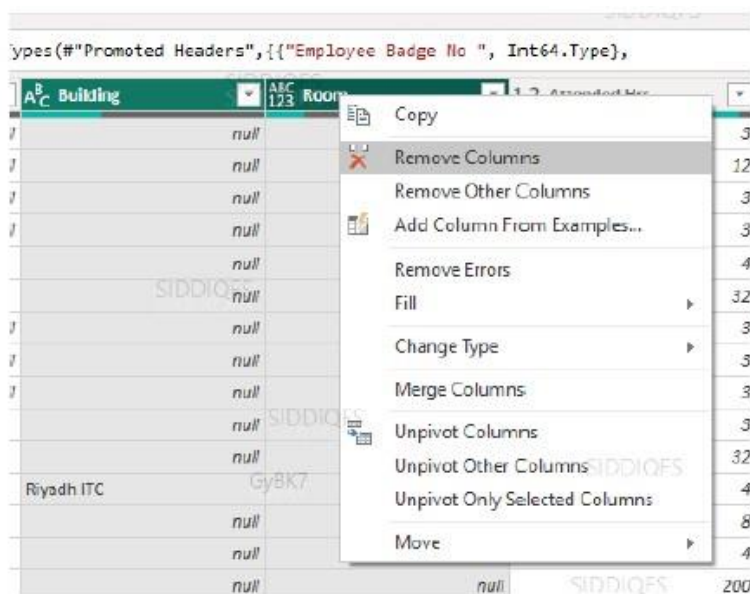
7. From Transform tab, click on **Use First Row as Headers→Use Row as Headers**.



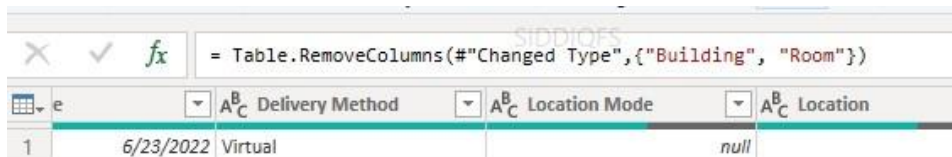
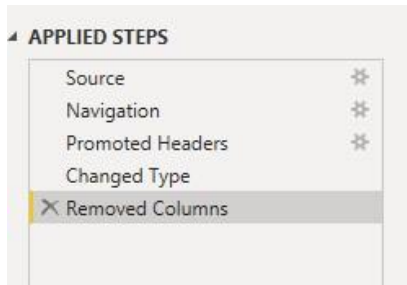
8. First row will become the header as shown below.

	Employee Badge No	Employee Name	Gender	Business Line	
1	105134	XXXXXXXXXX	M	Technical Services	U
2	105134	XXXXXXXXXX	M	Technical Services	U
3	105423	XXXXXXXXXX	M	Technical Services	U
4	105437	XXXXXXXXXX	M	Technical Services	U
5	105468	XXXXXXXXXX	M	HR & Corporate Services	H

9. Similarly, there are unnecessary columns which are not required in our dashboard. We can remove them as well. Select **Building & Room** → Right click → Remove Columns. (Use **Ctrl key** to select multiple columns).



10. The columns are removed. In Power Query, every step recorded as a **Marco**. Also, you see it as a function in **formula bar**.

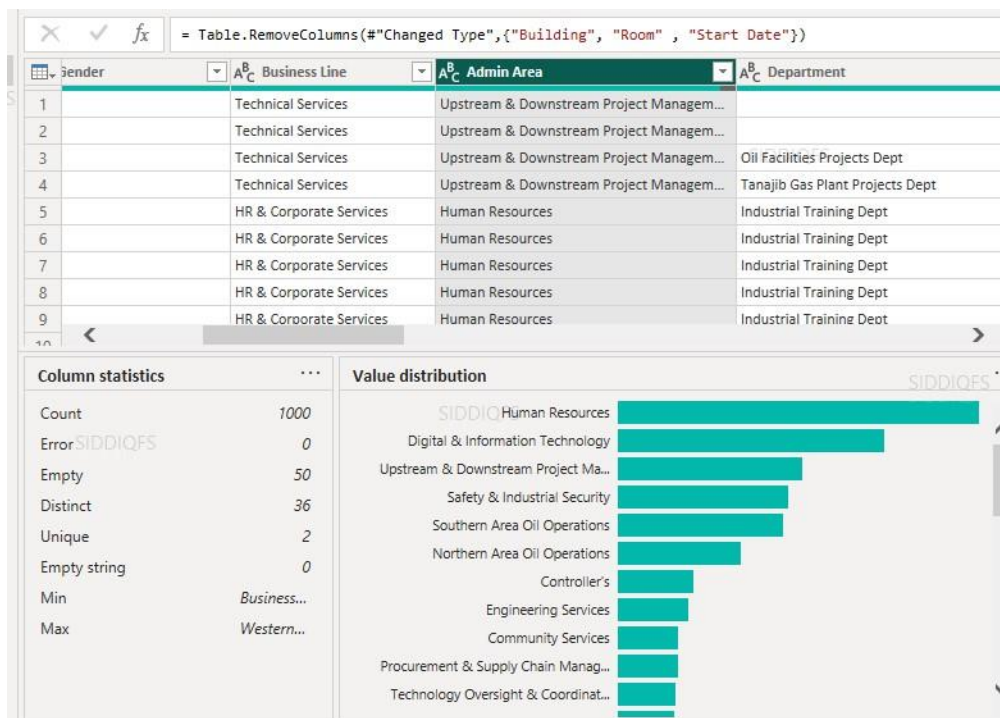


11. Modify the above formula & add **Start Date** to remove it as well.

Note: You can easily delete or modify any step from the applied steps.

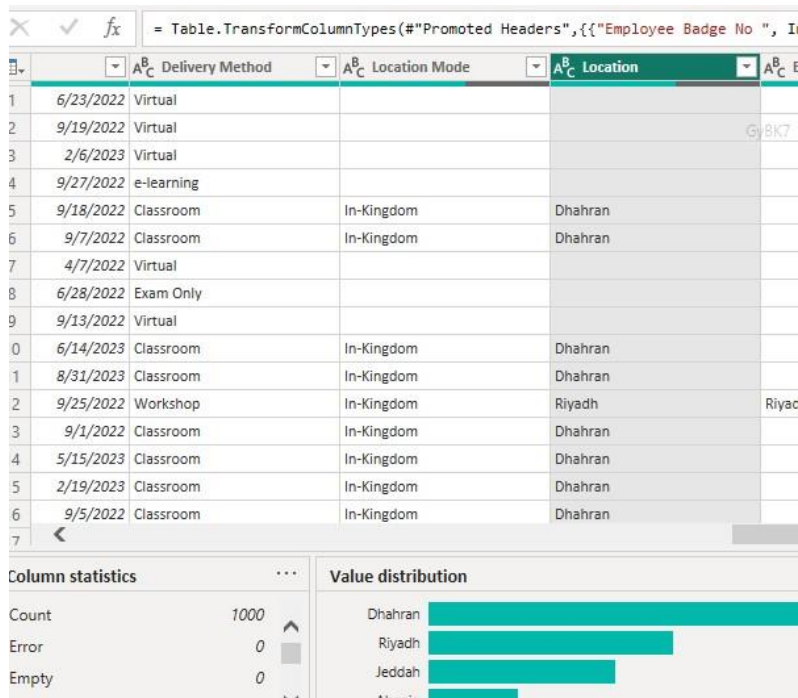
12. From View  check **column profile**.

13. Select any column, you can see details like count, empty values etc.

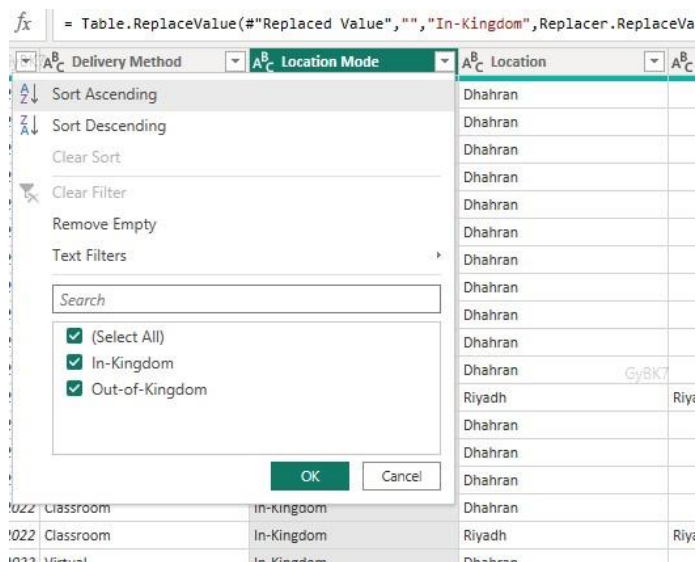


Note: Similarly, you can check column distribution, column quality from the view tab.

14. Select **Location** column. You will notice that there are many **blank** values. When the course delivery is **virtual** then location field is **empty**. Same applied for **Location Mode**.

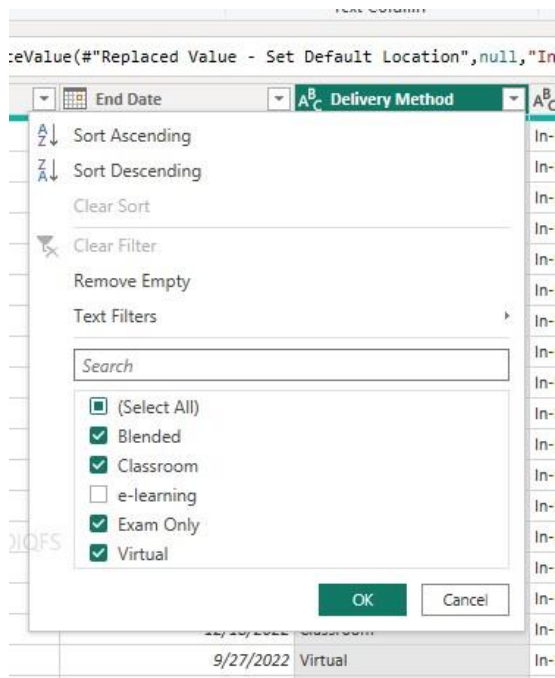


- Challenging Task:** For reporting purposes, if the location field is **empty** then the user wants to set **Dhahran** as default location. Use the **replace** function to replace all blank values in location column to **Dhahran**. In applied steps, rename this step to **replace blank locations**. Repeat the same for **Location Mode** column. Replace all blank values in Location Mode with the default value of **In-Kingdom**. Check both columns to make sure there are no empty values.

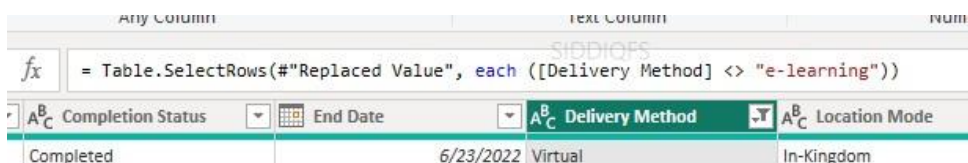
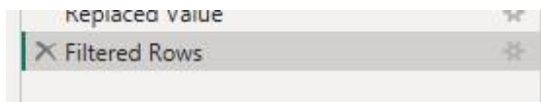


- When we extract the data from SuccessFactors, we are getting **e-learning** courses as well. However, we are required to show the data only for Virtual, Classroom, Blended & Exam in our dashboard. Select the arrow next to

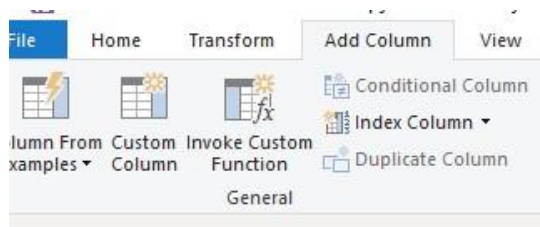
Delivery Method column & choose all delivery methods except e-learning as shown below.



17. Click OK. It will create **Filtered Rows** step using Table.SelectRows() function. (You can rename the step if you want).



18. Users are interested to analyze data by **Regions**. However, there is no region in the source data. We can utilize **custom column** for that purpose. Go to Add Column ➔ Custom Column.

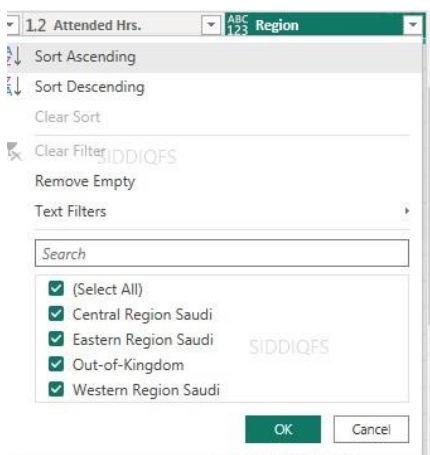


19. Region will be based on Location column as shown below. (where training sessions physically conducted. Mainly for classroom courses both for in-kingdom & out of kingdom).

- **Western Region Saudi** : Jeddah, Yanbu
- **Central Region Saudi** : Riyadh
- **Eastern Region Saudi** : Dhahran, Ras Tanura, Abqaiq
- **Out-of-Kingdom** : All International cities.

20. **Challenging Task:** Partial formula is given below. You need to **modify** the formula to create all **4 Regions** mention above. Name the new column & the step **Region**.

if List.Contains({"Jeddah","Yanbu"}, [Location]) then "Western Region Saudi" else "Out-of-Kingdom"



Note: Power Query using **M Language**. To learn more about M. visit the link below.

[Power Query M formula language reference - PowerQuery M | Microsoft Docs](#)

About **List.Contains()** function, visit the below link.

[List.Contains - PowerQuery M | Microsoft Docs](#)

Note: You can also create above formula using **conditional column** in Power query OR using **DAX** in Power BI Report Interface.

21. Click on **Home** ➔ **Advance Editor**. Power BI generates all transformation as script. if you want you can modify the script directly as well.

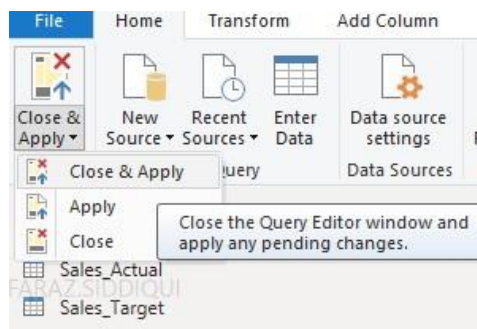
```

let
    Source = Excel.Workbook(File.Contents("C:\Users\siddiqs\Desktop\Courses\Analyzing & Visualizing Data with Power BI\Training Utilization Dashboard\Training Data.xlsx"), null, true),
    #"Training Data_Sheet" = Source[[Item="Training Data",Kind="Sheet"]][Data],
    #"Promoted Headers" = Table.PromoteHeaders(#"Training Data_Sheet", [PromoteAllScalars=true]),
    #"Changed Type" = Table.TransformColumnTypes(#"Promoted Headers",{{"Employee Badge No ", Int64.Type}, {"Employee Name", type text}, {"Gender", type text}, {"Business Line", type text}},),
    #"Removed Columns" = Table.RemoveColumns(#"Changed Type",{"Building", "Room", "Start Date"}),
    #"Filtered Rows" = Table.SelectRows(#"Removed Columns", each ([Delivery Method] <> "e-learning")),
    #"Replaced Value - Location" = Table.ReplaceValue(#"Filtered Rows",null,"Dhahran",Replacer.ReplaceValue,{"Location"}),
    #"Replaced Value - Location Mode" = Table.ReplaceValue(#"Replaced Value - Location",null,"In-Kingdom",Replacer.ReplaceValue,{"Location Mode"}),
    #"Added Custom - Region" = Table.AddColumn(#"Replaced Value - Location Mode", "Region", each if List.Contains({"Jeddah","Yanbu"},[Location]) then "Western Region"
    else if List.Contains({"Dhahran","Ras Tanura","Abqaiq"},[Location]) then "Eastern Region"
    else "Out-of-kingdom"),
    #"Renamed Columns" = Table.RenameColumns(#"Added Custom - Region",{{"Completion Status", "Completion Status Old"}},),
    #"Added Conditional Column" = Table.AddColumn(#"Renamed Columns", "Completion Status", each if [Completion Status Old] = "Attended Passed" then "Attended Passed" else if [Completion Status Old] = "Not Attended" then "Not Attended" else "Not Completed"),
    #"Added Conditional Column"
in
    #"Added Conditional Column"

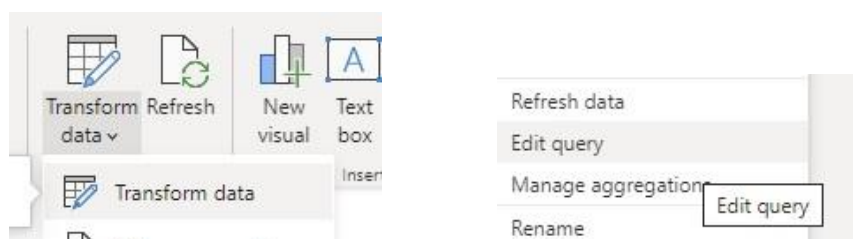
```

Note: You can also utilize the generated script in other Power BI Reports. For example, you can copy the script from advance editor, open a new Power BI report & paste the code in the advance editor of a new blank report. Power BI generate the steps for you.

22. Once done with all transformations, apply them and close power query editor.



23. We have done all the required cleaning and transformation on our source data. However, you can go back easily to power query editor any time and apply or change any steps based on the requirements. In order to go back to query editor, from **Queries & Transform Data** (Also, you can access by right click over the table name & select **Edit Query** from **Data**).



24. Save your work.

For Practice Only: (Optional. Do it only if time permits)

25. **Explore yourself:** Power Query is a very rich tool for data transformation & processing. spend some time to explore & practice with some other features of the tool. For example, Apply **Group by** on Course Category & Business Line and calculate Total hours & No. of Participants based on that grouping. The data is transformed into the summarized view as below.

fx = Table.Group(#"Added Custom - Region", {"Course Category", "Business Line"}, {"Total Hours",

	A ^B _C Course Category	A ^B _C Business Line	1.2 Total Hours	1.2 ₃ No. of Participants
1	General IT Training	Technical Services	20555.46	2709
2	General IT Training	HR & Corporate Services	35883.98	4530
3	General SAP Training	Technology & Innovation	15520	188
4	General IT Training	Technology & Innovation	7362.46	850
5	Digital Technologies	Technical Services	1680	46
6	General IT Training	Upstream	14171.46	1742
7	SAP Certifications	Technology & Innovation	2080	48
8	Advance Information Security	Technical Services	1360	133
9	General IT Training	Downstream	950	102
10	Digital Technologies	Technology & Innovation	4472	145
11	General IT Training	Finance	7862.48	815
12	SAP Certifications	HR & Corporate Services	1400	28
13	General SAP Training	HR & Corporate Services	2320	78
14	Advance Information Security	Technology & Innovation	6143	195
15	Advance Information Security	Finance	200	26
16	General IT Training	Strategy & Corporate Develo...	1527.48	173
17	Office Applications	Technology & Innovation	2272	44
18	Advance Digital Courses	Technology & Innovation	400	11
19	Office Applications	Strategy & Corporate Develo...	528	14
20	General SAP Training	Finance	2440	83
21	Special Program	Technology & Innovation	8	2
22	Office Applications	Upstream	1480	34
23	Office Applications	Finance	1104	27

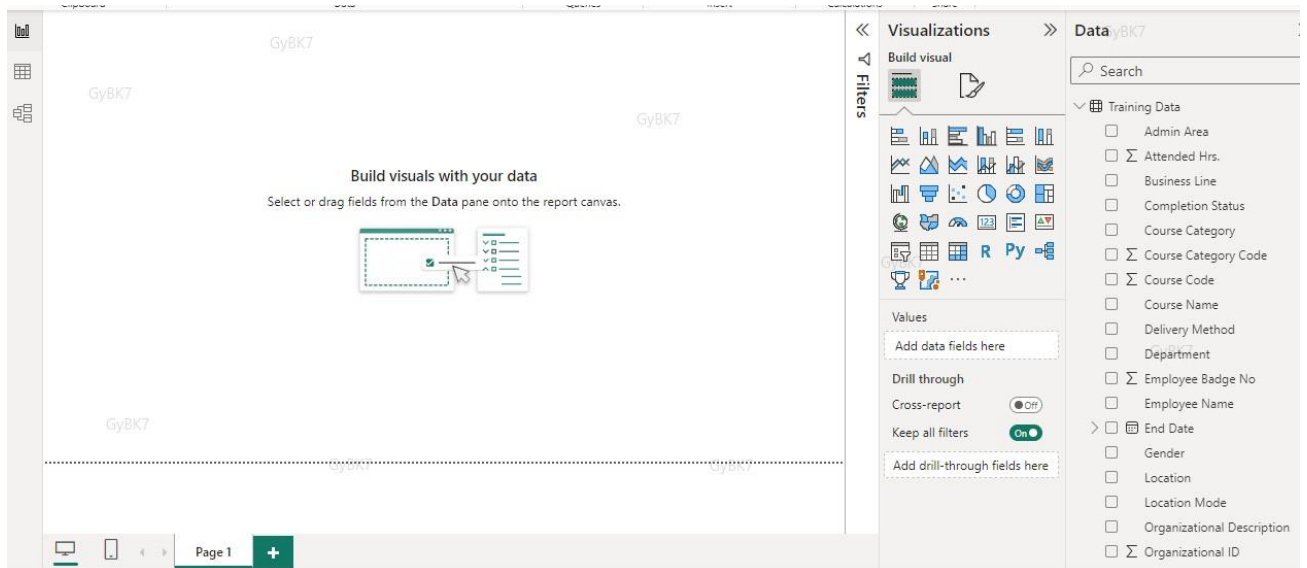
26. Modify the generated **Table.Group** function & add **Gender** to group.

	A ^B _C Course Category	A ^B _C Business Line	A ^B _C Gender	1.2 Total Hours	1 ² ₃ No. of Participants
1	General IT Training	Technical Services	M	14216.46	1951
2	General IT Training	HR & Corporate Services	M	21707	2940
3	General SAP Training	Technology & Innovation	M	8976	122
4	General IT Training	Technology & Innovation	M	4139.5	531
5	Digital Technologies	Technical Services	M	952	27
6	General IT Training	Upstream	F	2488.98	309
7	SAP Certifications	Technology & Innovation	M	1600	36
8	General IT Training	Upstream	M	11682.48	1433
9	Advance Information Security	Technical Services	M	1320	128
10	General IT Training	Downstream	M	392.5	37
11	Digital Technologies	Technology & Innovation	M	3096	101
12	General IT Training	Finance	F	3559	364
13	General IT Training	Technical Services	F	6339	758
14	General IT Training	HR & Corporate Services	F	14176.98	1590
15	General IT Training	Downstream	F	557.5	65
16	SAP Certifications	HR & Corporate Services	M	560	15
17	General SAP Training	HR & Corporate Services	M	1152	47
18	Advance Information Security	Technology & Innovation	M	3446	122
19	General IT Training	Finance	M	4202.48	451

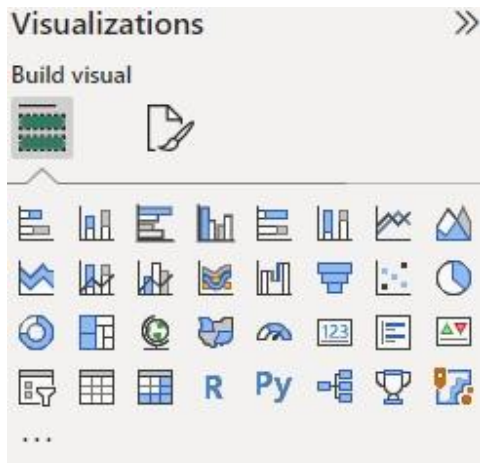
27. Once done, delete the above grouping & close the editor. You will back to the **Report View**.
28. Save your work.

Exercise 3: Creating Visualizations

Report view contains the page (canvas), Data panel, filter panel & visualization library. You can design the dashboard by inserting the visuals into the page from visualization and drag & drop data elements from Data panel into them.



Power BI provides variety of ways to present the data both visually or in tabular form.



To learn more about visualization types in Power BI, visit the below link.

[Visualization types in Power BI - Power BI | Microsoft Docs](#)

Power BI also support Custom Visuals. You can download & install custom visuals from Microsoft App Source. Explore the custom visual library from the link below:

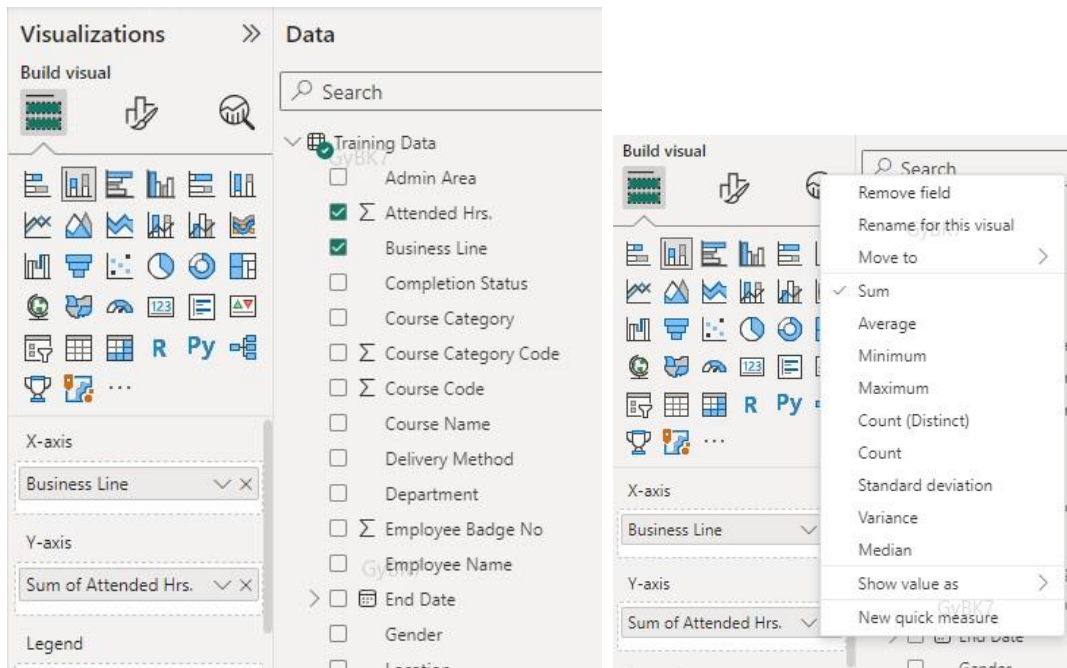
[Business Apps – Microsoft AppSource](#)

Steps:

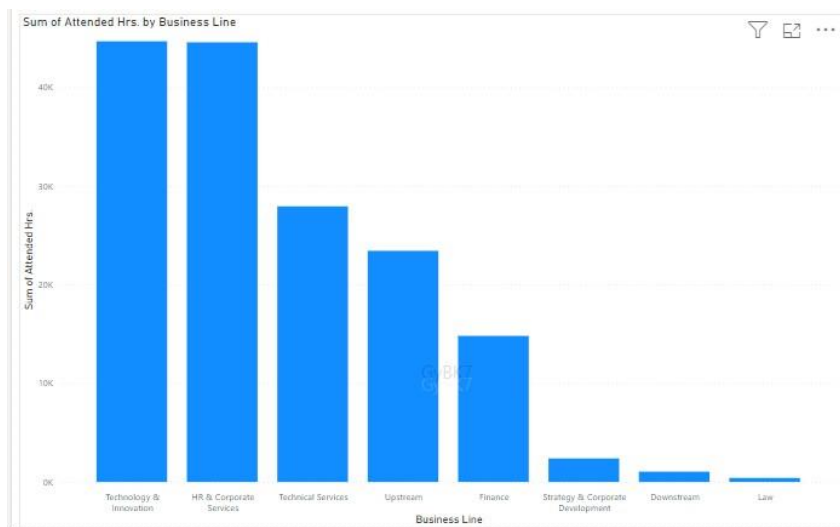
1. While in Report view, double click on Page 1 and rename it to **Practice**.



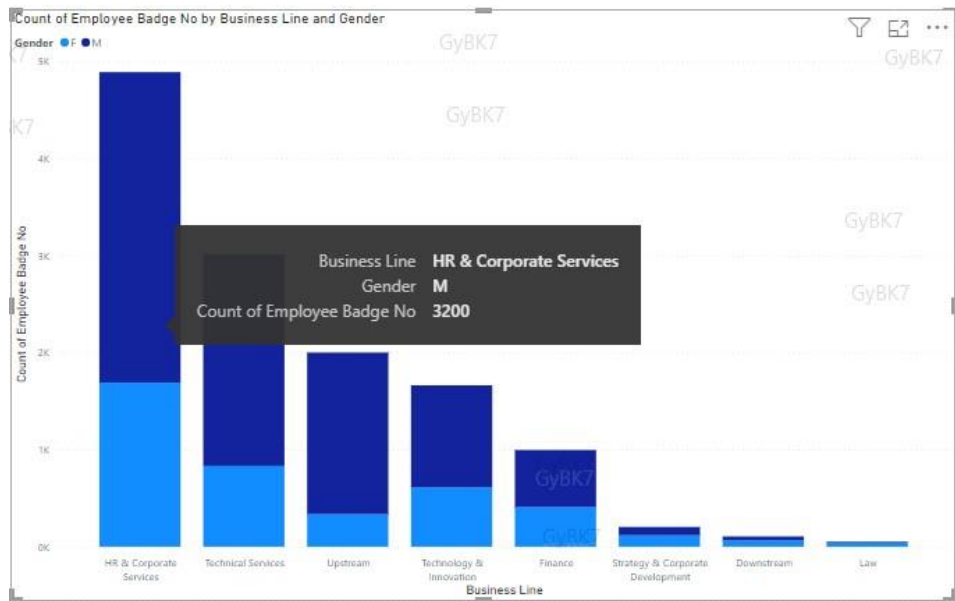
2. Insert **Stacked Column Chart** into the page.
3. From the Data, insert **Business Line** into X-axis and **Attended Hrs.** into Y-axis. Make sure Attended Hours display as Sum.



4. The graph will be populated as shown below. *(Showing Total Training Hours by Business Line)*



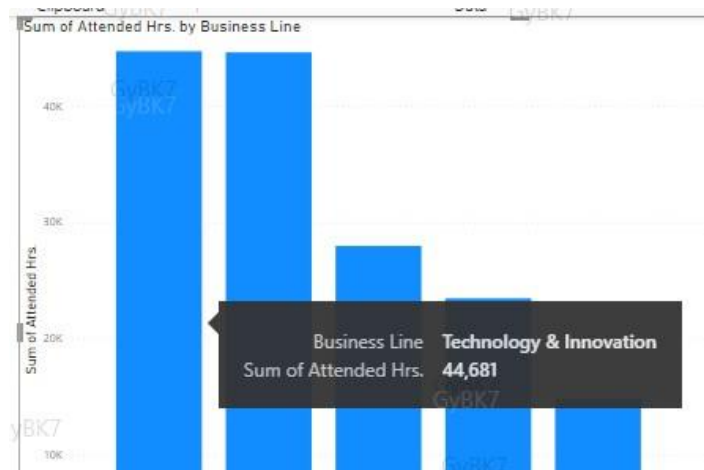
5. Insert **Gender** into legend and analyze the data.
6. Remove Attended Hrs. from Y-axis & insert **Employee Badge No.** instead. Make sure it should calculate as count.
7. Hover the mouse over the visual to see the values.



8. We can also set column properties one time using **Column Tools**. For example, select the Attended Hrs. from the Data panel, the column tools will be appeared on top.

Name Attended Hrs.		Format Decimal number		Summarization Sum	
Data type Decimal number		Formatting 0		Data category Uncategorized	
Structure		Formatting		Properties	

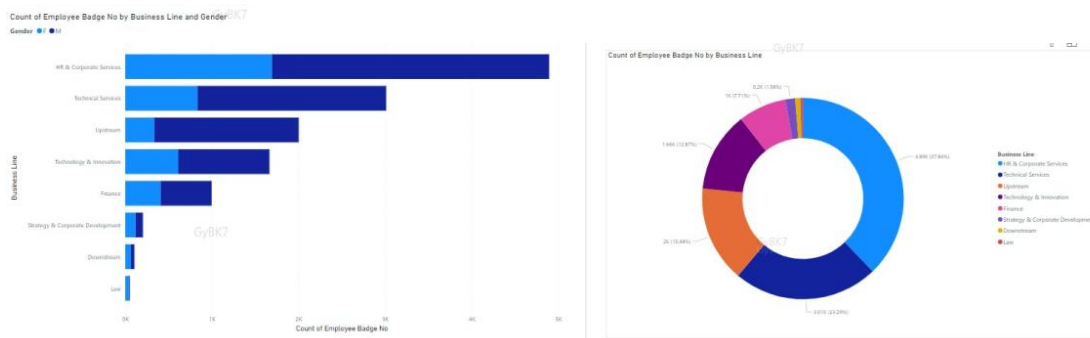
9. Select the thousand separator and round values to 0 decimal places. Now whenever we insert Attended Hrs. on any visual it will display values always with the settings we configured using column tools.



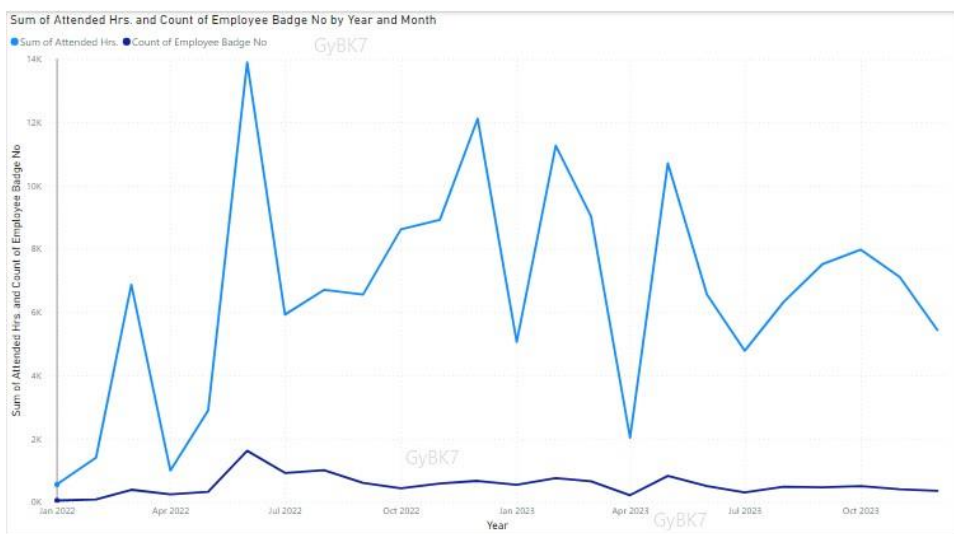
10. **Explore yourself:** Spend some time to explore **Table\Column tool Options**. Click on any field under Data panel so table & column tools are visible on main toolbar.

Note: Table/column/Measures tools in Report view help to apply some quick changes on your data without need to go to Power Query Editor.

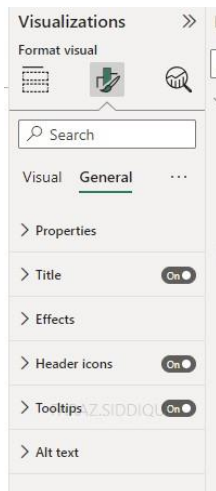
11. Select the above graph & from Visualizations, choose **stacked bar chart**. It will convert to a bar chart. Similarly, convert it into **donut chart**. Remove Gender from the graph.



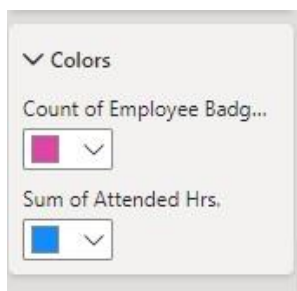
12. **Explore yourself.** Explore different types of charts. Spend some time to practice with them.
13. **Challenging Task:** Users are interested to see Attended Hrs. and No. of Participants over Time (End Date). Insert a new page and create a graph similar to below.



14. Most of the chart properties, you can set from **format visual** tab.



15. Select the above line chart & then from the format your visual ⑦ Visual ⑦ Legend ⑦ set the position Center Right. Also increase Text size from Legend ⑦ Text. Make the text **Bold**.
16. Open the Colors panel from Visual ⑦ Lines & apply different colors if you want.

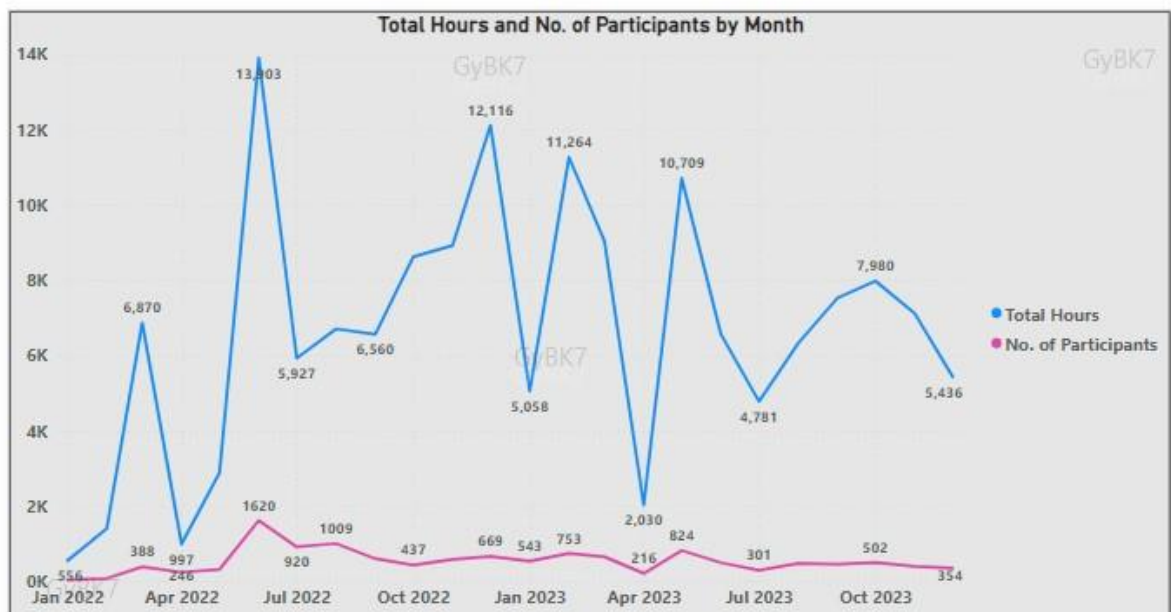


17. Enable **Data Labels**. Increase text size available under the values. Select Display units to display numbers in Millions, Thousands or choose None to show as real number.
18. Increase Text font size of both X-axis & Y-axis if needed. Make the text **Bold** if needed.
19. General ⑦ Effects ⑦ Shadow On.
20. Visual Border on.
21. Change Chart Background color if needed.
22. Make the title in the middle & call it **Total Hours & No. of Participant by Month**.
23. Under **Y-axis**, double click the columns & change the column labels to Total Hours & No. of Participants respectively. *(This is applied to selected visual only).*

X-axis	
End Date	▼ X
Year	X
Month	X

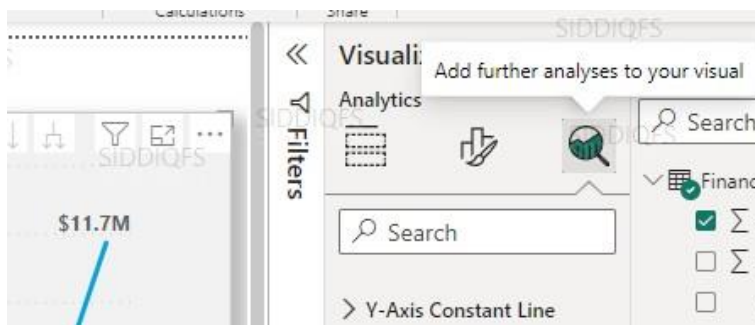
Y-axis	
Total Hours	▼ X
No. of Participants	▼ X

24. You will see the impact on a graph as shown below.

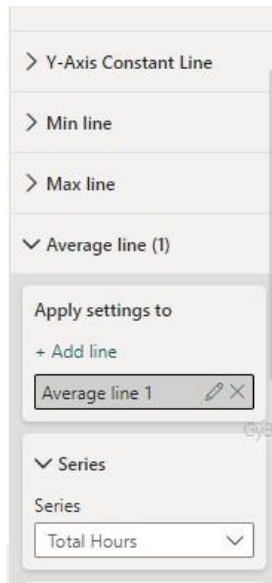


25. **Explore yourself.** Explore some other options from the **format visual** tab.

26. While selecting the above visual, click on **Add further analysis tab** (next to format visual tab).



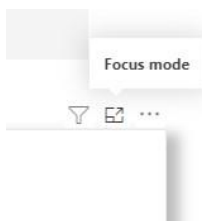
27. You can add more analysis. For example, create **Average line** over Total Hours.



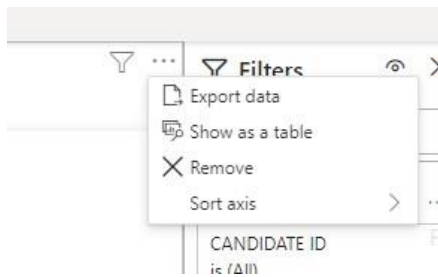
28. The Average line will be added as shown below.



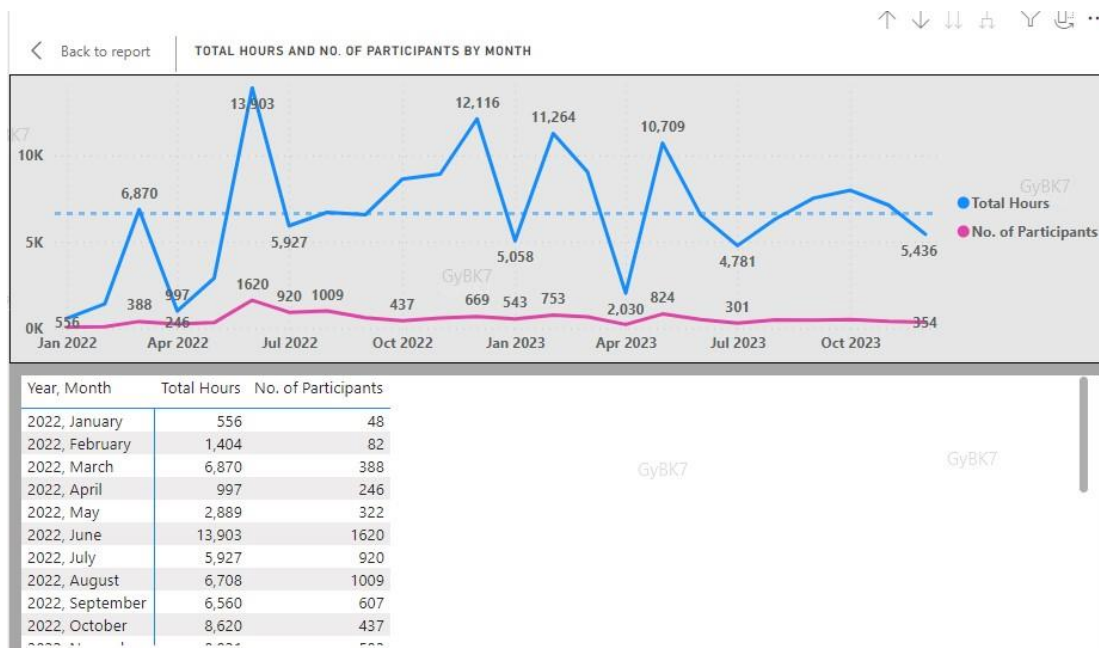
29. On the right corner of the graph, click on **focus mode**. the graph will show in full screen mode.



30. Click on 3 dots... to see some more options.



Note: For example, you can export the data behind this visual by selecting **Export data** option or **Show as a table** to display graph data as table.

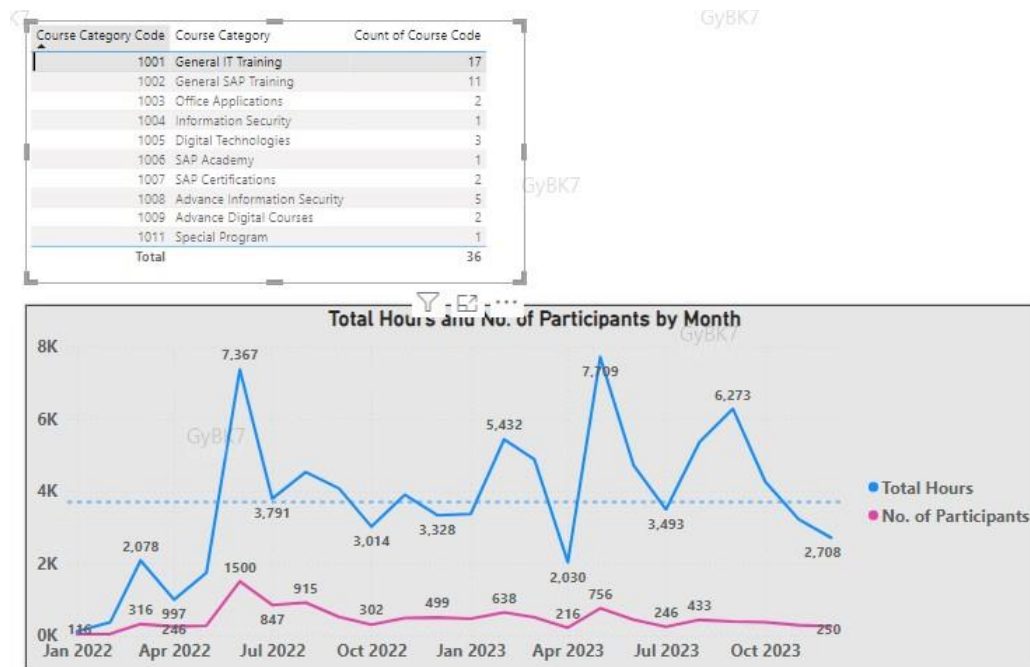


31. Click on **back to report** to back to the report view.
32. **Explore yourself.** Explore different visuals & their properties. Spend some time to practice with them.
33. Insert **Table** visual into the page. populate it with **Course Category Code**, **Course Category** & **Course Code (as Count Distinct)**.

Course Category Code	Course Category	Count of Course Code
1001	General IT Training	17
1002	General SAP Training	11
1003	Office Applications	2
1004	Information Security	1
1005	Digital Technologies	3
1006	SAP Academy	1
1007	SAP Certifications	2
1008	Advance Information Security	5
1009	Advance Digital Courses	2
1011	Special Program	1
Total		36

34. Select any Course Category from the table, the data in the graph will be filtered accordingly or vice versa. This is because of default **interaction feature** between the visuals in Power

BI.

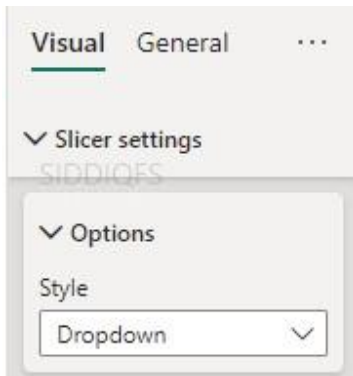


Note: You can disable the default interaction option from Format Menu Edit Interactions select **None**. OR enable it by clicking on **filter** icon.

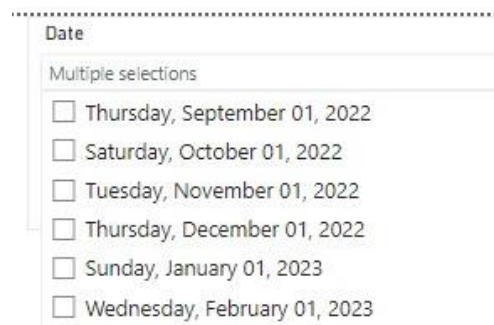
35. Insert **slider** into the canvas & populate it with **End Date**. You can filter the data either using calendar or date slider.

End Date

36. Select the slicer and from format tab, change the style to **Dropdown** from Visual  Slicer settings.



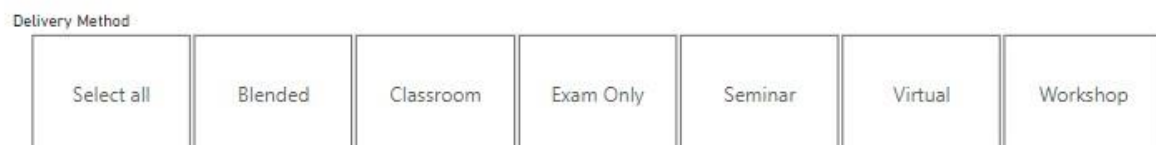
37. Date will appear as dropdown. Select multiple dates to filter the data.



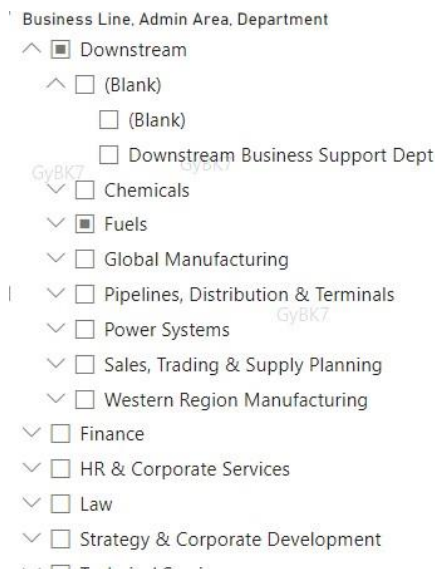
38. Convert it back to **Slider** view by selecting between style.
39. **Explore yourself:** Create slicer on some other fields & explore style & other properties.

Note: Slicer is the most common way to filter the data in power BI. Other options include page filters, interactions and drill-down.

40. **Challenging Task:** Create a slicer which shows **Delivery Method** as below. Also make sure that there is a **Select all** option to select all values at once.



41. **Challenging Task:** Create a **hierarchical slicer** so users can able to filter the data by selecting Business Line, Admin Area & Department within the same slicer.



Map Visualizations:

Power BI support built-in maps. These maps link with Microsoft Bing Map Services. You can also use custom maps as custom visuals.

To learn more about maps in Power BI, click on like below:

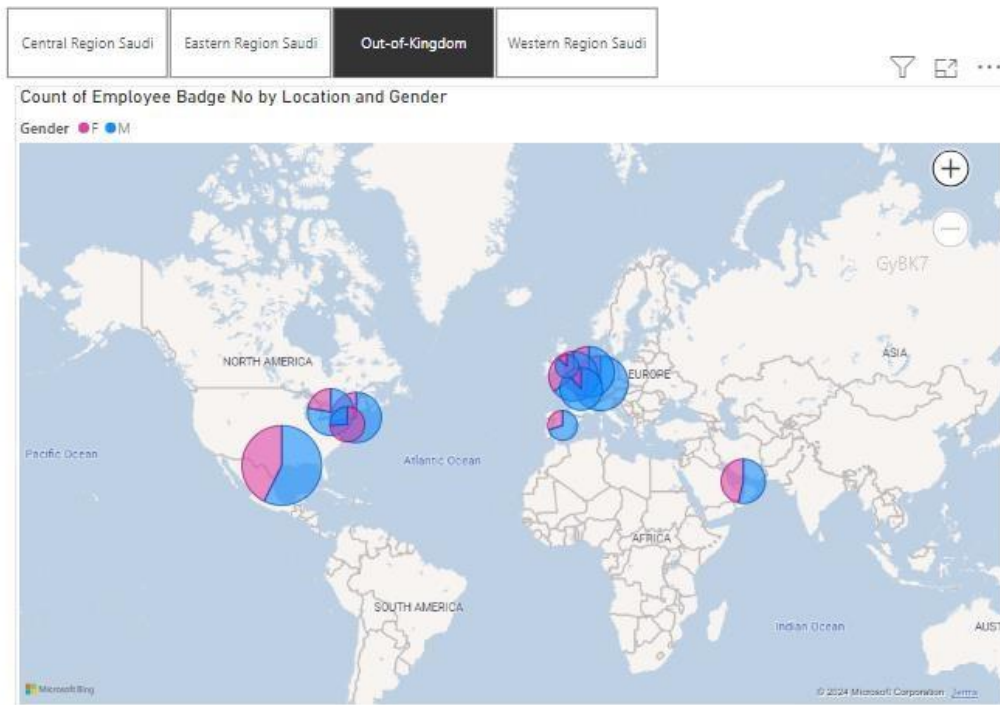
[Use Shape maps in Power BI Desktop \(Preview\) - Power BI | Microsoft Docs](#)

42. Insert new page.
43. From Visuals, insert **Map**.
44. Insert **Location** into Location & **Employee Badge No (as count)** in Bubble Size. This will show No. of Participants per City (*both In-kingdom & out-of-Kingdom*).

Count of Employee Badge No by Location

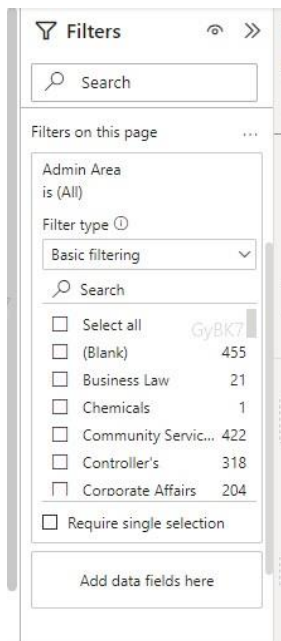


45. **Explore yourself:** Explore some of the properties of Map visual from **Format Visual** tab. **Note:** you can also use map latitude/longitude instead of location description.
46. **Challenging Task:** Update the above map to add **gender** & apply some formatting from format visual tab. Also insert **Region** as slicer so users can filter the map accordingly.



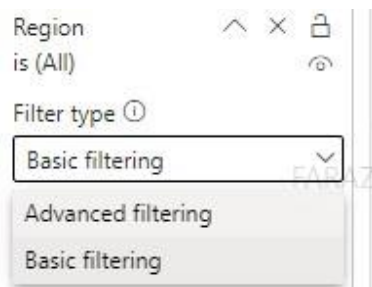
Page Filters:

47. From View, click on **Filters** to show page filters if not visible.
48. Drag & drop **Admin Area** into **filters on this page**. It will filter all visuals available at **page level**.

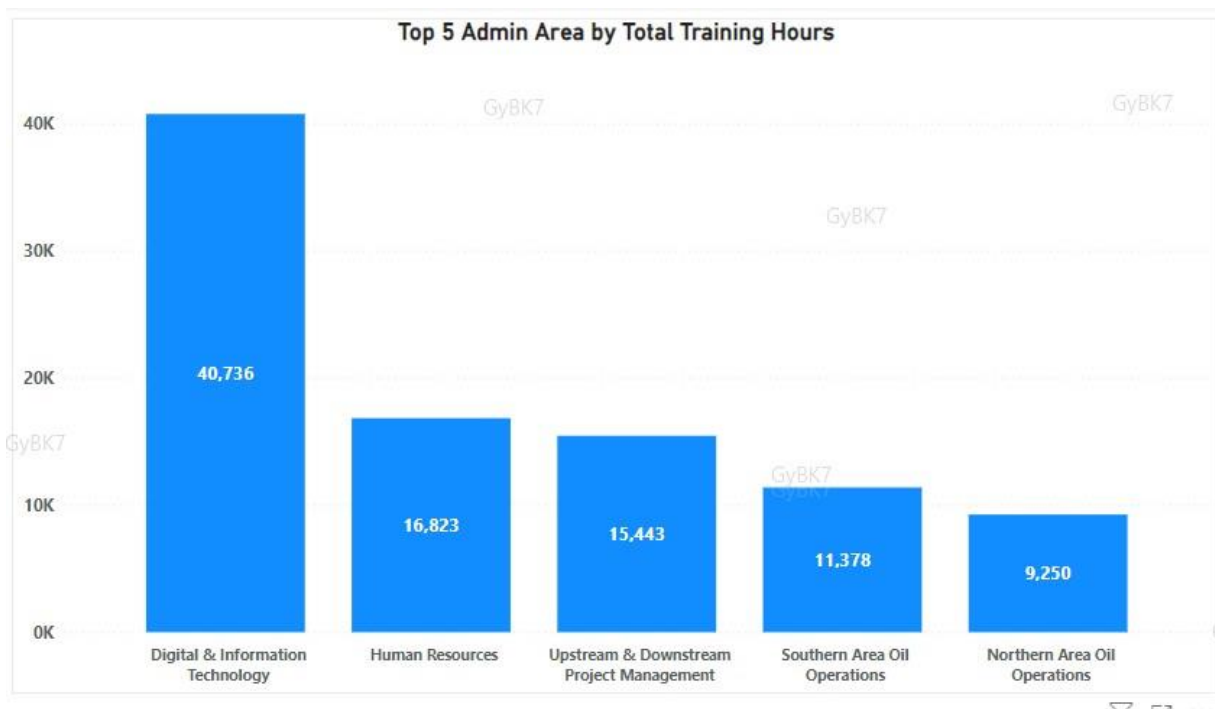


Note: you can also apply filters at visual level or all pages within the report as well.

49. **Explore yourself.** Explore various properties of filters particularly the advance filtering options.



50. **Challenging Task:** Users are interested to see **Top 5 Admin Area by Total Training Hours (Attended Hrs)**. Apply the **Ranking** from the filter panel to fulfill the requirements. Also make sure it will be sorted on descending order & display values as shown below.



Conditional Formatting:

Through Conditional formatting, you can highlight data based on conditions & rules.

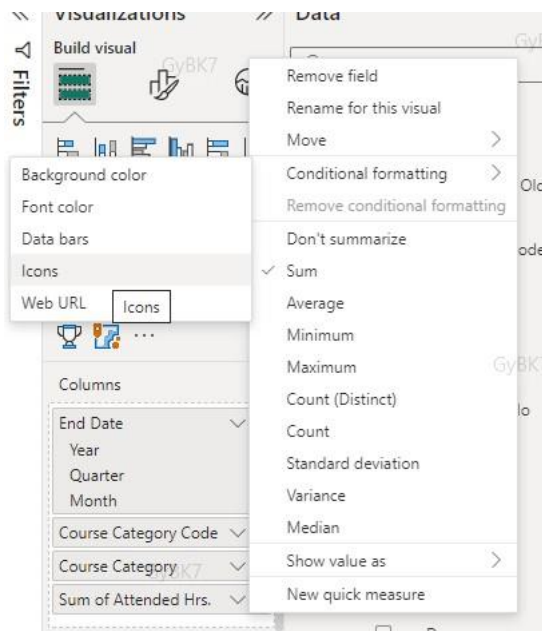
To learn more about Conditional Formatting in Power BI, visit the link below:

[Conditional table formatting in Power BI Desktop - Power BI | Microsoft Docs](#)

51. Insert new page and insert a table to show detailed report as below.

Year	Quarter	Month	Course Category Code	Course Category	Sum of Attended Hrs.
2022	Qtr 1	January	1001	General IT Training	116
2022	Qtr 1	January	1005	Digital Technologies	440
2022	Qtr 1	February	1001	General IT Training	372
2022	Qtr 1	February	1003	Office Applications	768
2022	Qtr 1	February	1005	Digital Technologies	264
2022	Qtr 1	March	1001	General IT Training	2,078
2022	Qtr 1	March	1002	General SAP Training	2,872
2022	Qtr 1	March	1003	Office Applications	1,176
2022	Qtr 1	March	1004	Information Security	288
2022	Qtr 1	March	1005	Digital Technologies	456
2022	Qtr 2	April	1001	General IT Training	997
2022	Qtr 2	May	1001	General IT Training	1,737
2022	Qtr 2	May	1002	General SAP Training	192
2022	Qtr 2	May	1003	Office Applications	280
2022	Qtr 2	May	1004	Information Security	288
2022	Qtr 2	May	1005	Digital Technologies	280
2022	Qtr 2	May	1008	Advance Information Security	112
2022	Qtr 2	June	1001	General IT Training	7,367
2022	Qtr 2	June	1002	General SAP Training	3,120
2022	Qtr 2	June	1003	Office Applications	840
2022	Qtr 2	June	1004	Information Security	200

52. Select Columns ⑦Attended Hrs.⑦Conditional formatting⑦Icons



53. Choose **Shape** icons from the Style drop down.

54. Specify the following Rules. Make to use **Number** instead of percentage.

- If Attended Hrs. more than 3000 then green
- Between 1000 & 3000 yellow

c. Less than 1000 red

Icons - Sum of Attended Hrs.

Format style

Rules

Apply to

Values only

What field should we base this on?

Sum of Attended Hrs.

Summarization

Sum

Icon layout

Left of data

Icon alignment

Top

Style

Custom

Rules

Reverse icon order

New rule

If value	>	3000	Number	and	<=	Max	Number	then		↑ ↓ ×
If value	>=	1000	Number	and	<=	3000	Number	then		↑ ↓ ×
If value	>=	0	Number	and	<	1000	Number	then		↑ ↓ ×

[Learn more about conditional formatting](#)

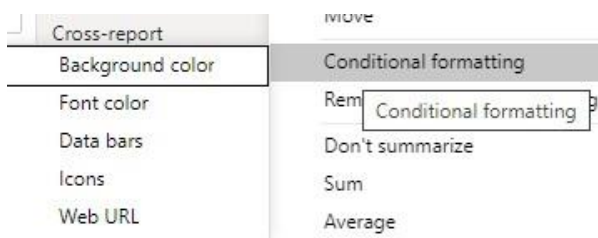
OK

Cancel

55. The table will be updated as shown below:

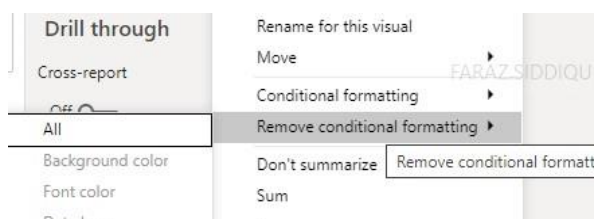
Year	Quarter	Month	Course Category Code	Course Category	Sum of Attended Hrs.
2022	Qtr 1	January	1001	General IT Training	116
2022	Qtr 1	January	1005	Digital Technologies	440
2022	Qtr 1	February	1001	General IT Training	372
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2022	Qtr 2	June	1001	General IT Training	7,367
2022	Qtr 2	June	1002	General SAP Training	3,120
2022	Qtr 2	June	1003	Office Applications	840
2022	Qtr 2	June	1004	Information Security	200
2022	Qtr 2	June	1005	Digital Technologies	800

56. **Explore yourself.** Explore & practice with other options for highlighted data including **data bars**, **background color** etc. For example, apply **background color** based on same rules.

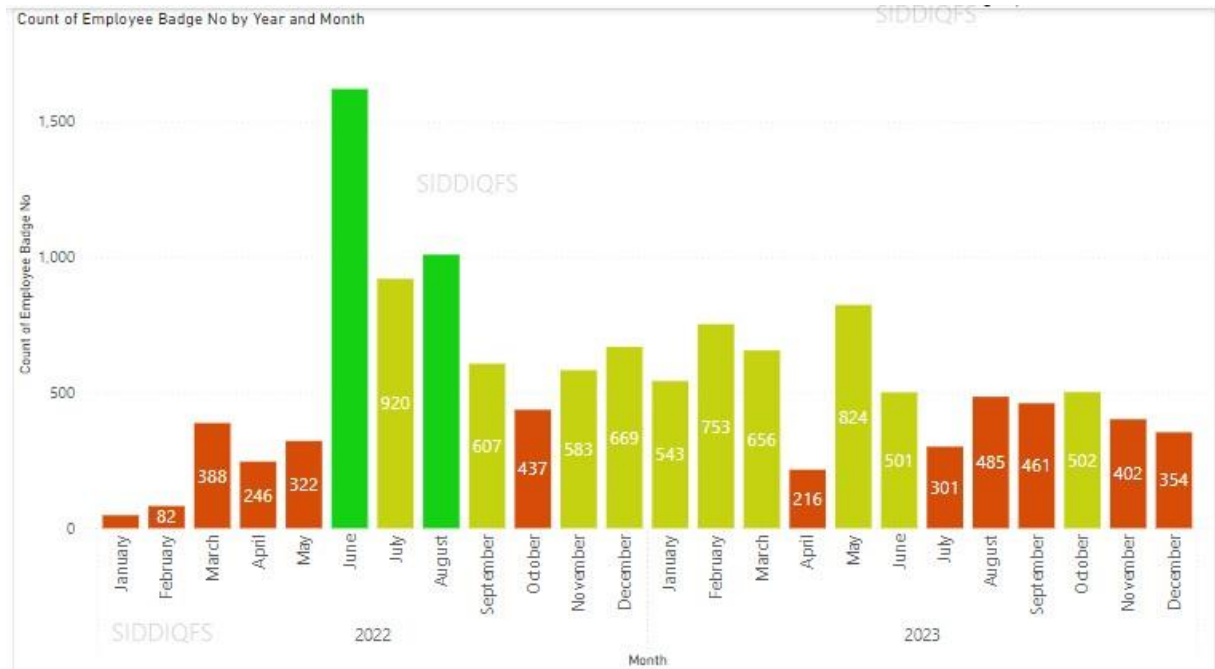


Year	Quarter	Month	Course Category Code	Course Category	Sum of Attended Hrs.
2022	Qtr 1	January	1001	General IT Training	116
2022	Qtr 1	January	1005	Digital Technologies	440
2022	Qtr 1	February	1001	General IT Training	372
2022	Qtr 1	February	1003	Office Applications	768
2022	Qtr 1	February	1005	Digital Technologies	264
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2022	Qtr 1	March	1002	General SAP Training	2,872
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2022	Qtr 2	June	1001	General IT Training	7,367
2022	Qtr 2	June	1002	General SAP Training	3,120
2022	Qtr 2	June	1003	Office Applications	840
2022	Qtr 2	June	1004	Information Security	200
2022	Qtr 2	June	1005	Digital Technologies	800
2022	Qtr 2	June	1008	Advance Information Security	1,216

57. In order to remove any conditional formatting, click on **Remove conditional formatting** (all or particular formatting).



58. **Challenging Task.** You can also apply conditional formatting on graphs. Insert a new chart with Year/Month and No. of Participants. Apply rules based on No. of Participants. (**Hint:** Format visual Columns Color).



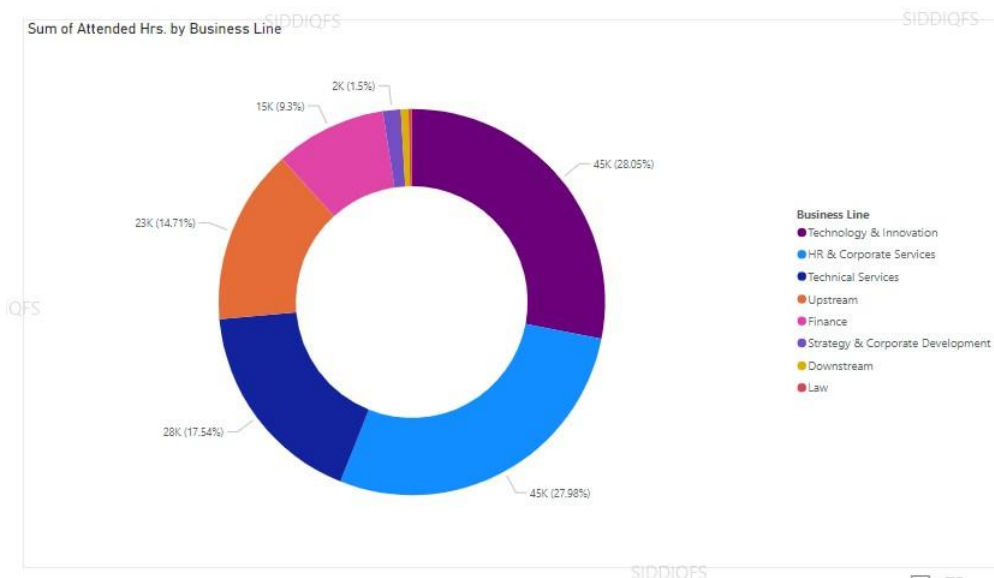
Drill Mode in Power BI:

Using drill down and drill up on your data points, you can explore in-depth details about your data.

To learn more about Drill mode in power bi, visit the link below:

[Drill down and drill up in a visual - Power BI | Microsoft Docs](#)

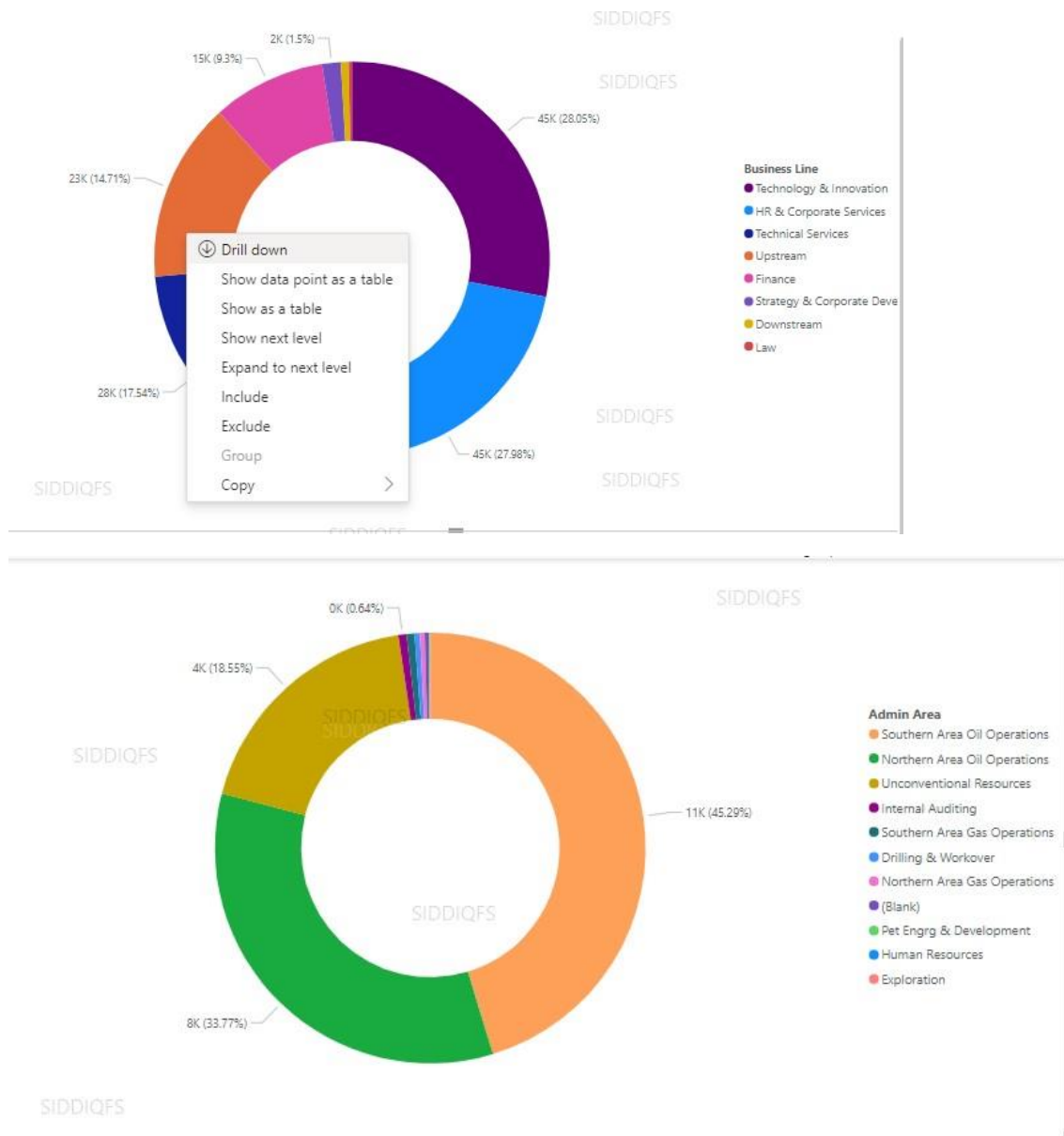
59. Insert **donut chart** & populate with Business Line & Attended Hrs.



60. Insert **Admin Area** next to Business Line under Legends. You will notice that drill options enabled. Click on **turn on Drill down**.

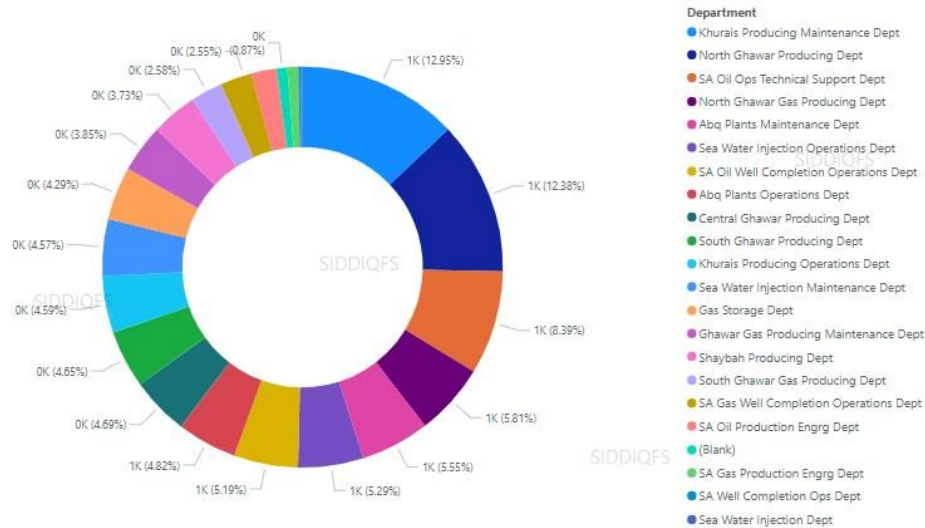


61. Right click over Upstream Business line and select **Drill down**. Graph will display the data now at Admin Area level.

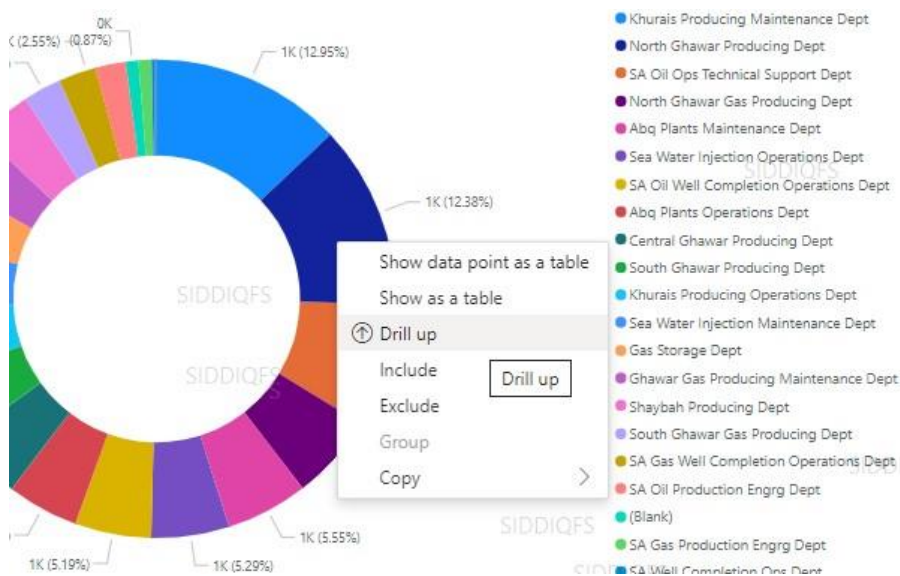


62. Add **Department** below Admin Area under Legend. click on drill down again to show the data at department level for selected Business Line & Admin Area.

Sum of Attended Hrs. by Business Line, Admin Area and Department



63. Similarly, you can **drill up** to jump to higher level.



64. Save your work.

Note: using drill menu, you can apply other drill options as well.

65. **Challenging Task.** Drill mode is built-in in matrix visual. Insert **Matrix** visual and try to present the data as shown below. Users can expand & collapse the fields to see the data in different levels.

Total Hours by BL/Admin Area/Dept. and Delivery Method

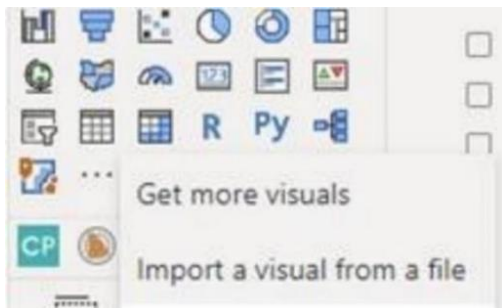
Business Line	Blended	Classroom	Exam Only	Seminar	Virtual	Workshop	Total
Downstream	32	755	6	54	136	64	1,046
Finance	200	11,251	57	241	2,419	647	14,814
HR & Corporate Services	1,032	27,804	261	1,331	10,919	3,216	44,563
Law		196	6	7	167	32	408
Strategy & Corporate Development	104	1,529	3	183	376	188	2,383
Technical Services	1,064	14,562	246	1,466	8,750	1,859	27,947
		438		50	105	32	625
Digital & Information Technology		68	3		203		274
Engineering Services	296	2,812	39	309	1,262	320	5,037
Infrastructure & JV Project Management	168	1,581	18	248	1,237	396	3,647
Internal Auditing		4			32		36
Procurement & Supply Chain Management	152	1,422	18	75	787	128	2,581
Project Management	8	252	3		42		305
Upstream & Downstream Project Management	440	7,986	165	784	5,085	983	15,443
Technology & Innovation	168	39,230	54	1,780	1,659	1,791	44,681
Upstream	672	13,328	99	1,523	4,890	2,928	23,439
Total	3,272	108,655	732	6,585	29,315	10,725	159,283

Using Custom Visuals:

Power BI support Custom Visuals. You can download & install custom visuals from Microsoft App Source:

[Business Apps – Microsoft AppSource](#)

66. Select **Import a visual from a file** from Visualizations tab.

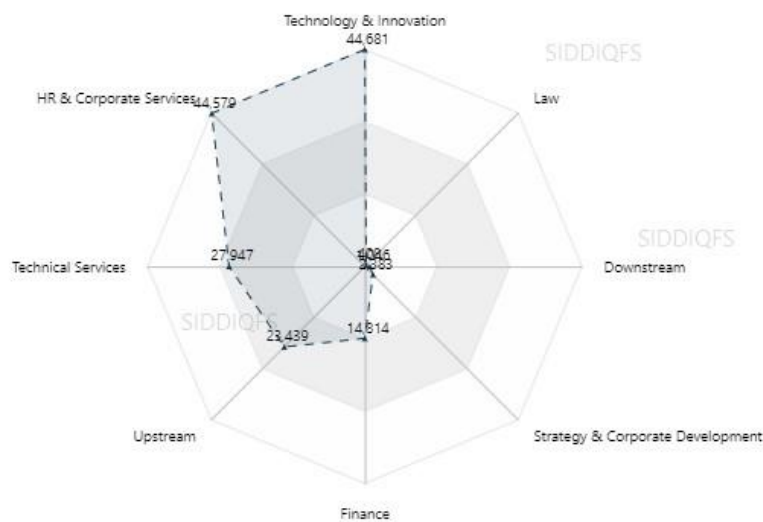


67. Browse the provided custom visual files. (*custom visuals available in .pbviz file format*).
68. For example, search the **radar chart** & import into the library.
69. Custom Visuals will be added under Visualizations.



70. You can use custom visuals same like built-in visuals.

Sum of Attended Hrs. by Business Line



71. Try to explore some other custom visuals as well.

72. Save your work.

Exercise 4: Build Data Model

As mention in use case overview, **Training Data** file contains the completion data for all training courses. However, training cost & budget details are maintained separately in **Training Cost** data source. For calculating our KPI's, we need to load the data from both files.

Your Task: You are required to load the data from **Training Cost** file & build the data model so we can able to calculate the KPI's.

Steps:

1. Load the new Data File called **Training Cost (Get Data🔗 Text/CSV)** from the provided dataset folder.
2. Insert **Table** visual into a new page.
3. Drag & drop **Business Line & Attended Hrs.** from **Training Data** and **Plan Budget** from **Training Cost**.

Business Line	Sum of Attended Hrs.	Sum of Plan Budget
Downstream	1,046	\$12,644,336
Finance	14,814	\$12,644,336
HR & Corporate Services	44,563	\$12,644,336
Law	408	\$12,644,336
Strategy & Corporate Development	2,383	\$12,644,336
Technical Services	27,947	\$12,644,336
Technology & Innovation	44,681	\$12,644,336
Upstream	23,439	\$12,644,336
Total	159,283	\$12,644,336

4. Values are repeated for Plan Budget. Why?

That's because the data comes from two different sources and there is no relationship exists between them so Power BI is doing **Cross-join** by default. To solve this problem, we need to build **Data Model**.

To learn more about data modeling in Power BI, visit the link below:

[Model data in Power BI - Learn | Microsoft Docs](#)

Note: There are several techniques to build data models. It is not necessary that what technique used in this scenario can exactly work with your requirement. Here, we are going to use a well-known data modeling technique called **Star Schema**.

Star Schema is a Data Warehousing technique where Fact tables (Transactional tables) are linked with one or more Dimension tables (Master tables) but the fact tables are not linked with each other.

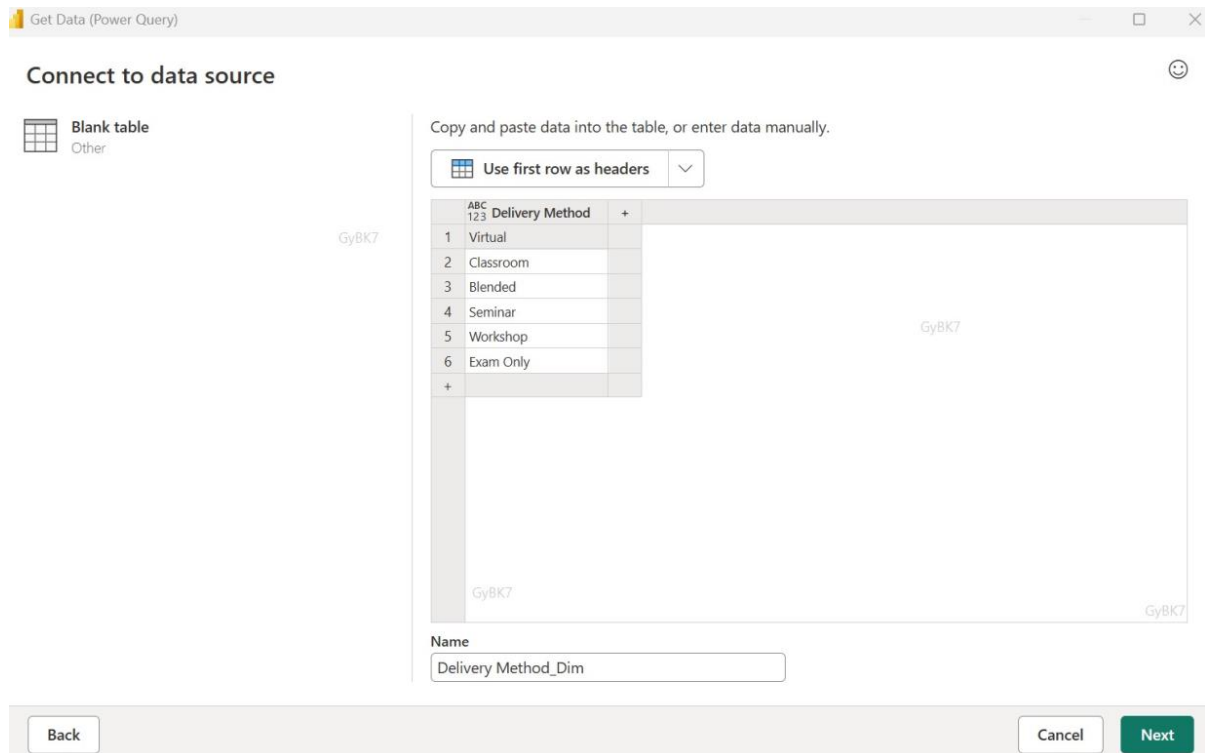
If dimension tables (master data) already exist in the source then you can simply bring them into your model. In our scenario, we don't have master data available separately so we have to create them in Power BI.

There are **4** ways to create dimension tables in Power BI. We will apply all in our use case.

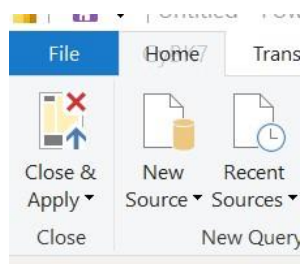
- Create Dimension table from scratch.
 - a. Using Enter Data.
 - b. Using DAX
- Generate Dimension table from Fact table.
 - a. Using Power Query.
 - b. Using DAX.

Create Dimension table from scratch (Using Enter Data):

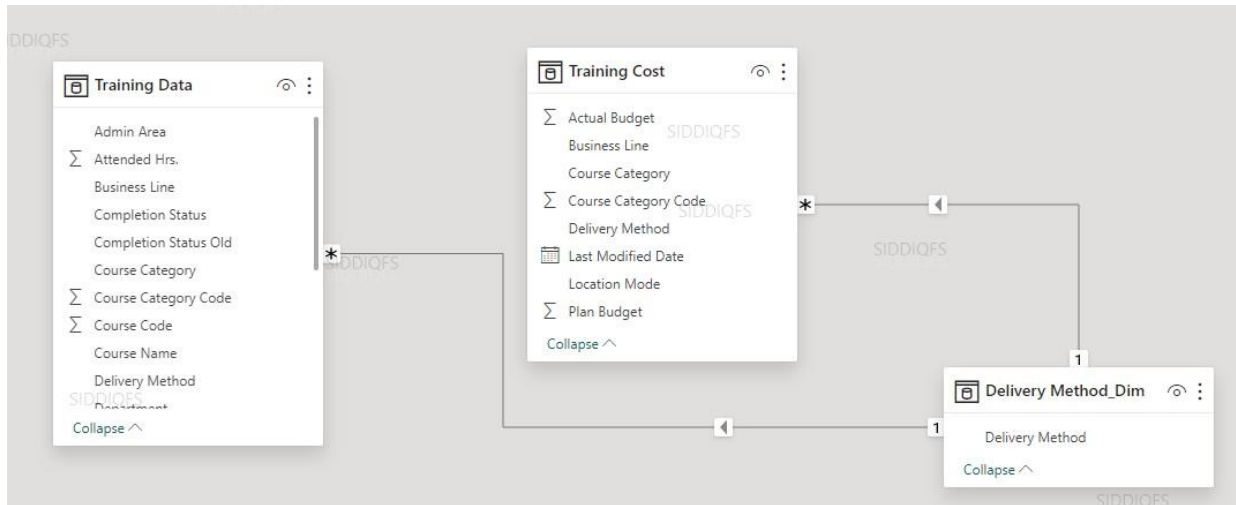
5. From Home, click on **Enter Data**.
6. Name the table as **Delivery Method_Dim** and write values in a column (Virtual, Classroom, Blended, Seminar, Workshop, Exam Only). Change column name to **Delivery Method**. Make sure **Use first row as headers** is selected. Click **Next**.



7. It will open **Power Query Editor**. Click on **Close & Apply**.



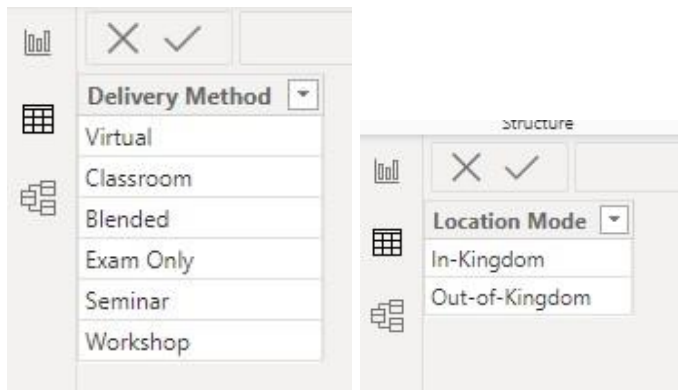
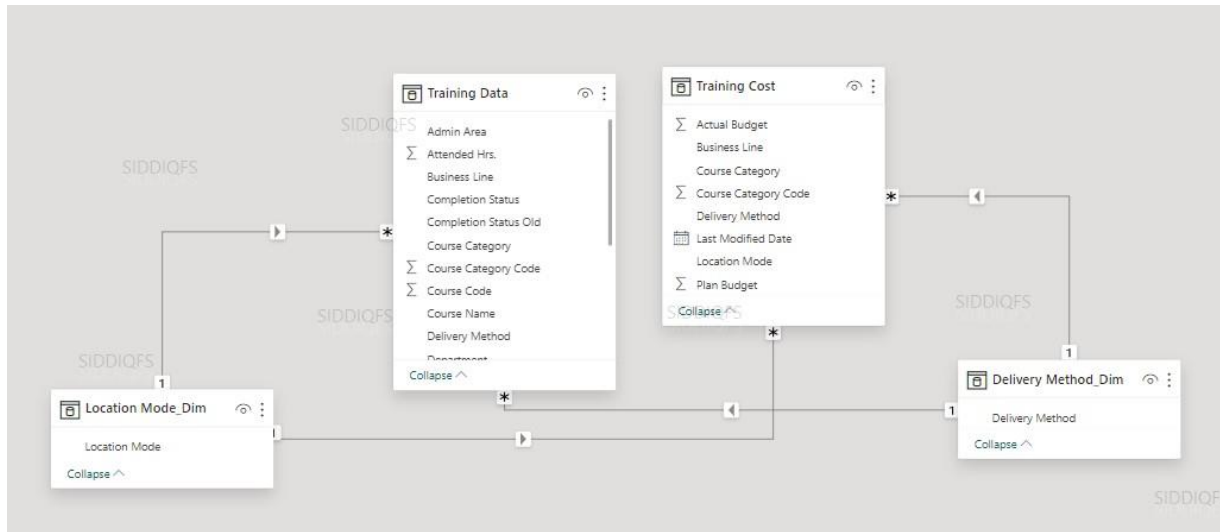
8. Power BI creates a new table & also create joins by itself. (if column names are same. To see the joins, click on **Model view**).



9. Repeat the above steps (from 5 to 8) & create dimension table for **Location Mode** called Location Mode_Dim. (Values : In-Kingdom , Out-of-Kingdom).

Note: It's a good practice to add **_Dim** in your dimension tables and **_Fact** in your fact tables.

10. Click on Model & Table view to make sure joins and values are correct.



Generate Dimension table from Fact table (Using Power Query):

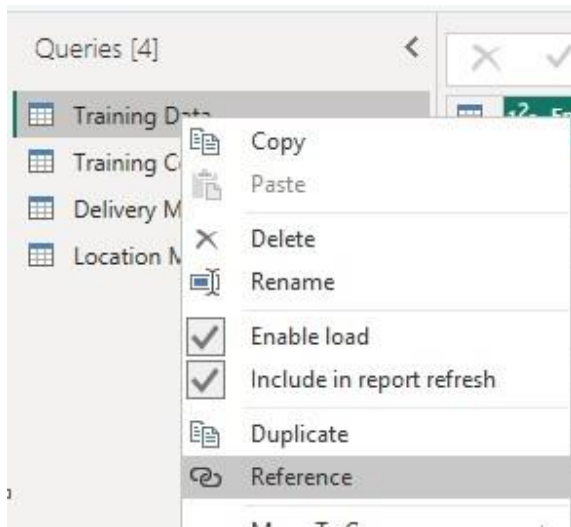
As there are multiple values for **Business line**, instead of creating it from scratch, we can generate a dimension table called **Business line _Dim** directly from the fact table (*Training Data*).

Note: We can generate dimension tables from fact tables either through **Power Query** or **DAX**. Below steps for Power Query. Creating dimension tables through DAX will cover in **Exercise 5**.

11. From Queries, click on **Transform data** to go to Power Query Editor.



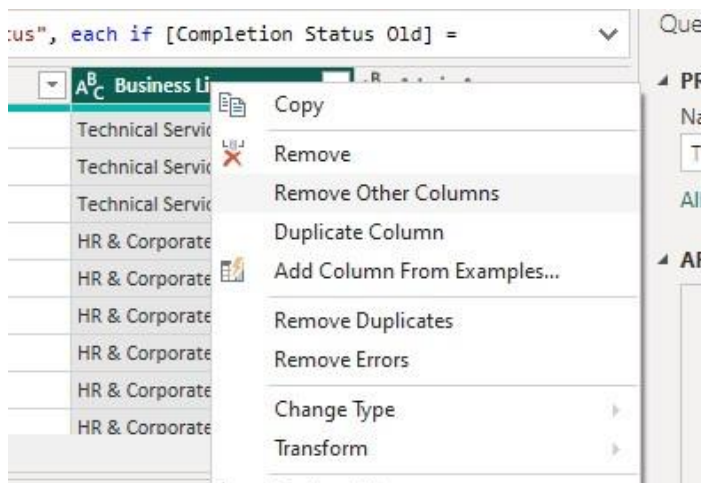
12. Select Training Data ⚙️ Right Click ⚙️ Reference.



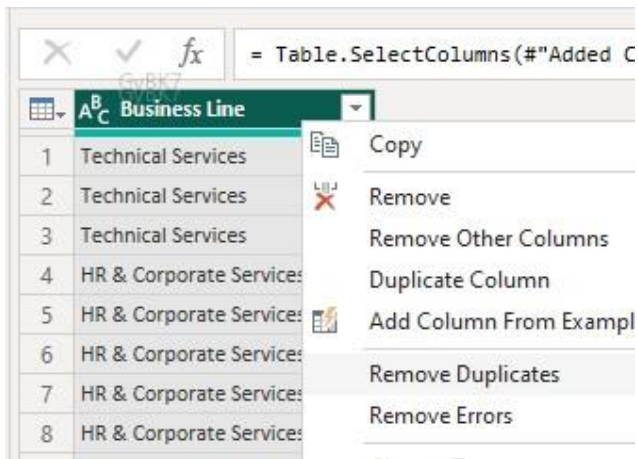
Note: Referenced table link with the main table. If main table updates, referenced table will also update automatically.

13. Rename the new table to **Business Line_Dim**.

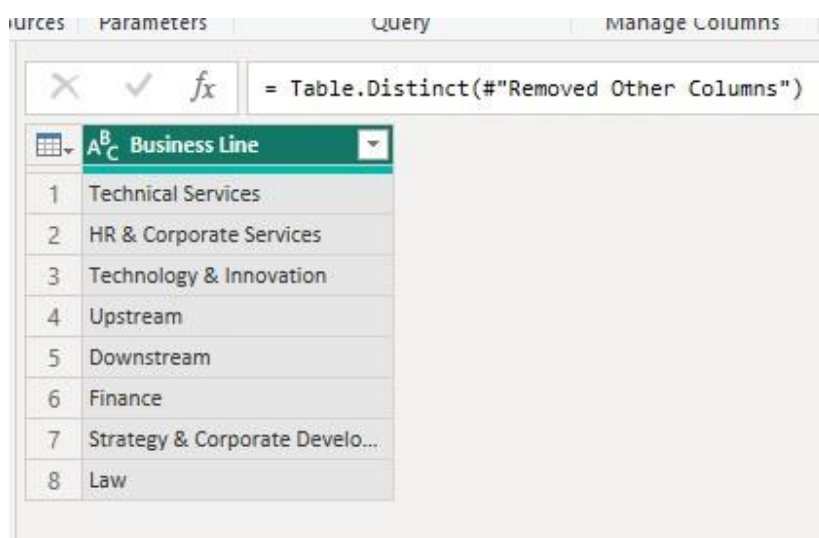
14. Select Business Line (in new table) ⚙️ Right Click ⚙️ Remove Other Columns.



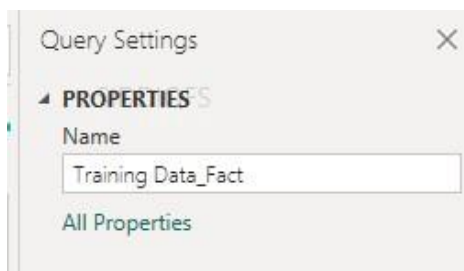
15. Select Business Line ⚙️ Right Click ⚙️ Remove Duplicates.



16. The Business Line table will now contain unique values.

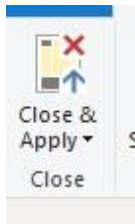


17. Rename Training Data to **Training Data_Fact** and Training Cost to **Training Cost_Fact**.

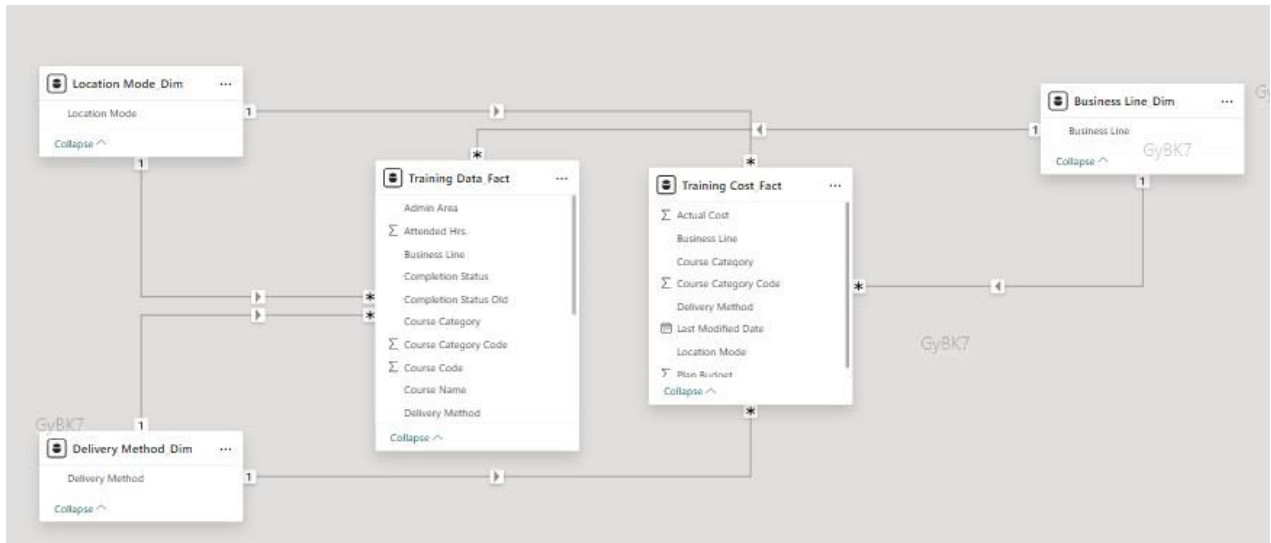


Note: It's a good practice to add **_Dim** in your dimension tables and **_Fact** in your fact tables.

18. Click on Close & Apply.



19. Click on **Model view**. Arrange it in such a way that the fact tables are in the middle surrounded by dimension tables (**Star Schema**).



20. Power BI automatically create joins if column names are same. However, you can also create joins by yourself.
21. Double click the join, you can edit relationship if needed.

Edit relationship

Select tables and columns that are related.

Training Cost_Fact

Course Category Code	Course Category	Business Line	Last Modified Date	Delivery Method	Lo
1001	General IT Training	HR & Corporate Services	Monday, January 31, 2022	Classroom	In
1002	General SAP Training	Finance	Monday, January 31, 2022	Classroom	Ok
1002	General SAP Training	Technology & Innovation	Monday, January 31, 2022	Classroom	In

Business Line_Dim

Business Line

- Technical Services
- HR & Corporate Services
- Technology & Innovation

Cardinality: Many to one (*:1)

Cross filter direction: Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

OK Cancel

Note: Cardinality is very important. Cardinality between fact & dimension is always **many to one**. (For example, many rows of business line column in a fact table link with only one row of same business line in a dimension table).

To learn more about joins & relationships, visit the links below.

[Understand star schema and the importance for Power BI - Power BI | Microsoft Docs](#)

[Model relationships in Power BI Desktop - Power BI | Microsoft Docs](#)

It is very important that all joins are correctly define otherwise you will not get the right data.

- In a table created above (step 3), update the **Business Line** & read now from the dimension table. The data will appear correctly.

Business Line	Sum of Attended Hrs.	Sum of Plan Budget
Downstream	1,046	\$866,759
Finance	14,814	\$1,107,238
HR & Corporate Services	44,563	\$2,079,742
Law	408	\$210,000
Strategy & Corporate Development	2,383	\$1,679,200
Technical Services	27,947	\$2,342,190
Technology & Innovation	44,681	\$2,482,050
Upstream	23,439	\$1,877,157
Total	159,283	\$12,644,336

23. Save your work.

Note: In order to analyze the data by **Time** (Year, Quarter, Month, Week etc.) and **Course Category**, we need to create their dimension tables. We will use **DAX** for that in next exercise.

Note: Once we have shared fields available in dimension tables, its good practice to hide them from the fact tables (*for example, hide Delivery method, location mode & business line from both fact tables as now we will insert them from dimension tables whenever needed*). In order to hide any field, right click over the field & select **hide**.

Exercise 5: Performing Calculations using DAX

In Power BI, you can perform simple to advance calculations using very powerful Data Analysis Expressions (DAX).

What is DAX?

Data Analysis Expressions (DAX) is a formula expression language used in Analysis Services, Power BI, and Power Pivot in Excel. DAX formulas include functions, operators, and values to perform advanced calculations and queries on data in related tables and columns in tabular data models.

To learn more about DAX, visit the below link.

[DAX overview - DAX | Microsoft Docs](#)

As mention in use case above, we need to calculate following KPI's & measures for our dashboard.

- **Total Hours.** (*Participant Contact Hours (PCH). The Hours credited to participants on course completion. Its calculated as course duration * Hours per day. For example, if course duration is 2 full days then Total hours will be $2 * 8 = 16$.*)
- **Total Participants.** (*No. of Participants successfully complete the training. We can calculate it by count the employee badge numbers from training data based on completion status column).*
- **Total Courses.** (*No. of distinct courses delivered. Again, its calculated from training data).*

- **Total Sessions.** (Same course can be delivered multiple times that's called **Session**. To calculate number of sessions, we need to group the data at **course level** & then do the required calculation based on **End Date** column).
- **Plan Budget.** (Initial Budget allocated for training. This is coming from Training Cost file).
- **Actual Cost.** (Actual cost of training).
- **Budget Variance** (Difference between Plan vs. Actual. This is very important KPI as management wants to minimize the gap between what was plan vs. actual cost).
- **Budget Consumption %.** (How much budget is consumed in %).
- **Avg. Cost Per Participant.** (Actual Cost divided by Total Participants. This is another very important KPI).

Your Task: You are required to create remaining dimension tables & calculate the above KPI's using DAX.

Steps:

Generate Dimension table from Fact table (Using DAX):

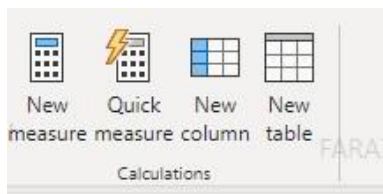
1. Course Category Dimension:

- In order to analyze the data by Course Category, we need to create Course Category dimension table. We will use DAX **Calculated table** feature for that.

To learn more about **Calculated tables**, visit the link below.

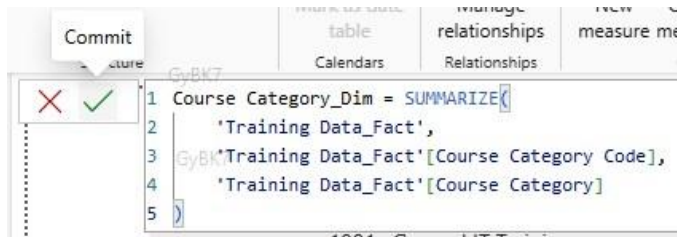
[Using calculated tables in Power BI Desktop - Power BI | Microsoft Learn](#)

- From Main Menu, click on **Modeling** tab then click **New table** from Calculations panel.



- In formula bar, write the following expression & click Commit.

```
Course Category_Dim = SUMMARIZE(
    'Training Data_Fact',
    'Training Data_Fact'[Course Category Code],
    'Training Data_Fact'[Course Category] )
```



Note: Summarize or Distinct functions are used to get unique values from source tables.

To learn more about **Summarize function**, visit the link below.

[SUMMARIZE function \(DAX\) - DAX | Microsoft Learn](#)

To learn more about **Distinct function**, visit the link below.

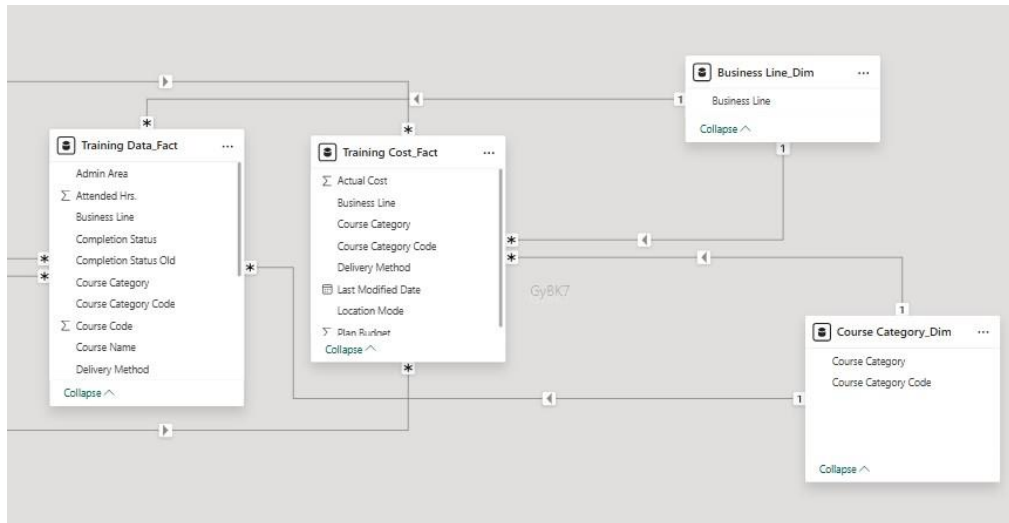
[DISTINCT \(table\) function \(DAX\) - DAX | Microsoft Learn](#)

d) Click on **Table View** to see the Course Category dimension.

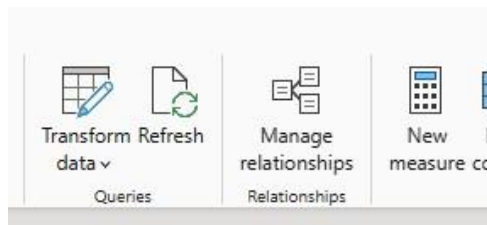
Course Category Code	Course Category
1001	General IT Training
1002	General SAP Training
1005	Digital Technologies
1007	SAP Certifications
1008	Advance Information Security
1003	Office Applications
1009	Advance Digital Courses
1011	Special Program
1004	Information Security
1006	SAP Academy

e) Click on **Model view**. Drag & drop **Course Category Code** from **Course**

Category_Dim to **Course Category Code** of **Training Data_Fact** in order to create the relationship. Repeat the same for **Training Cost_Fact**.



Note: Alternatively, we can use **Manage relationships** under Model view to create joins.



f) **Challenging Task: (Optional. Do it only if time permits)**

In above expression, we are getting course categories from Training Data_Fact table while some categories exist only in second fact table (*Training Cost_Fact*). Modify the above expression to make sure that we will get **unique** course categories from both fact tables. The final dimension table should look like below. (*Hint: Use **Union()** function along with **Distinct()** function*).

Course Category Code	Course Category
1001	General IT Training
1002	General SAP Training
1003	Office Applications
1004	Information Security
1005	Digital Technologies
1006	SAP Academy
1007	SAP Certifications
1008	Advance Information Security
1009	Advance Digital Courses
1010	Digital Courses for Leaders
1011	Special Program

Create Dimension table from scratch (Using DAX):

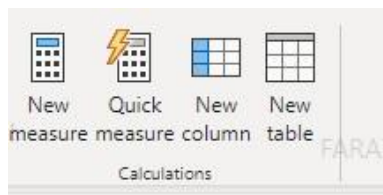
2. Date Dimension:

- a) In order to analyze the data by Time (Year, Quarter, Month, Week etc.), we need to create Date Dimension. We will use DAX **Calendar** function for that.

To learn more about **Calendar function**, visit the link below.

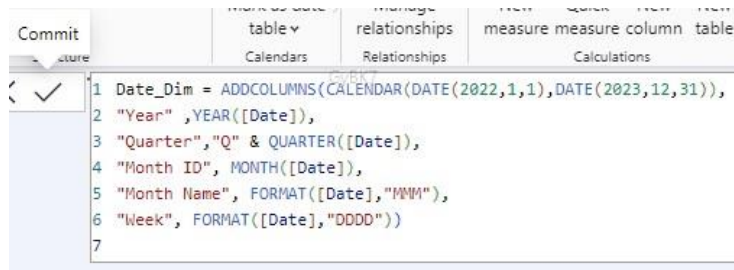
[CALENDAR function \(DAX\) - DAX | Microsoft Learn](#)

- b) From Main Menu, click on **Modeling** tab then click **New table** from Calculations panel.



- c) In formula bar, write the following expression & click Commit.

```
Date_Dim = ADDCOLUMNS(CALENDAR(DATE(2022,1,1),DATE(2023,12,31)),  
"Year", YEAR([Date]),  
"Quarter", "Q" & QUARTER([Date]),  
"Month ID", MONTH([Date]),  
"Month Name", FORMAT([Date], "MMM"),  
"Week", FORMAT([Date], "DDDD"))
```



Note: We have data for two years (2022,2023) so we provide the range accordingly. You can create date dimension by simply providing any required range as start date and end date.

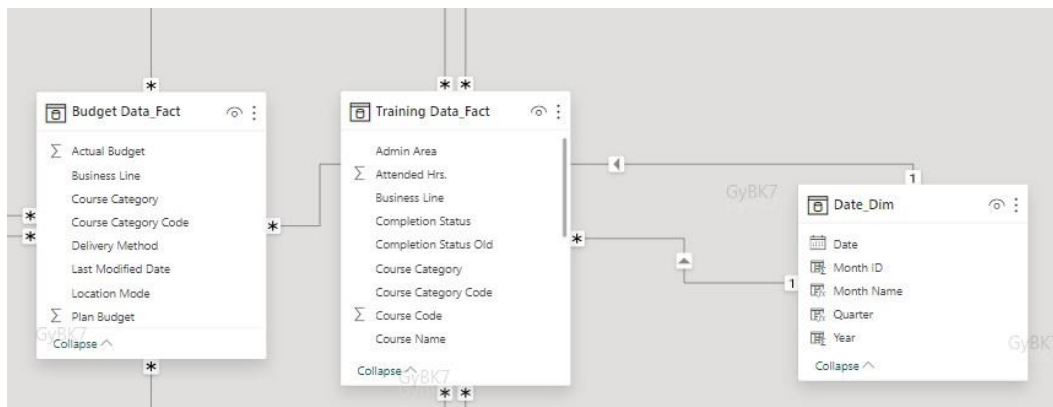
- d) The Date Dimension table will be created.



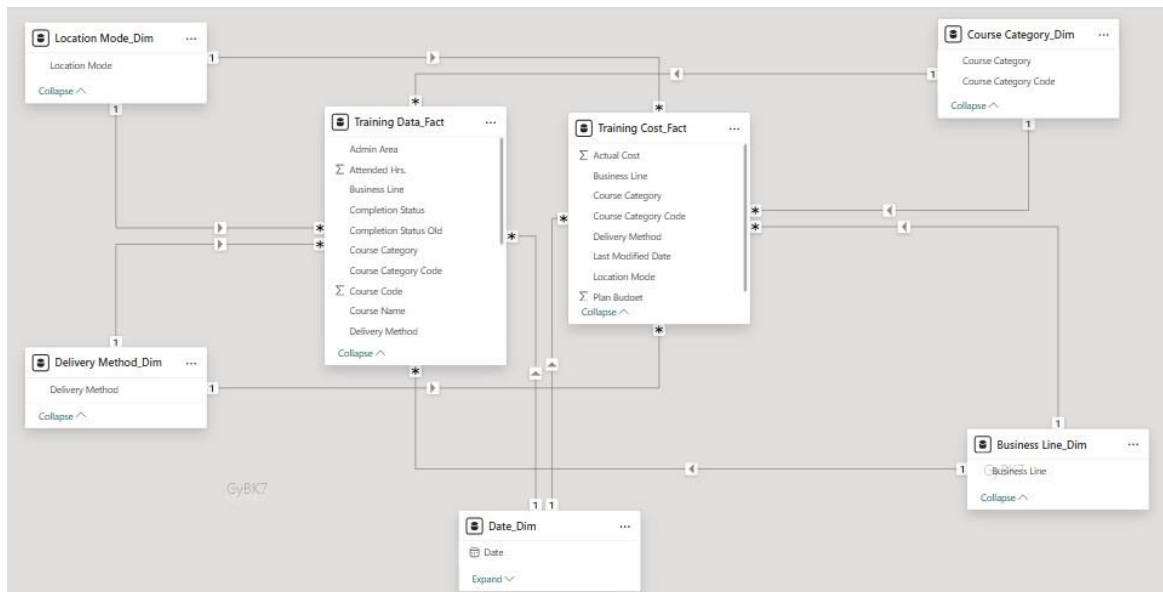
- e) Click on **Table View** to see the Date Dimension.

Date	Year	Quarter	Month ID	Month Name	Week
7/1/2022 12:00:00 AM	2022	Q3	7	Jul	Friday
7/2/2022 12:00:00 AM	2022	Q3	7	Jul	Saturday
7/3/2022 12:00:00 AM	2022	Q3	7	Jul	Sunday
7/4/2022 12:00:00 AM	2022	Q3	7	Jul	Monday
7/5/2022 12:00:00 AM	2022	Q3	7	Jul	Tuesday
7/6/2022 12:00:00 AM	2022	Q3	7	Jul	Wednesday
7/7/2022 12:00:00 AM	2022	Q3	7	Jul	Thursday
7/8/2022 12:00:00 AM	2022	Q3	7	Jul	Friday
7/9/2022 12:00:00 AM	2022	Q3	7	Jul	Saturday
7/10/2022 12:00:00 AM	2022	Q3	7	Jul	Sunday
7/11/2022 12:00:00 AM	2022	Q3	7	Jul	Monday
7/12/2022 12:00:00 AM	2022	Q3	7	Jul	Tuesday
7/13/2022 12:00:00 AM	2022	Q3	7	Jul	Wednesday
7/14/2022 12:00:00 AM	2022	Q3	7	Jul	Thursday
7/15/2022 12:00:00 AM	2022	Q3	7	Jul	Friday
7/16/2022 12:00:00 AM	2022	Q3	7	Jul	Saturday
7/17/2022 12:00:00 AM	2022	Q3	7	Jul	Sunday

- f) Click on **Model view**. Drag & drop **Date** from Date_Dim to **End Date** of Training Data_Fact in order to create join. Do the same to **Last Modified Date** of Training Cost_Fact.

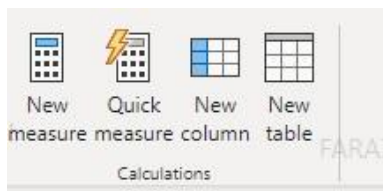


- g) Our Data Model is now complete. Make sure the model view looks similar as below.

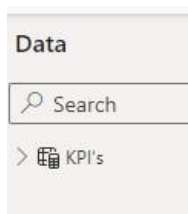


3. Total Hours:

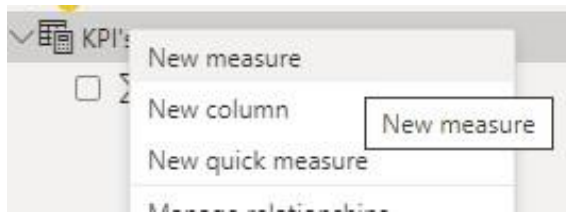
- We can create measures & KPI's in the same table where source fields are located. However, as we have to create multiple calculated measures, it's always recommended to create **new blank table** and keep all calculations in one place.
- From Main Menu, click on **Modeling** tab then click **New table** from Calculations panel. rename it to **KPI's** & click Commit.



- KPI's table will be created under the **Data** panel.

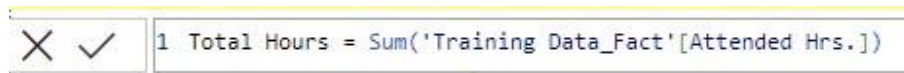


- Right click on **KPI's** then choose **New measure**.



e) In formula bar, write the following expression.

Total Hours = `Sum('Training Data_Fact'[Attended Hrs.])`



f) To make sure that calculation is correct, insert card in practice page and drag & drop the above measure into it.

159,283

Total Hours

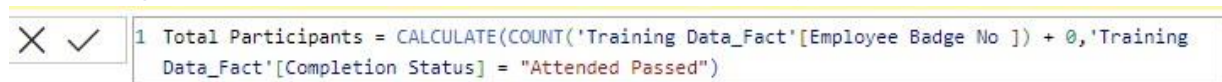
g) Right click on KPI's table & **hide** the default column.

4. Total Participants:

a) Under the Data, right click on **KPI's** then choose **New measure**.

b) We need to count only those participants who complete their training successfully (*completion status is **Attended Passed***). We can use DAX **calculate function** in order to count the rows plus apply the filter. In formula bar, write the following formula.

Total Participants = `CALCULATE(COUNT('Training Data_Fact'[Employee Badge No ]) + 0, 'Training Data_Fact'[Completion Status] = "Attended Passed")`



To learn more about **Calculate function**, visit the link below.

[CALCULATE function \(DAX\) - DAX | Microsoft Docs](#)

c) Insert a card & make sure the value should match with below.

12,537

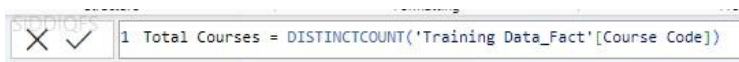
Total Participants

Note: Optionally, you can add + 0 in an expression to show the value as 0 if value is blank or null.

5. Total Courses:

Total unique courses offered for all course categories. We can calculate by applying **distinct count** on Course Code.

Total Courses = `DISTINCTCOUNT('Training Data_Fact'[Course Code])`



36

Total Courses
SIDDIQFS

6. Total Sessions:

Challenging Task:

To calculate Total sessions, we need to group the data at **course level** and then **count distinct** sessions based on **End Date** (*when the session is complete*). For example, course **1475** conducted **5 times** as shown below.

Course Code	Course Name	End Date
1475	XXXXXXXXXX	03/24/2022
1475	XXXXXXXXXX	06/09/2022
1475	XXXXXXXXXX	10/27/2022
1475	XXXXXXXXXX	12/22/2022
1475	XXXXXXXXXX	02/19/2023

Explore the following DAX functions which help to build the required logic.

SUMX & VALUES

[SUMX function \(DAX\) - DAX | Microsoft Learn](#)

[VALUES function \(DAX\) - DAX | Microsoft Learn](#)

Make sure to filter the expression and select rows when the completion status is **Attended Passed** only. Total Sessions should be **694**.

694

Total Sessions

7. Plan Budget & Actual Cost:

- Now we need to calculate measures from Training Cost table.
Calculate Plan Budget & Actual Cost using following expressions.

Plan Budget = `Sum('Training Cost_Fact'[Plan Budget])` Actual

Cost = `Sum('Training Cost_Fact'[Actual Cost])`

\$12,644,336

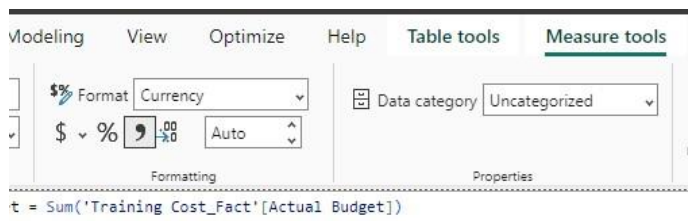
Gy8K7

Plan Budget

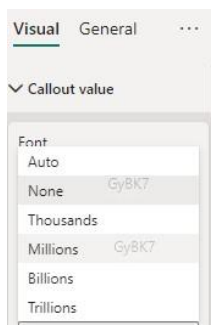
\$12,477,652

Actual Budget

- b) In order to show the value with currency sign and thousand separator, select the measure from the Data panel and then from **Measures Tools**, apply the required changes.



- c) If you would like to show the values in Millions then select the card visual & from visual properties, change **display unit** to Millions.



\$12.64M

Plan Budget

\$12.48M

Actual Budget

7. Budget Variance, Budget Consumption % & Avg. Cost Per Participant:

- a) Calculate Budget Variance using following expression.

Budget Variance = [Plan Budget] - [Actual Cost]

\$166,684

Gy8K7

Budget Variance

- b) Calculate Budget Consumption % using following expression.
Make sure to show the value as % (*Using Measure Tools*).

Budget Consumption % = $\frac{[\text{Actual Cost}]}{[\text{Plan Budget}]}$

99%

Budget Consumption %

Challenging Task: There is a possibility that plan budget is 0 while actual cost is not. In that case, the above formula can generate divide by 0 error. Modify the above formula & incorporate the logic to avoid divide by 0 error. (*Hint: Divide Function*).

- c) Calculate Avg. Cost Per Participant using following expression.

Avg. Cost Per Participant = $\frac{[\text{Actual Cost}]}{[\text{Total Participants}]}$

\$995

Avg. Cost Per Participant

Quick Measures: (Optional. Do it only if time permits)

You can use **quick measures** to quickly and easily perform common, powerful calculations. A quick measure runs a set of Data Analysis Expressions (DAX) commands behind the scenes, then presents the results for you to use in your report

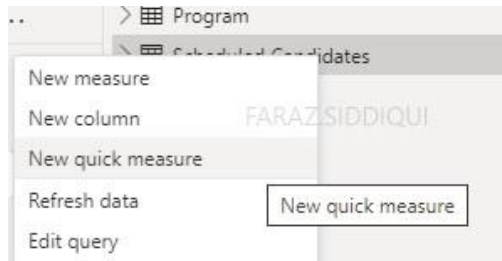
To learn more about **Quick Measures**, visit the link below.

[Use quick measures for common and powerful calculations - Power BI | Microsoft Docs](#)

8. Running Total:

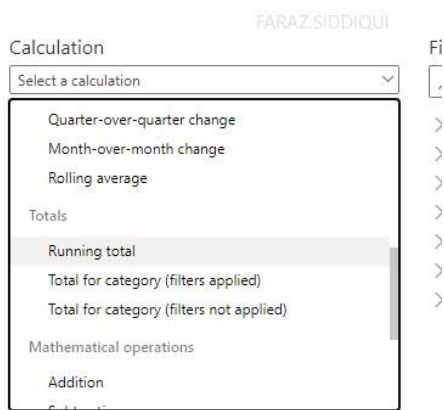
We would like to calculate running total for Actual Budget. We can utilize quick measure for that purpose.

- a) Select Month Name under Date_Dim table and apply sorting from Column tools ➡ Sort by column ➡ Month ID.
- b) Right click on KPI's table & choose **New Quick Measures**.

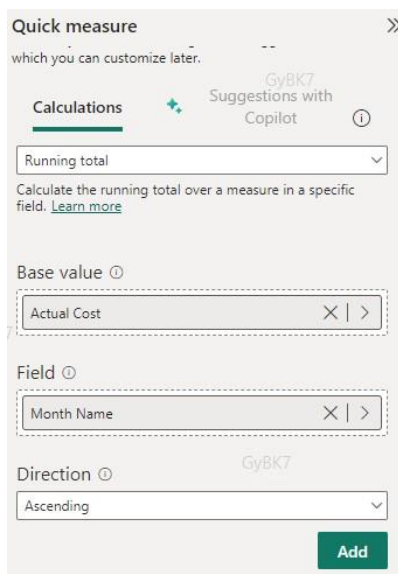


- c) From the built-in calculations, select **Running Total**.

Quick measures



- d) Set Actual Cost as **Base value**, Month Name as **Field** (From Date Dimension table) and direction **Ascending** as shown below. Click Add.



- e) New measure will be created under KPI's table. Click on the measure to see the generated code.

```

1 Actual Cost running total in Month Name =
2 CALCULATE(
3     [Actual Cost],
4     FILTER(
5         CALCULATETABLE(
6             SUMMARIZE('Date_Dim', 'Date_Dim'[Month ID], 'Date_Dim'[Month Name]),
7             ALLSELECTED('Date_Dim')
8         ),
9         ISONORAFTER(
10            'Date_Dim'[Month ID], MAX('Date_Dim'[Month ID]), DESC,
11            'Date_Dim'[Month Name], MAX('Date_Dim'[Month Name]), DESC
12        )
13     )
14 )

```

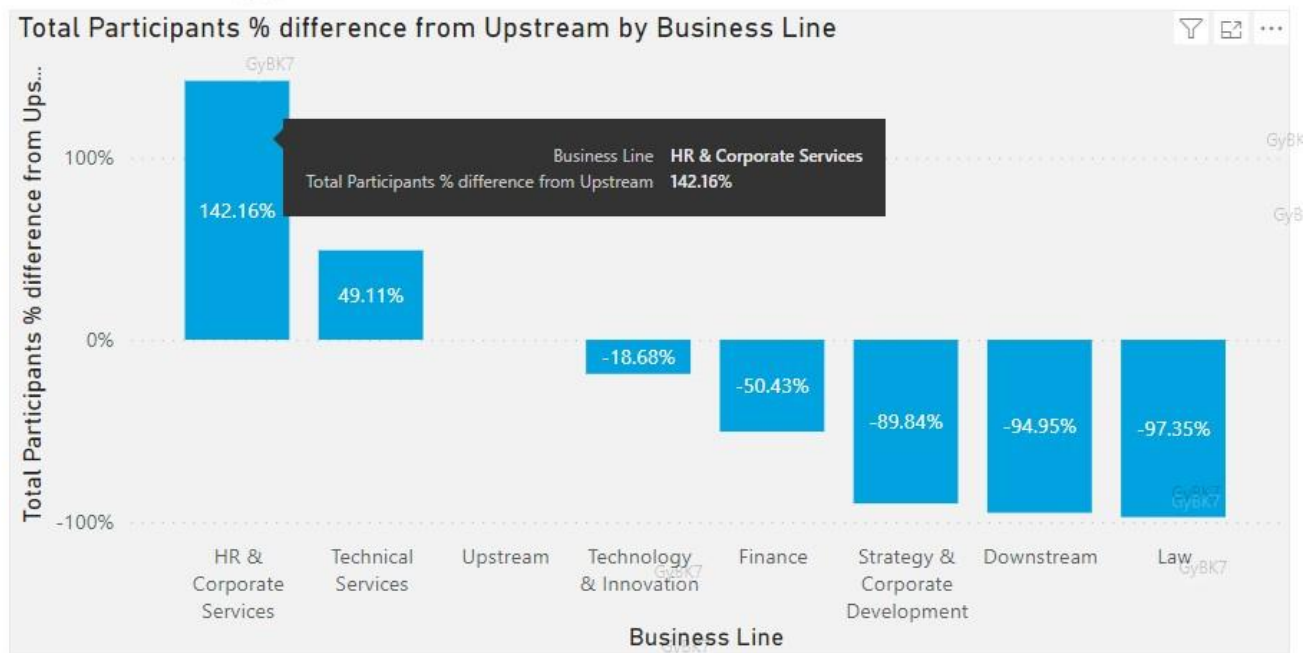
Note: You can modify the generated code if needed.

- f) In practice page, insert **table** visual. populate it with Month name, Actual Cost and newly created Running Total measure. Actual Cost will show as cumulative sum or running total in RT measure.

Month Name	Actual Cost	Actual Cost running total in Month Name
Jan	\$904,829	\$904,829
Feb	\$1,192,655	\$2,097,484
Mar	\$1,114,161	\$3,211,645
Apr	\$747,903	\$3,959,548
May	\$1,187,210	\$5,146,758
Jun	\$1,165,248	\$6,312,006
Jul	\$932,977	\$7,244,983
Aug	\$1,061,376	\$8,306,359
Sep	\$1,055,980	\$9,362,339
Oct	\$1,032,683	\$10,395,022
Nov	\$1,034,458	\$11,429,480
Dec	\$1,048,172	\$12,477,652
Total	\$12,477,652	\$12,477,652

- g) Save your work.

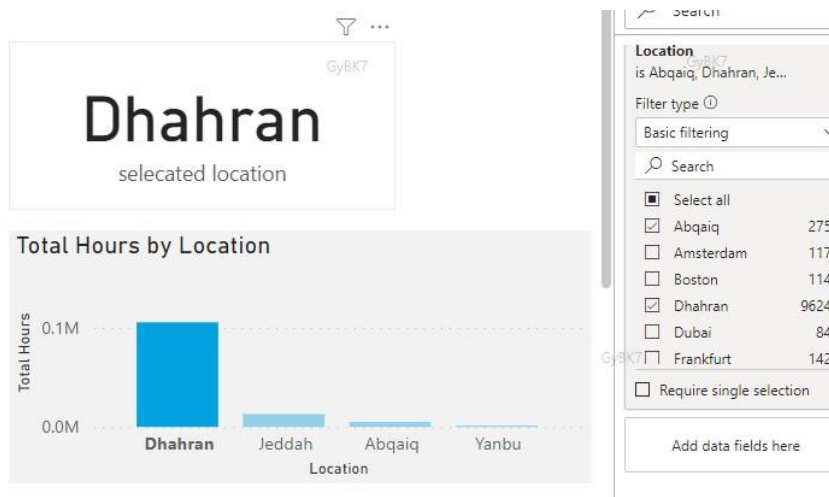
Challenging Task: Users are interested to compare Total Participants in other Business Lines while considering **Upstream** as base value. You are required to use quick measure called **Percentage difference from filtered value** in order to compare Total Participants % (+/-) in other business lines with Upstream. Use **Stacked column chart** to visualize the result as well as shown below.



9. **For Practice Only: (Optional. Do it only if time permits)**

Capture Filter Value:

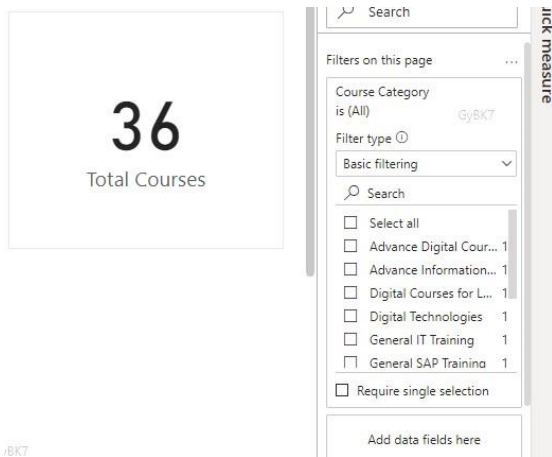
- In any practice page, insert Location as page filter. Also insert any visual and populate with Location & Total Hours.
- We want to **capture** the location either selected through page filter or through visual & then print the selected value in the report as shown below:



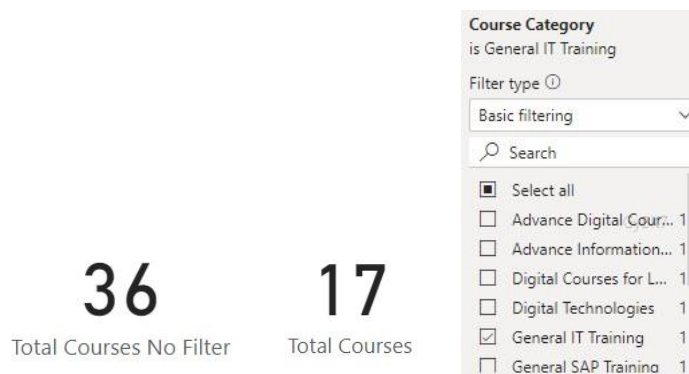
- Create a **DAX measure** to achieve the above requirement. Also make sure that if there is no filter then the measure should print **All Locations**.

Skip Page Filters:

- a) Insert new page as practice, insert Course Category as page filter. Insert card visual and populate it with Total Courses.



- b) Now choose any course category from page filter, the data will be filtered as expected. However, we would like to **skip page filter** so when the user selects any course category, the data will not be filtered in our visual.
- c) Create new DAX measure called **Total Courses No Filter** & apply a logic to bypass any kind of page filters ([Hint: Explore the ALL Function](#)). For example, when we choose General IT Training, Total Courses filtered but the new measure not impacted by that filter.



Extract Part of Text:

- a) Create a new measure called **network id** & write the following formula.

network id = USERPRINCIPALNAME()

- b) Insert card to display the new measure.

siddiqfs@aramco.com
network id

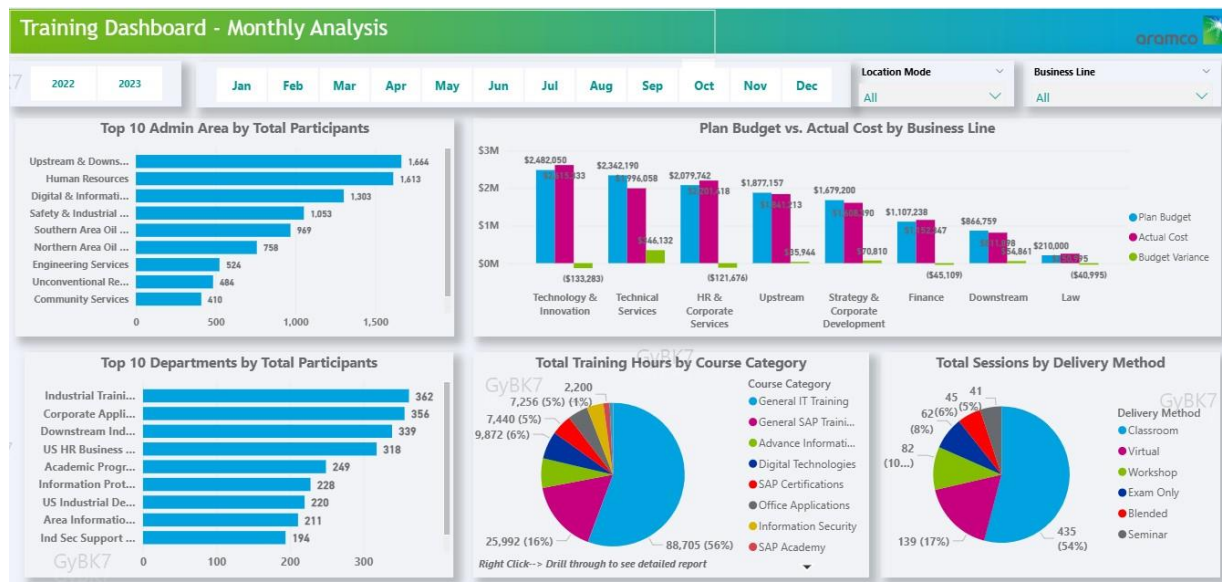
- c) **USERPRINCIPALNAME()** function is used to retrieve the user id with full email address of the user connected to the machine. **Modify** the above formula to retrieve only user id.

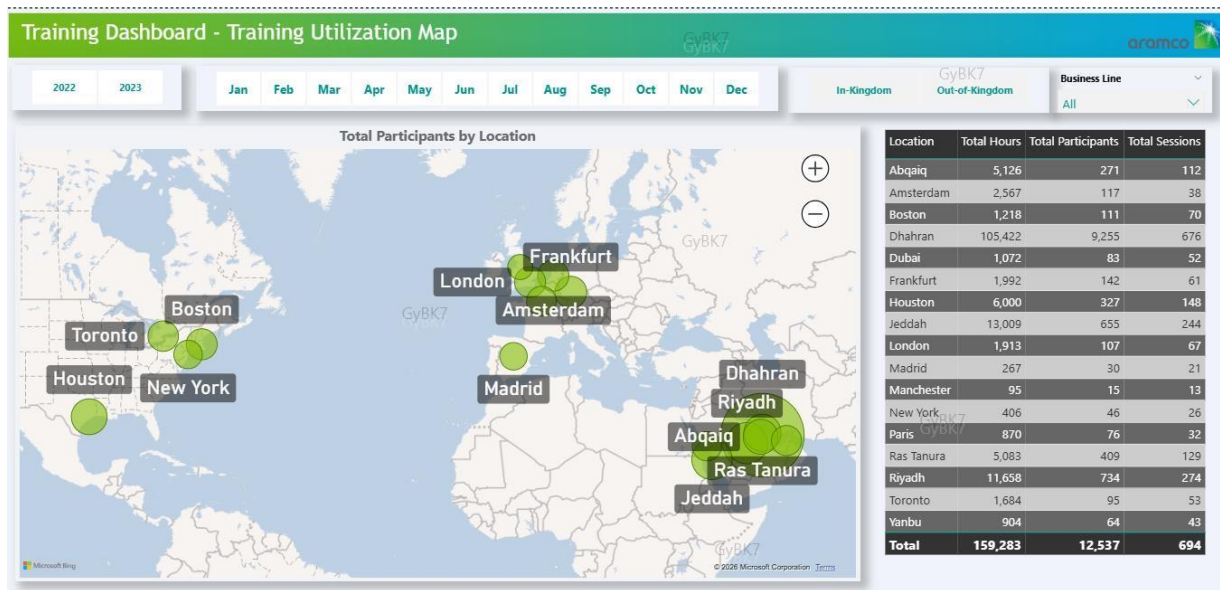
siddiqfs
network id

Exercise 6: Design the Dashboard

We have load the & transform the source data, build data model & calculate our KPI's & measures. Now it's a time to design the dashboard.

At the end of this exercise, your dashboard should look similar to below. *(It contains three pages QBR Summary, Monthly Analysis & Training Utilization Map).*





Steps:

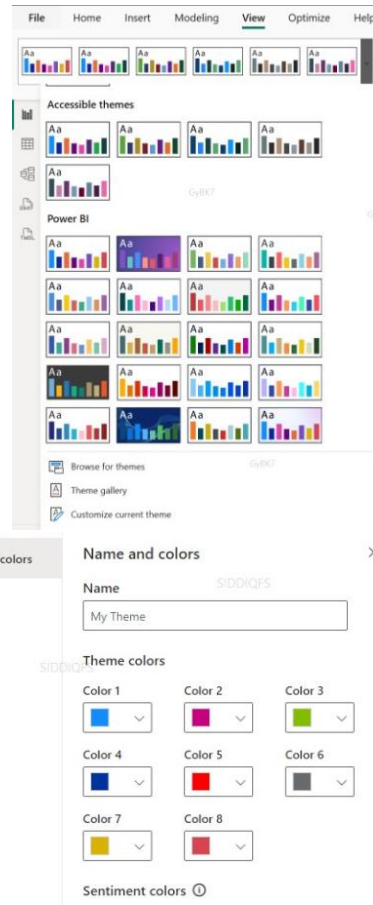
1. Color Theme:

It's not required that you should follow the exact same color themes, font size & other details mention in the document. you can use your own theme if you want.

However, if would like to follow the same theme then below are the details.

- Page Background & Wall paper: f0f4fa
- Main colors for visuals: 00A3E0, C6007E, 84BD00, 0033A0, FF0000, 676A6E
- Dashboard Header: Font: Trebuchet MS, Size: 20, Color: White
- Visual Background color: F2F2F2
- Visual Heading: 676A6E, Font: Segoe UI, Size: 13, Bold

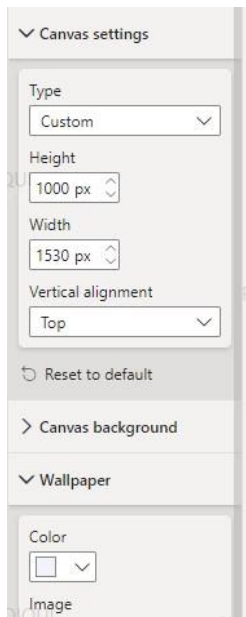
Optionally, you can set your own color theme, font size etc. by customizing the current theme and save it with a new name from **View Themes Customize current theme**.



Note: Some specific formatting details are also mention within the visuals below.

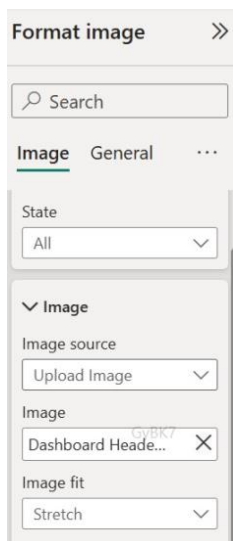
2. Page Settings:

- Insert new page. Rename it to **QBR Summary**.
- From format page, change page size. Height **720**, Width **1500**.
- Change Wall paper Color as mention above (*or use your own colors*).

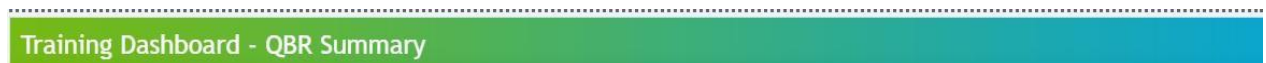


3. Dashboard Heading:

- a) From menu ⑦ Insert ⑦ Image. Then from **Format image** browse the **Dashboard Header** image available in icon folder. Under Image fit, set **Stretch**.



- b) Arrange the image on top of the page.
- c) Insert **Text box**. Remove its background from properties. Write **Training Dashboard - QBR Summary**. Put it on top of dashboard header image & change font, color as mention above under Color themes section. Should look like below:



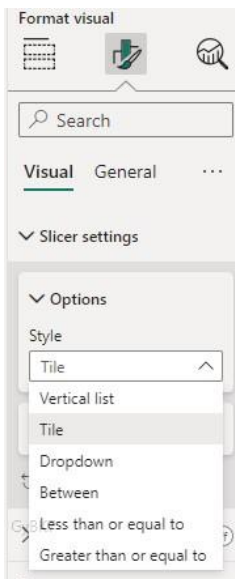
- d) Save your work.

4. Design the Visuals:

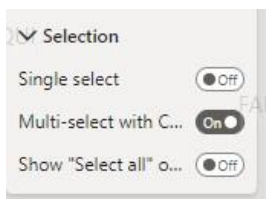
Users would like to filter the data by Year, Quarter, Location Mode, Business Line & Course Category etc. We will use **Slicer** for that purpose.

4.1. Slicers (Year, Quarter, Location Mode, Business Line, Course Category):

- Insert slicer just below the heading on left side.
- Insert **Year** into the field (*from Date Dimension*).
- Change the style to **Tile** from Format visual  Visual  Slicer settings.



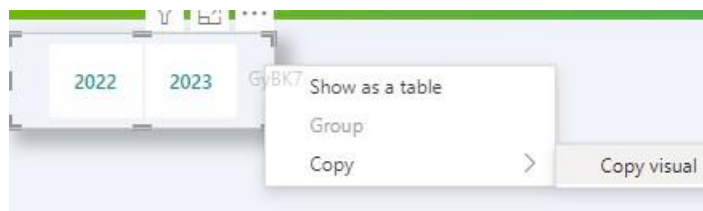
- From Slicer settings , set the **selection** as below:



- Set some other properties from Format visual tab (*if needed*).
 - Visual  Slicer Header off.
 - Visual  Values  Font color 00968f, Font Segoe UI Semibold & Size 11.
 - Visual  Values  Background color white.
 - General  Effects  Background **Off** & Shadow **On**.
 - General, set the height & width (**approx. 60,210**).
- Should look similar to below.



- g) Select the slicer ⑦ Right click ⑦ Copy Visual. Paste it (using Ctrl-V command) next to it.



- h) Populate it with **Quarter**. Expand to show all quarters. Set font size 11 and Height & Width (**approx. 60,442**).
- i) Copy & paste again & populate with **Location Mode**. Height & Width (**approx. 60, 278**).
- j) Copy & paste again & populate with **Business Line** & set the following properties.
- Visual ⑦ Slicer settings ⑦ Style ⑦ Dropdown.
 - Visual ⑦ Values ⑦ Background color #F2F2F2.
 - Visual ⑦ Slicer header ⑦ On
 - Visual ⑦ Slicer header ⑦ Text ⑦ Bold
 - General, set the height & width (**approx. 60, 266**).
- k) Copy & paste the Business line slicer and populate with Course Category. Set the same properties as the business line slicer.
- l) Try to match as shown below.



Note: Populate Year, Quarter, Location Mode, Business Line & Course Category from their respective **dimension tables** & not from the fact (transaction) tables.

4.2. Card Visuals (Total Hours, Total Participants, Total Sessions, Plan Budget, Actual Cost, Avg.

Cost Per Participant):

- Insert **Card Visual** (just below the Year slicer on the left).
- Populate with **Total Hours** (from KPI's).



- c) Set following properties of the card visual from format visual tab.
- Approx. Height & Width **(148,260)**.
 - Callout value color **#84BD00** (from the theme or your own).
 - Callout value font **DIN** and font size **40**.
 - Callout Value label **On** & size **16. Bold**.
 - Callout Value Display Unit **None**.
 - Visual ☒ Cards ☒ Border **off**.
 - General ☒ Effects ☒ Background **On**. Background color **#F2F2F2** (or any other color).
 - General ☒ Effects ☒ Shadow **On**. Shadow Color **#84BD00** (or any other color).
- d) Should look like below.



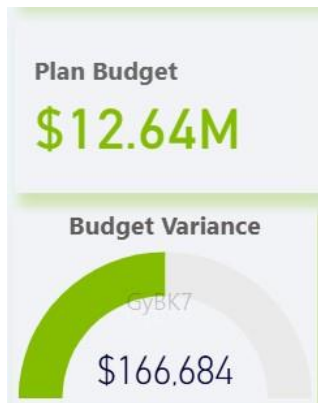
- e) Copy & paste the above card to show **Total Participants, Total Sessions, Plan Budget, Actual Cost & Avg. Cost Per Participant**. Just change the KPI under the Fields. Arrange it so it should look like below:



4.3. Gauge (Budget Variance & Budget Consumption %):

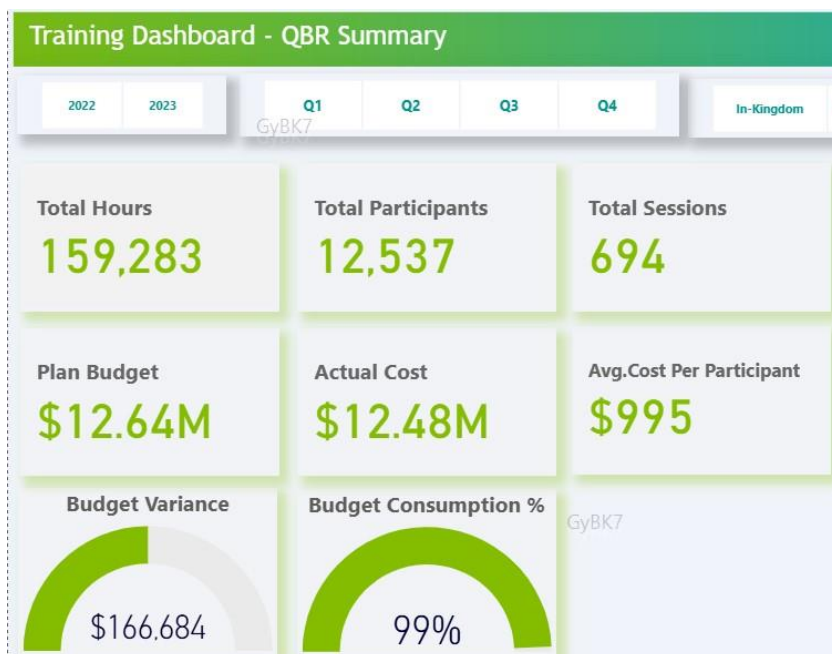
To display Budget Variance and Budget Consumption %, we would like to use **Gauge** visual.

- Insert **Gauge** & populate with **Budget Variance** (*just below the plan budget card. Approx. height & width 174,260*).
- Apply same theme as with other visuals & try to match with following:



- Repeat the above steps to populate **Budget Consumption %** (*copy & paste visual*).

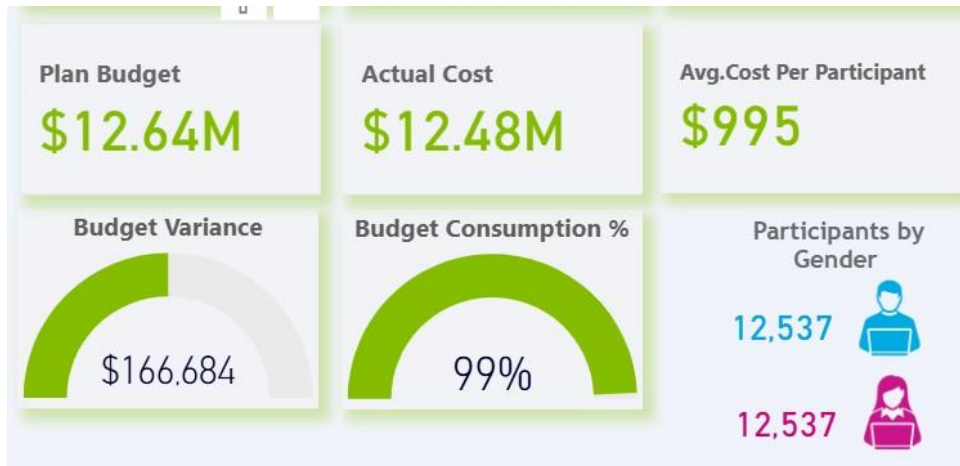
should look like below.



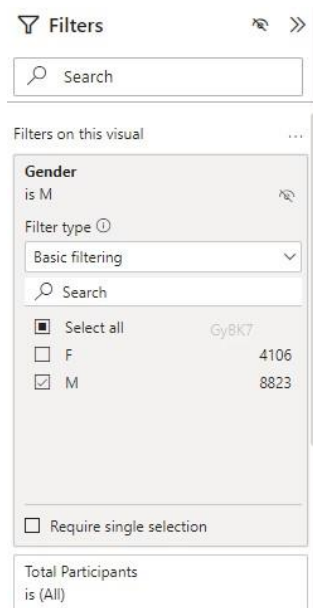
4.4. Participants by Gender (Using Images):

- To Show Total Participants by Gender, we would like to use **custom images**.
- Insert male & female icons as **images** (*Just below the Avg. Cost Per Participant Card*). Insert **Card** visuals next to the images & populate with **Total Participants**. (*Optionally, copy/paste existing cards*). Insert the **text**

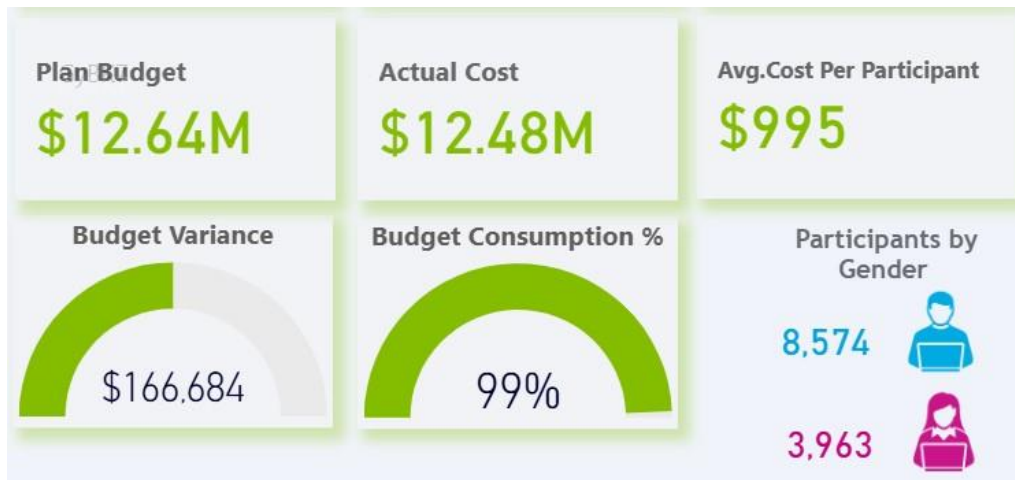
box and write **Total Participants by Gender**. Arrange & apply the necessary formatting. Try to match with following.



- c) To filter the visual, we can apply **filter on this visual** option.
- i) Select card visual for Male, insert Gender from Training Data_Fact under **Filters on this visual** Select **M**. Make sure to **hide** the filter as well.

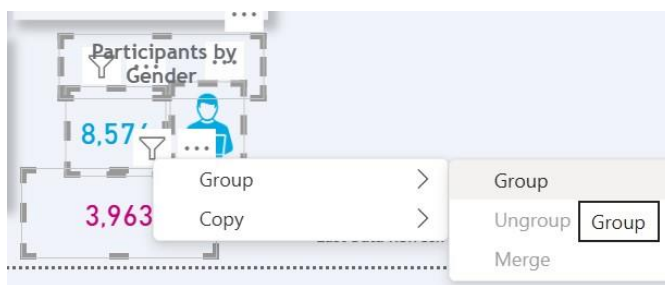


- j) Repeat the above step for Female Card. Make sure values are correct as shown below.

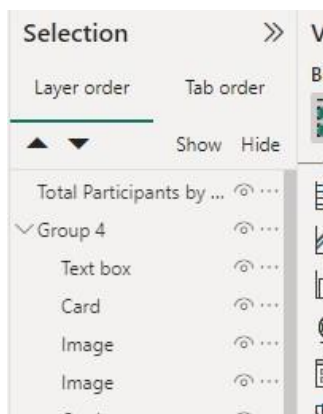


Note: Alternatively, you can create **calculated measures** for both male & female visuals.

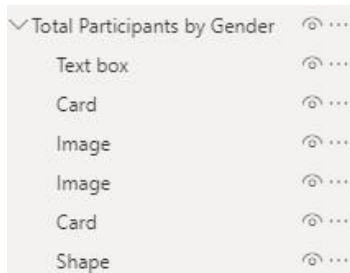
- k) Optionally, select images, card visuals and text by pressing **Ctrl** key. Right click & select **Group** **Group**. Grouping helps to move multiple objects easily.



- l) From View, click **Selection**. Selection panel make it easy to select visuals from canvas particularly if you have done the grouping.



- m) Optionally, you can rename the group. For example, rename the above created group to **Total Participants by Gender**.



Note: it's a good practice to give proper name to groups in **selection** panel.

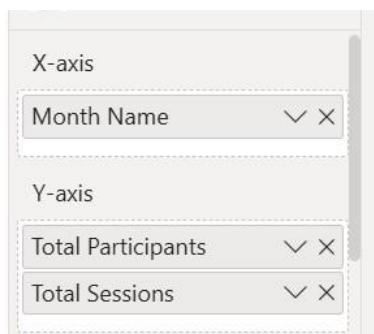
4.5. Trend Analysis (Total Participants & Total Sessions by Month):

To visualize **Total Participants** and **Total Sessions** by Month, we will use the **Line** Visual.

- Insert **Line chart** next to card visuals and below the slicers. (*Approx. height & width*

245,634)

- Populate it with Month Name, Total Participants & Total Sessions.



- From the format visual tab, apply necessary formatting (color theme, title, background, Data Labels, shadow etc.)
- Try to match with following.



4.6. Trend & Comparative Analysis (Plan Budget vs. Actual Cost by Month):

To visualize and Compare the **Plan Budget vs. Actual Cost** by Month, we would like to use the **Line and Clustered Column Chart**.

- e) Insert **Line and Clustered Column Chart** just below the above graph.
(Approx. height & width 249,634)
- f) Populate it with Month Name, Plan Budget & Actual Cost.

X-axis

Month Name

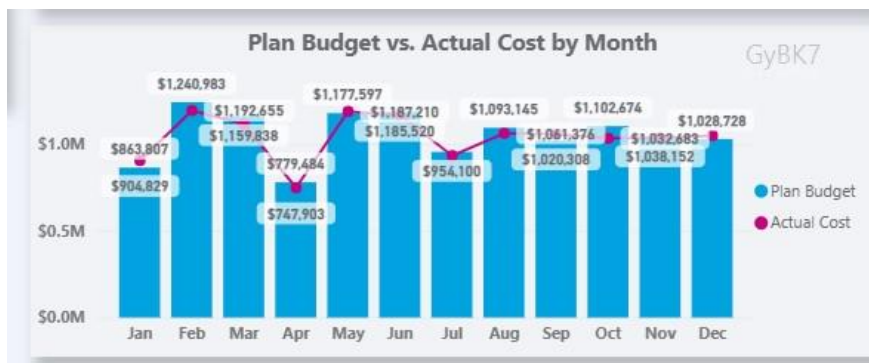
Column y-axis

Plan Budget

Line y-axis

Actual Cost

- g) From the format visual tab, apply necessary formatting (color theme, title, background, Data Labels, shadow etc.)
- h) Try to match with following.



- i) On the right bottom of the page, insert text box & write **Design by: (your network id)**.
- j) Insert Aramco logo at right corner of the page header.
- k) Finally, the **QBR Summary** page looks similar to below. Save your work.



4.7. Challenging Tasks (Trend icon and Gauge Targets. Do it only if time permits)

We would like to create trend icon, apply conditional formatting and apply target over the gauge visuals.

- To show **trend indicator** for Avg. Cost Per Participant, create new DAX measure and write following expression.

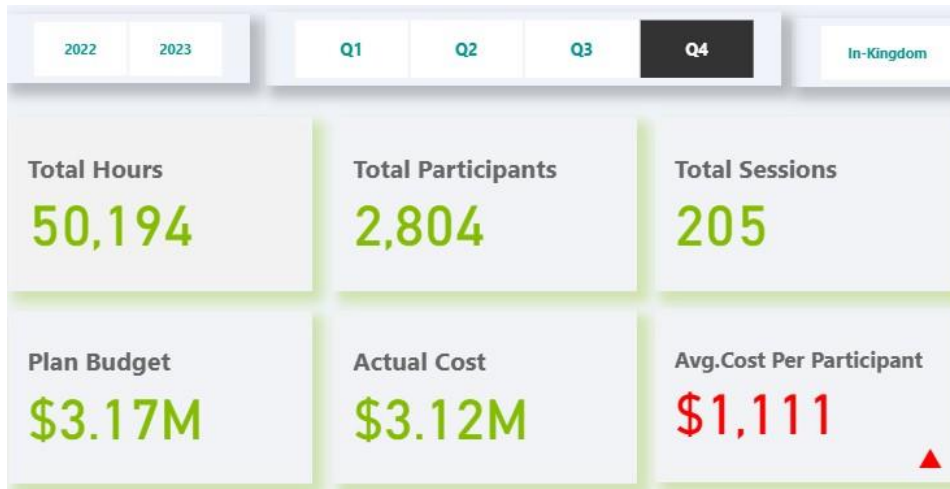
Avg. Cost Indicator = `if([Avg. Cost Per Participant]>=1000,UNICHAR(9650),UNICHAR(9660))`

Note: If Avg. Cost >= 1,000 then upward trend otherwise downward.

- Insert a card and populate with above measure. Remove label, background and card border. Set callout value **18**. Callout value color **#84BD00** (or your own). Adjust and arrange over the **Avg. Cost per Participant** card created above. (Step 4.2.e).

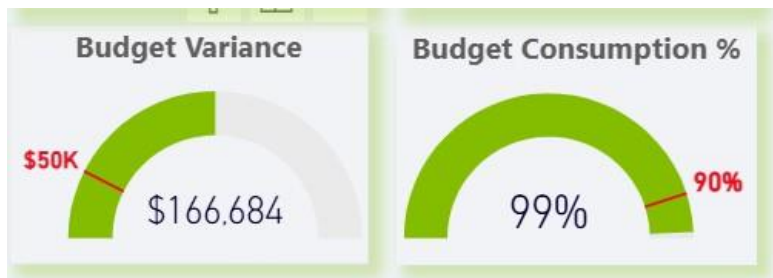


- Apply conditional formatting over Value & Indicator icon.
 - Avg. Cost Per Participant >= 1000 then Red otherwise same.



Gauge Targets:

- d) Set the **Target line** using gauge properties as below. • Budget Consumption is 90% • Budget Variance is 50,000.

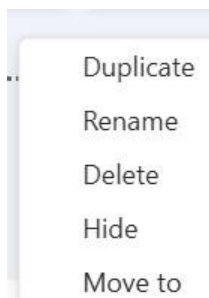


Monthly Analysis Page. (This page is optional to design. Do it only if time permits):

4.8. Top 10 Admin Areas/Departments by Total Participants:

In order perform monthly analysis from various perspectives, we would like to create a new page.

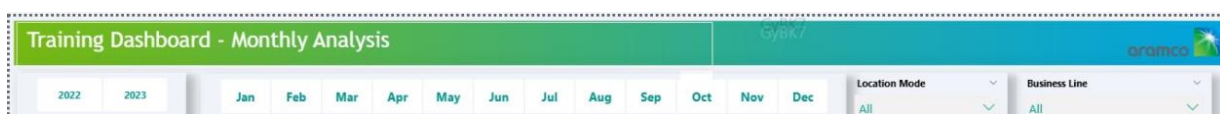
- a) Right click over QBR Summary page and select **Duplicate**.



- b) Rename the page to **Monthly Analysis**.
c) While in Monthly Analysis page, delete all visuals except line graph and the slicers as shown below.



- d) Rename the page title to **Training Dashboard – Monthly Analysis**.
e) Replace the **Quarter** Slice with **Month Name**. Change **Location Mode** slicer style to **Dropdown**. Remove Course Category Slicer.



- f) Select the line graph and change its type to **Stacked bar chart**. Replace the existing values with following:

Y-axis

Admin Area

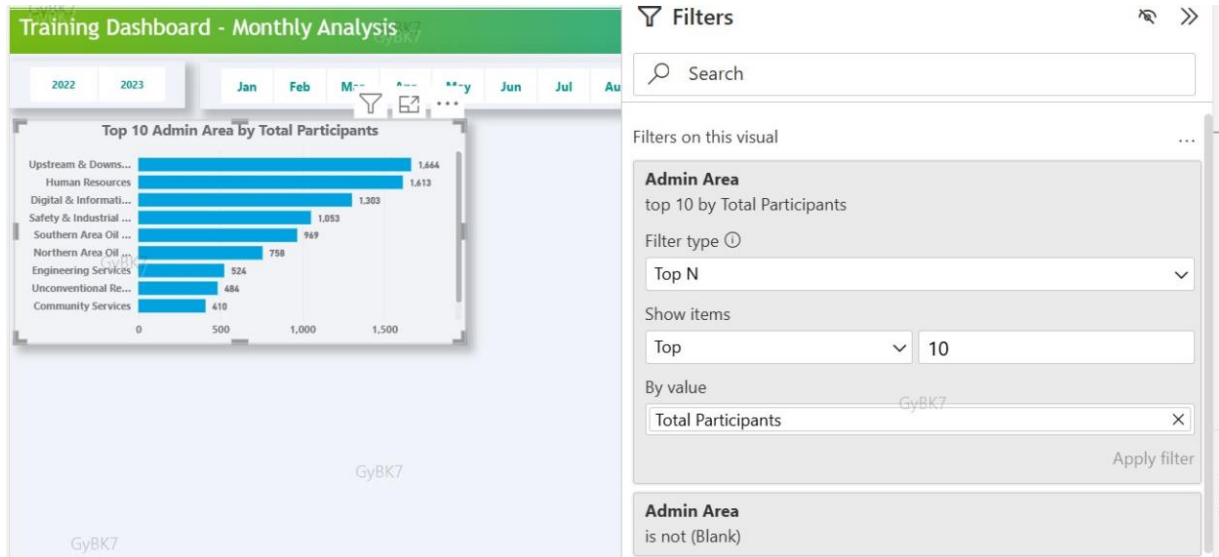
X-axis

Total Participants

- g) Move the graph to put it below Year & Month Slicers. Give it a title **Top 10 Admin Area by Total Participants**. set Approx. Height & Width **(266,535)**. Apply necessary formatting if needed and try to match with following:



- h) We would see only **Top 10 Admin Areas**. Select the above visual and from Filters **Filters on this visual**, apply the **ranking function** as shown below. Also make sure to exclude the **Null values**.



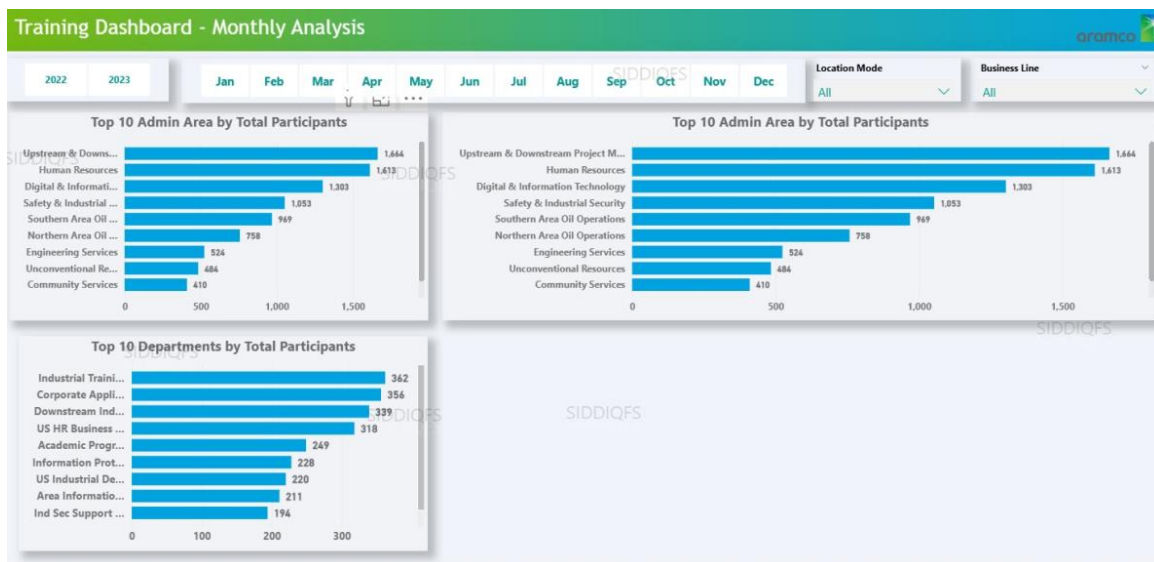
- i) Repeat the above steps to create a similar graph for **Top 10 Departments by Total Participants**. Arrange it below the above graph as shown below. (Just copy/paste the above graph and change the values and filter conditions).



4.9. Plan Budget vs. Actual Cost & Budget Variance by Business Line:

In order to analyze the actual budget consumption vs. allocated budget for training courses, we would like to use the **Clustered column chart**.

- a) Copy **Top 10 Admin Area by Total Participants** chart created above and paste next to it and expand it to the right as shown below.



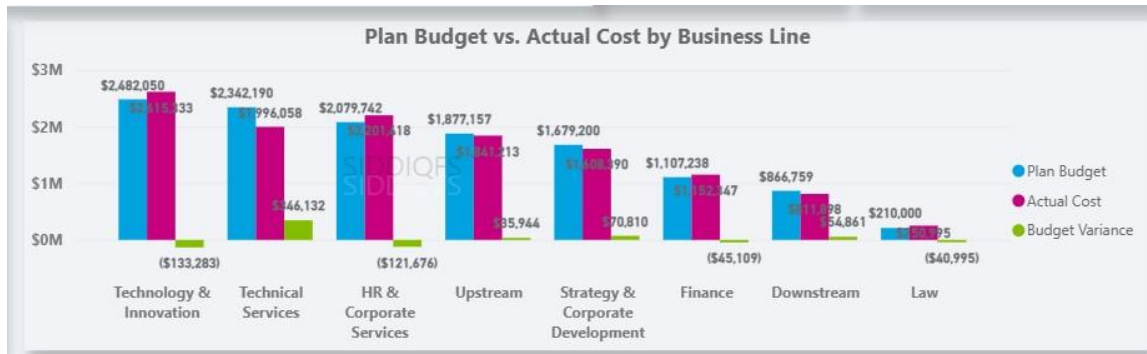
- b) Select and convert it into **Clustered column chart** and populate it with following values:

The screenshot shows a configuration dialog box for the chart. It has two main sections: 'X-axis' and 'Y-axis'.

X-axis: The dropdown menu is set to 'Business Line'.

Y-axis: The dropdown menu is set to 'Plan Budget', 'Actual Cost', and 'Budget Variance'.

- c) Apply necessary formatting if needed and try to match with following:'



4.10. The Pie Charts (Total Hours by Course Category & Total Sessions by Delivery Method):

- Copy any existing graph and convert into a **Pie chart**.
- Populate it with following fields:

Legend

Course Category

Course Code

Values

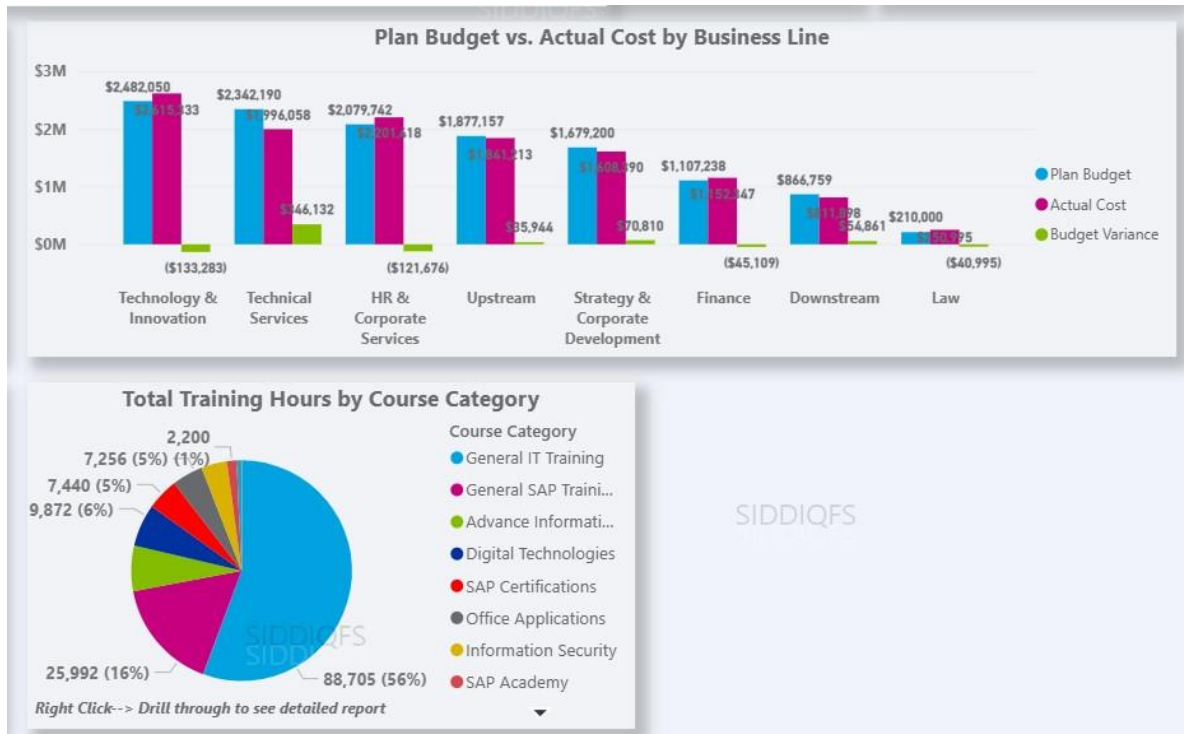
Total Hours

Note: Adding Course code below Course Category allow users to perform **drill-down** if needed.

- Add a text box at the bottom of the graph and type *Right Click--> Drill through to see detailed report.* (Set font size to 9).

Note: We will implement **Drill through** feature in Exercise 8.

- Apply necessary formatting and try to match as below.



- e) To show **Total sessions by Delivery method**, copy and paste the above graph next to it and populate with following:

Legend

Delivery Method

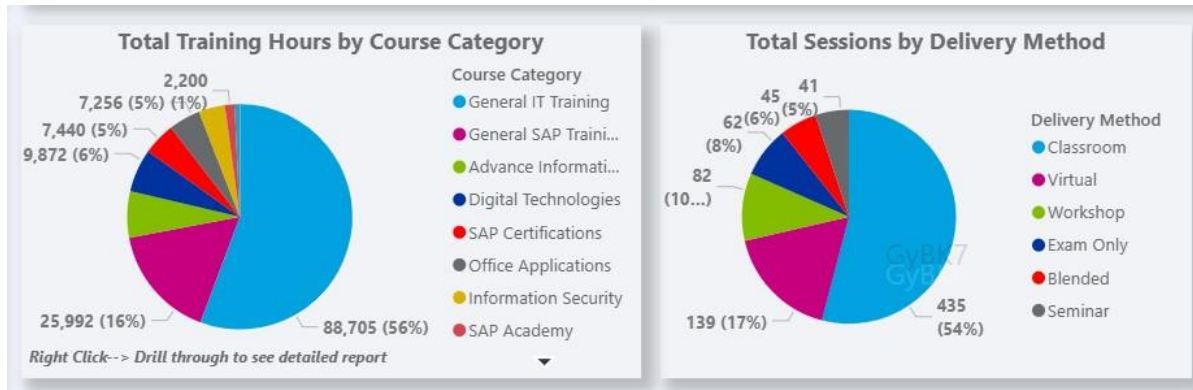
Values

Total Sessions

Details

Add data fields here

- f) Apply necessary formatting and try to match as below.



g) This is how the **Monthly Analysis** page look like finally.

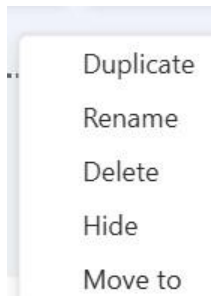


Training Utilization Map page. (This page is optional to design. Do it only if time permits):

4.11. The Map:

In order to analyze the No. of Participants trained in various locations (In-Kingdom, Out-of-Kingdom), we would like to use the **Map Visual**.

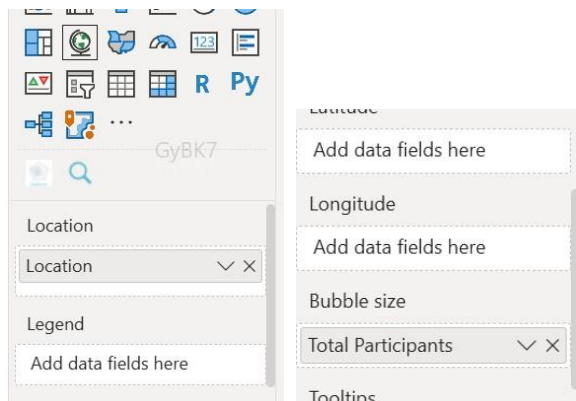
- j) Right click over **Monthly Analysis** page and select **Duplicate**.



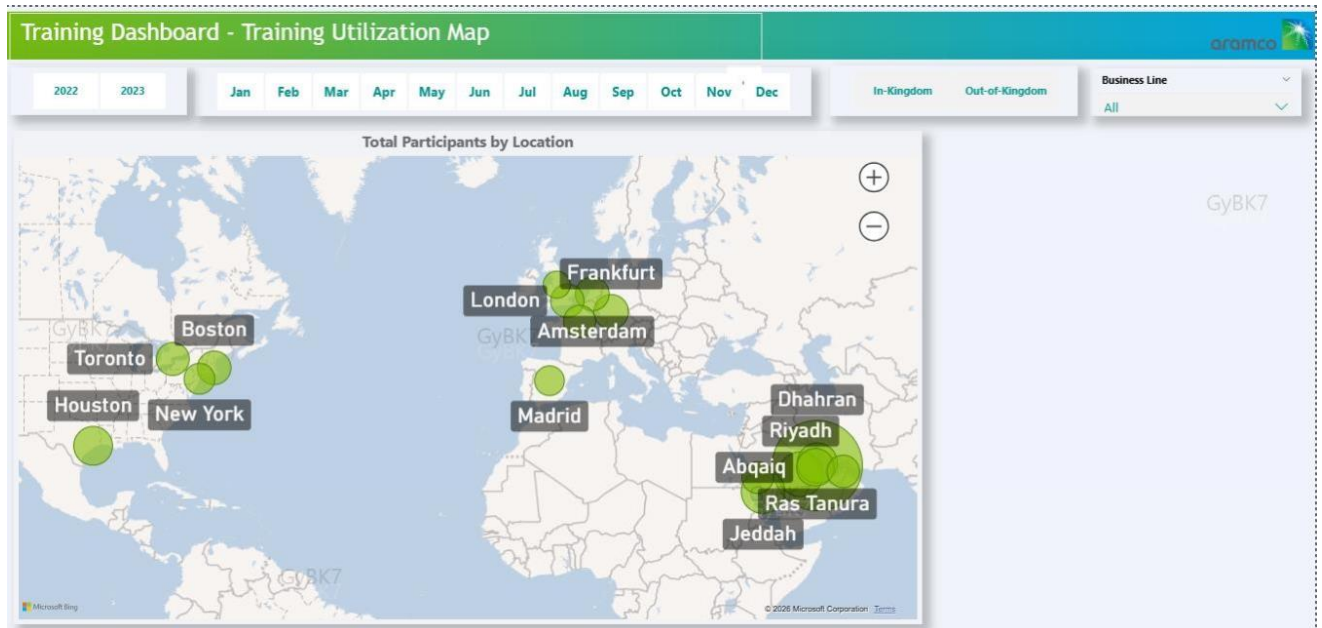
- k) Rename the page to **Training Utilization Map**.
- l) While in Training Utilization Map page, delete all visuals except **Plan Budget vs. Actual Cost by Business Line**. Also, don't delete the slicers.



- m) Rename the page title to **Training Dashboard – Training Utilization Map**.
- n) Change **Location Mode** slicer style to **Tile**.
- o) Convert the existing graph into **Map**.
- p) Set following values for Map.



- q) Apply necessary formatting and try to match with below: (Approx. height & width 562,1037)



4.12. The Table Visual:

We would like to display important KPI's by location. We will use **Table** visual for that.

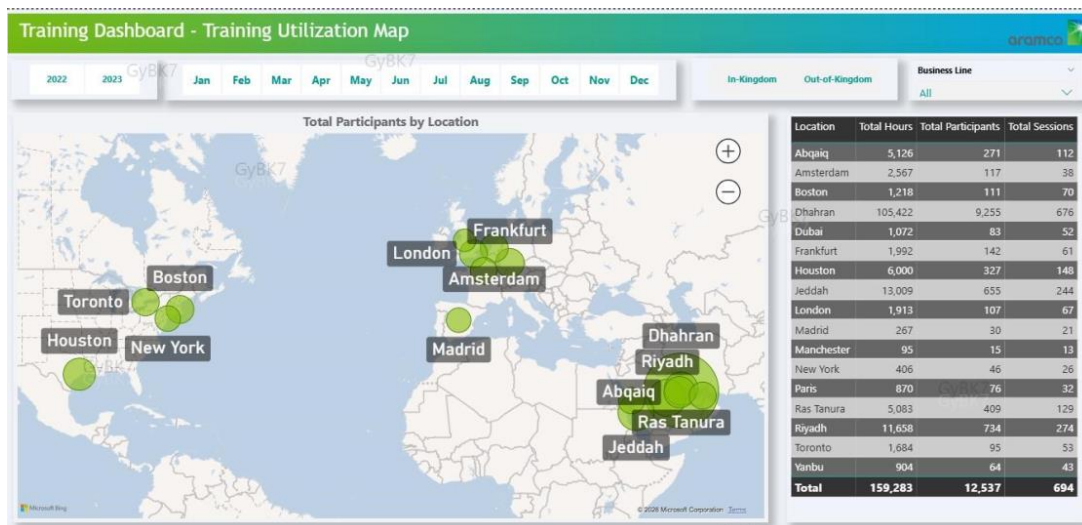
- Insert **Table visual** next right to the Map. (Approx. height & width 562,407).
- Populate it with following values.

Columns	
Location	▼ ×
Total Hours	▼ ×
Total Participants	▼ ×
Total Sessions	▼ ×

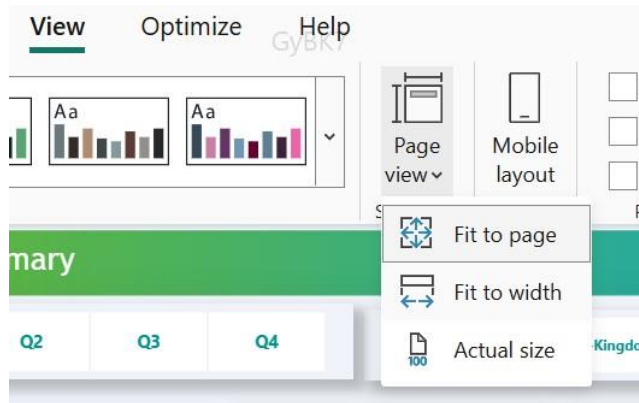
- Apply necessary formatting and try to match as below:

Location	Total Hours	Total Participants	Total Sessions
Abqaiq	5,126	271	112
Amsterdam	2,567	117	38
Boston	1,218	111	70
Dhahran	105,422	9,255	676
Dubai	1,072	83	52
Frankfurt	1,992	142	61
Houston	6,000	327	148
Jeddah	13,009	655	244
London	1,913	107	67
Madrid	267	30	21
Manchester	95	15	13
New York	406	46	26
Paris	870	76	32
Ras Tanura	5,083	409	129
Riyadh	11,658	734	274
Toronto	1,684	95	53
Yanbu	904	64	43
Total	159,283	12,537	694

d) This is how the **Training Utilization Map** page look like finally:



- e) Select values from different slicers to see how filter applied in your visuals. Also try to filter the data using interactions within visuals.
- f) Hide all the practice pages created before.
- g) Select **QBR Summary** page and from View  Page view, make sure **Fit to page** is selected.



h) Save your work.

4.13. Last Refresh Date:

a) **Challenging Task:** Users would like to know when the data is last updated in the dashboard. Create a **date object** & insert in report bottom (*QBR Summary*) so whenever user click on **refresh** or scheduler execute the job the date will update accordingly as shown below. (*Hint: Use Power Query*).



Exercise 7: Publishing to Server

As we design the dashboard completely, now it's a time to publish into **Power BI Report Server** so users can able to access it through the browser.

What is Power BI Report Server:

Power BI Report Server is an on-premises report server with a web portal in which you display and manage reports and KPIs. Along with it come the tools to create Power BI reports, paginated reports, mobile reports, and KPIs. Your users can access those reports in different ways: viewing them in a web browser or mobile device, or as an email in their in-box.

To learn more about it, visit the link below:

[What is Power BI Report Server? - Power BI | Microsoft Docs](#)

Note: If you don't have a folder in Power BI report Sever then coordinate with IT to get one. If folder is already existing for your organization & you need to just access certain reports or dashboards then coordinate with a responsible group or ISA within your organization.

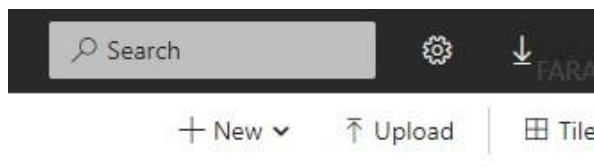
Note: Power BI report server is only support **Edge Browser**.

Steps:

1. In Edge browser, type the below URL. [https://cw-](https://cw-002795.aramco.com/Reports/browse/TD_AdminArea/PowerBI_Training_Sessions/Trainings)

[002795.aramco.com/Reports/browse/TD_AdminArea/PowerBI_Training_Sessions/Trainings](https://cw-002795.aramco.com/Reports/browse/TD_AdminArea/PowerBI_Training_Sessions/Trainings) 2. Click

on **upload** & select the dashboard you just developed.



Note: To avoid conflict with other participants, make sure to embed your network id in file name.

3. Click on the dashboard so it will be open in the browser.
4. **Explore yourself:** Explore the options available in toolbar.



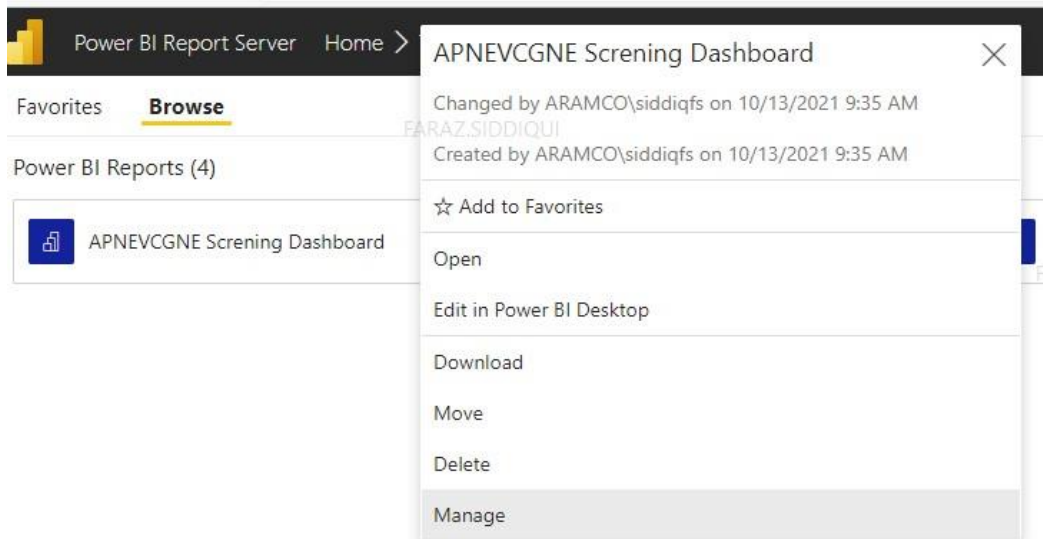
Note: To view the dashboard in full screen, add the following text at the end of the URL string. ?rs:embed=true

Note: You can't perform below steps as it requires administrator rights on the server.

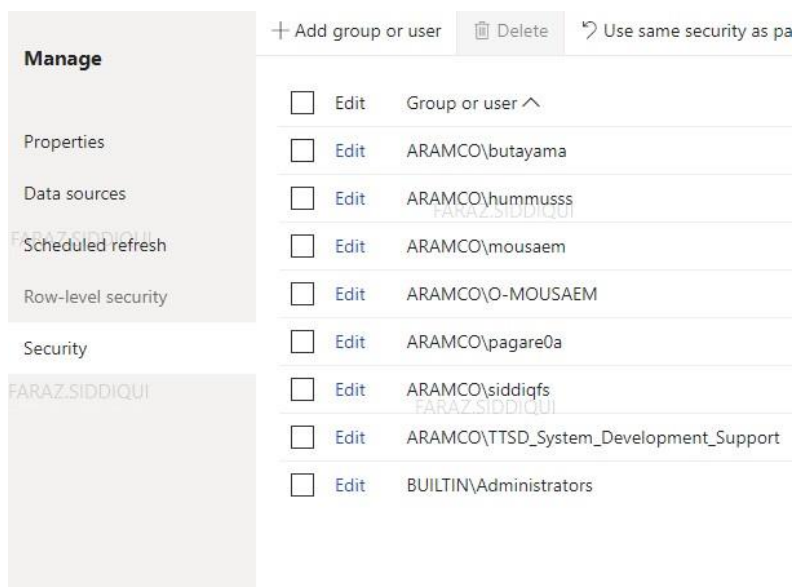
Give Access to Others:

If you have the required rights, you can give access to others.

5. From the main folder, select your dashboard **⚙️...⚙️** Manage.



6. Select Security **⚙️** Add group or user



7. Type either network id or group name (*OE groups from outlook*) & provide required access based on the user profile. For example, **ARAMCO\your network id**.

Use this page to assign role-based security for **APNEVCGNE Screening Dashboard**.

Group or user: FARAZ.SIDDIQUI

Select one or more roles to assign to the group or user.

☐ Role ^	Description
☐ Browser	May view folders, reports and subscribe to reports.
☐ Content Manager	May manage content in the Report Server. This includes folders, reports and resources.
☐ My Reports	May publish reports and linked reports; manage folders, reports and resources in a users My Reports folder.
☐ Publisher	May publish reports and linked reports to the Report Server.
☐ Report Builder	May view report definitions.

FARAZ.SIDDIQUI

To learn more about predefine roles, please visit the link below.

[Role definitions - predefined roles - SQL Server Reporting Services \(SSRS\) | Microsoft Docs](#)

Set the Refresh Schedule:

You can create schedule refresh so the dashboard will update the data in background based on schedule settings.

8. From the main folder, select your dashboard ⑦... ⑦Manage.
9. Click on Data Source & provide the required credentials in following format.
 - Authentication Type: Windows Authentication (*or choose based on application type*)
 - User name: aramco.com\network id
 - Password: Domain Password

Properties

Data sources

Scheduled refresh

Row-level security

Security

Type

SharePoint

Connection string [Learn more](#)

https://sharek.aramco.com.sa/orgs/30008311/30018908/30001431

Credentials

Log into the data source

Authentication Type

Windows Authentication

User name*

aramco.com\siddiqfs

Password*

Test connection

10. Click on Test Connection & make sure that connection is successful.

User name*

aramco.com\siddiqfs

Password*

Test connection

Connected successfully

11. Click on Scheduled Refresh **➕** New Scheduled refresh plan

FARAZ.SIDDIQUI

Manage

Properties

Data sources

Scheduled refresh

Row-level security

Security FARAZ.SIDDIQUI

+ New scheduled refresh plan

☐ Edit

12. Provide description for schedule and set the schedule details based on the requirements then click Apply.

Manage

FARAZ.SIDDIQUI

FARAZ.SIDDIQUI

FARAZ.SIDDIQUI

FARAZ.SIDDIQUI

Schedule details

FARAZ.SIDDIQUI

Choose whether to run the report on an hourly, daily, weekly, monthly, or one time basis.

① All times are expressed in (UTC+03:00) Arab Standard Time.

Hour

☒ Day

Week

Month

Once

Daily schedule

On the following days:

☒ Sun

☒ Mon

☒ Tue

☒ Wed

☒ Thu

☒ Fri

☒ Sat

Every weekday

☒ Repeat after this number of days: 1

Start time: 02 : 00 AM

Start and end dates

FARAZ.SIDDIQUI FARAZ.SID

Specify the date to start and optionally end this schedule.

Begin running this schedule on:

Oct 18, 2021

Stop this schedule on:

Apply

Cancel

13. Once executed, you can check the status of your scheduler.

+ New scheduled refresh plan	+ New from existing	▷ Refresh Now	🗑 Delete	Search...	
✓ Edit	Description ^	Schedule			FARAZ.SIDDIQUI Last run
✓ Edit	Test Plan	At 1:15 PM on 1/12/2023	FARAZ.SIDDIQUI		Jan 12, 2023 1:16:26 PM

14. You can run the scheduler anytime by clicking on **Refresh Now**. Also, you can edit the schedule if needed.

+ New from existing

▷ Refresh Now

Note: If your data source is Excel, CSV, TXT etc. then you need to first put your files into the **shared folder** before configure the above steps.

Refresh Schedule for Excel & other file-based Data Sources:

Scheduled Refresh can be configured for file-based sources (Excel, CVS, TXT, XML etc.) to automate the data refresh for your dashboard/reports by following the steps below.

15. Create the shared folder, put your files and make sure you can access it in below format.

\\aramco\data\Your Folder\Your file.

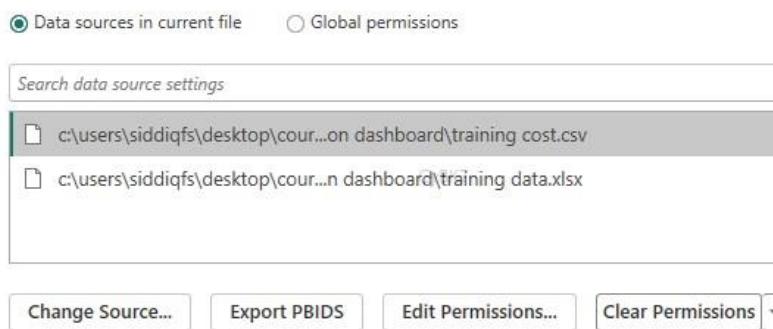
16. Open your power bi document in Power BI Desktop Application.

2026 Saudi Aramco | Non-Business Use

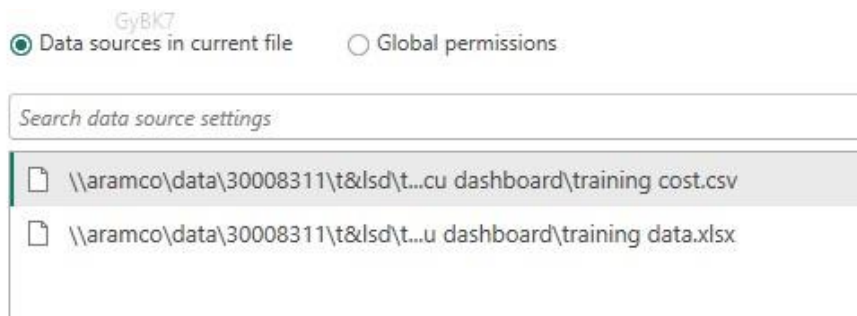
17. From transform data, click on **Data source settings**.



18. Select the file and click on **change source**. Change the location to read from the shared folder.



19. Repeat the above step for each file & make the location is set to shared folder.



20. Save & re-upload power bi file into the server.

21. Repeat the above steps **(8-10)** to configure the data source tab in the server as shown below.

Data Source 1:

Connection

Type GyBK7

File

Connection string [Learn more](#)

\\aramco\data\30008311\t&l&sd\t&l&sd_common\ttcu_dashboard\training cost.csv

GyBK7

Credentials

Log into the data source

Authentication Type

Windows Authentication

User name*

aramco.com\siddiqfs GyBK7

Password*

Test connection

✓ Connected successfully

Note: If there are multiple files then make sure to configure the connection for each file.

22. Repeat the above steps **(11-14)** to create schedule refresh.

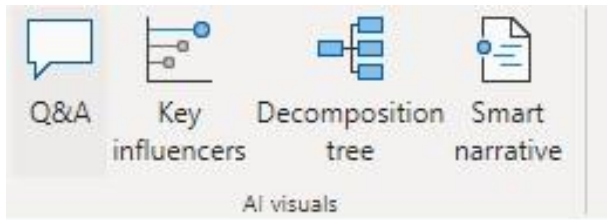
Exercise 8: Advance Features

In this exercise, you will learn some of the advance features of Power BI includes:

- AI Visuals
- Linking pages or reports
- What-if Analysis
- ML Models with R/Python (demo only)
- Row level Security (optional) **Steps:**

1. AI Visuals:

Power BI supports built-in AI visuals to perform advance analytics. This includes Q&A, Smart Narrative, Decomposition tree etc.



Note: Currently we have only Decomposition tree AI visual supported in our environment.

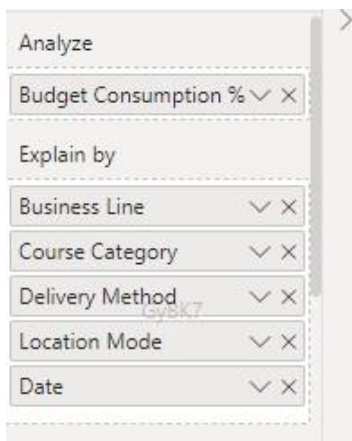
2.1 Decomposition Tree:

The **Decomposition Tree** visual automatically aggregates your data and lets you drill down into your dimensions so that you can view your data across multiple dimensions. Because Decomposition Tree is an AI visual, you can use it for improvised exploration and conducting **root cause analysis**.

To learn more about Decomposition Tree, visit the link below:

[Use the Decomposition Tree visual to break down a measure - Learn | Microsoft Docs](#)

- a) Insert a new page called **Advance Analytics**.
- b) Insert **Decomposition Tree** into Advance Analytics page.
- c) Insert following fields in **Analyze** and **Explain By**. *(Insert dimension fields from their respected dimension tables).*



- d) In the graph, click on + icon. As you click again, the measure will decompose accordingly.



2.2 Q&A:

The **Q&A** feature in Power BI lets you explore your data in your own words by allowing you to ask natural language questions and then providing you with answers to those questions.

Power BI records all questions that are asked, and you can use this information to set up the Q&A feature to be more effective. When the Q&A feature answers so many questions, you'll have fewer people turning to you for those answers.

To learn more about Q&A, visit the link below:

[Use the Q&A visual - Learn | Microsoft Docs](#)

2.3 Smart Narrative:

The smart narrative visualization helps you quickly summarize visuals and reports. It provides relevant innovative insights that you can customize.

To learn more about Smart Narrative, visit the link below:

[Smart narratives tutorial - Power BI | Microsoft Docs](#)

2. Page Navigation & Linking Reports:

In power BI, you can easily navigate to pages by inserting **Buttons**. You can link pages by passing values between as well as link two different reports published in the server by using **Cross-report Drill through** Feature.

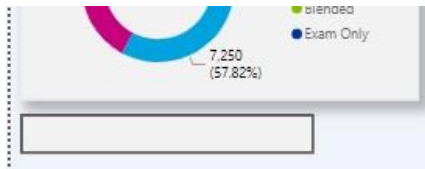
To learn more about Page Navigation, please visit the link below:

[Create buttons in Power BI reports - Power BI | Microsoft Docs](#)

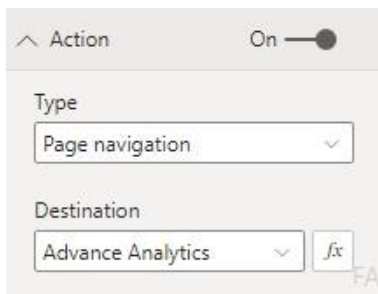
To learn more about cross-report, please visit the link below:

[Use cross-report drillthrough in Power BI Desktop - Power BI | Microsoft Docs](#)

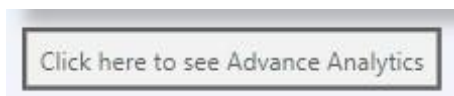
- a) While in **QBR Summary** page, click on Insert  Buttons  Blank to insert **blank button** at the bottom left.



- b) Select the button & from the format tab, go to **Action** & set the below options.



- c) Enable Style  Text & write **Click here to see Advance Analytics**.



- d) Now go to the Advance Analytics page and repeat the similar steps to link it **back** to the dashboard page. (*select back from Action*).



- e) Hide the Advance Analytics page.
f) Save your changes.
g) Re-upload your dashboard into the server & see how page linking will work.

Note: Page Navigation helps if you have multiple pages in your dashboard and you want users to navigate the pages through links instead of showing them all the pages. This will help to simplify your report or a dashboard.

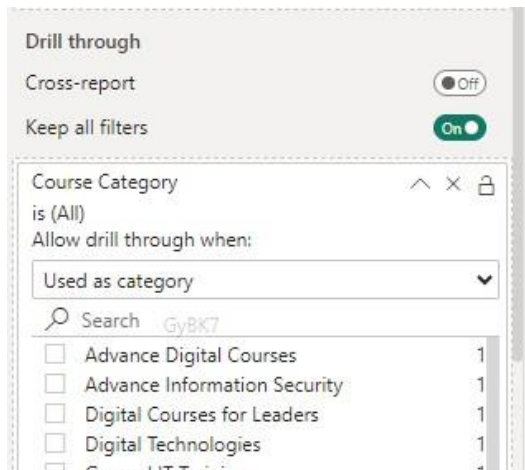
Passing Values between Pages:

Users would like to see detailed report by **Course Category**. As we have a graph to show Total hours by Course Category in Monthly Analysis page, we can create detailed report & link with that graph. Once user select any course category, the detailed report will be appeared. Otherwise hide. We can accomplish that through **drill-through** feature.

- Create a new page called **Detailed Report**.
- Insert table in new page and populate with data as shown below:

Course Category Code	Course Category	Business Line	Year	Month ID	Month Name	Delivery Method	Location Mode	Total Hours	Total Participants	Total Sessions	Total Courses
1002	General SAP Training	Finance	2022	7	Jul	Classroom	In-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2022	8	Aug	Classroom	In-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2022	9	Sep	Classroom	In-Kingdom	120	5	2	1
1002	General SAP Training	Finance	2022	10	Oct	Classroom	In-Kingdom	248	9	4	2
1002	General SAP Training	Finance	2022	11	Nov	Classroom	In-Kingdom	648	9	4	4
1002	General SAP Training	Finance	2022	12	Dec	Classroom	In-Kingdom	168	5	3	2
1002	General SAP Training	Finance	2023	1	Jan	Classroom	In-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2023	2	Feb	Classroom	In-Kingdom	48	2	1	1
1002	General SAP Training	Finance	2023	3	Mar	Classroom	In-Kingdom	96	4	2	1
1002	General SAP Training	Finance	2023	10	Oct	Classroom	In-Kingdom	384	16	3	1
1002	General SAP Training	Finance	2023	11	Nov	Classroom	In-Kingdom	72	3	2	1
1002	General SAP Training	Finance	2023	12	Dec	Classroom	In-Kingdom	240	8	4	2
1002	General SAP Training	Finance	2022	12	Dec	Classroom	Out-of-Kingdom	80	2	1	1
1002	General SAP Training	Finance	2023	3	Mar	Classroom	Out-of-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2023	11	Nov	Classroom	Out-of-Kingdom	240	10	2	1
1002	General SAP Training	HR & Corporate Services	2022	7	Jul	Classroom	In-Kingdom	96	4	2	1
1002	General SAP Training	HR & Corporate Services	2022	8	Aug	Classroom	In-Kingdom	120	5	2	1
1002	General SAP Training	HR & Corporate Services	2022	9	Sep	Classroom	In-Kingdom	72	3	1	1
1002	General SAP Training	HR & Corporate Services	2022	10	Oct	Classroom	In-Kingdom	112	4	3	2
1002	General SAP Training	HR & Corporate Services	2022	11	Nov	Classroom	In-Kingdom	224	7	2	2
1002	General SAP Training	HR & Corporate Services	2022	12	Dec	Classroom	In-Kingdom	320	10	4	3
1002	General SAP Training	HR & Corporate Services	2023	1	Jan	Classroom	In-Kingdom	96	4	1	1
1002	General SAP Training	HR & Corporate Services	2023	2	Feb	Classroom	In-Kingdom	400	2	2	2
1002	General SAP Training	HR & Corporate Services	2023	3	Mar	Classroom	In-Kingdom	72	3	2	1
1002	General SAP Training	HR & Corporate Services	2023	10	Oct	Classroom	In-Kingdom	72	3	2	1
Total								25,992	447	44	11

- While in detailed report, insert Course Category from Course Category dimension table into **Drill through**.

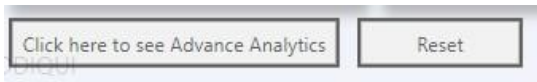


- d) Hide the detailed Report.
- e) Re-upload dashboard into the server.
- f) From the browser, right click over any course category. Detailed report will be appeared to show the data for the selected course category only.



Course Category Code	Course Category	Business Line	Year	Month ID	Month Name	Delivery Method	Location Mode	Total Hours	Total Participants	Total Sessions	Total Courses
1002	General SAP Training	Finance	2022	7	Jul	Classroom	In-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2022	8	Aug	Classroom	In-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2022	9	Sep	Classroom	In-Kingdom	120	5	2	1
1002	General SAP Training	Finance	2022	10	Oct	Classroom	In-Kingdom	248	9	4	2
1002	General SAP Training	Finance	2022	11	Nov	Classroom	In-Kingdom	648	9	4	4
1002	General SAP Training	Finance	2022	12	Dec	Classroom	In-Kingdom	168	5	3	2
1002	General SAP Training	Finance	2023	1	Jan	Classroom	In-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2023	2	Feb	Classroom	In-Kingdom	48	2	1	1
1002	General SAP Training	Finance	2023	3	Mar	Classroom	In-Kingdom	96	4	2	1
1002	General SAP Training	Finance	2023	10	Oct	Classroom	In-Kingdom	384	16	3	1
1002	General SAP Training	Finance	2023	11	Nov	Classroom	In-Kingdom	72	3	2	1
1002	General SAP Training	Finance	2023	12	Dec	Classroom	In-Kingdom	240	8	4	2
1002	General SAP Training	Finance	2022	12	Dec	Classroom	Out-of-Kingdom	80	2	1	1
1002	General SAP Training	Finance	2023	3	Mar	Classroom	Out-of-Kingdom	24	1	1	1
1002	General SAP Training	Finance	2023	11	Nov	Classroom	Out-of-Kingdom	240	10	2	1
1002	General SAP Training	HR & Corporate Services	2022	7	Jul	Classroom	In-Kingdom	96	4	2	1
1002	General SAP Training	HR & Corporate Services	2022	8	Aug	Classroom	In-Kingdom	120	5	2	1
1002	General SAP Training	HR & Corporate Services	2022	9	Sep	Classroom	In-Kingdom	72	3	1	1
1002	General SAP Training	HR & Corporate Services	2022	10	Oct	Classroom	In-Kingdom	112	4	3	2
1002	General SAP Training	HR & Corporate Services	2022	11	Nov	Classroom	In-Kingdom	224	7	2	2
1002	General SAP Training	HR & Corporate Services	2022	12	Dec	Classroom	In-Kingdom	320	10	4	3

- g) Power BI also create back button to navigate to main page.
- h) **Challenging Task:** Create a **Reset** button in QBR Summary page which allow users to reset dashboard by removing all applied filters.



- i) Upload again to the server & test reset functionality.

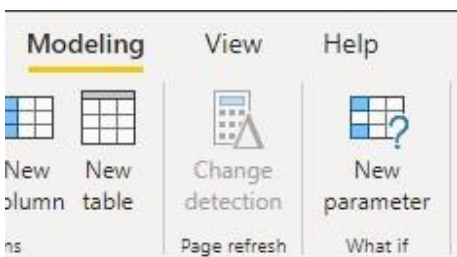
3. What-if Analysis:

You can create **what-if** variables for your reports, interact with the variable as a slicer, and visualize and quantify different key values in your reports.

To learn more about What-if Analysis, please visit the link below:

[Use what-if parameters to visualize variables - Power BI | Microsoft Docs](#)

- a) Insert a new page called **What-if Analysis**.
- b) From the Modeling tab, click on **New Parameter**.



- c) Set the following values in parameter screen & click OK.

What-if parameter GyBK7 X

Name
Additional Budget %

Data type
Decimal number

Minimum
0

Maximum
0.30

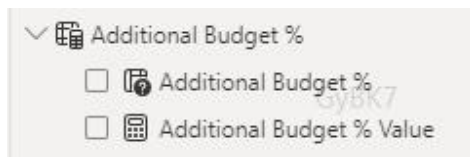
Increment
0.01

Default
0

☒ Add slicer to this page

OK **Cancel**

- d) Power BI automatically create new measures to visualize the parameter values.

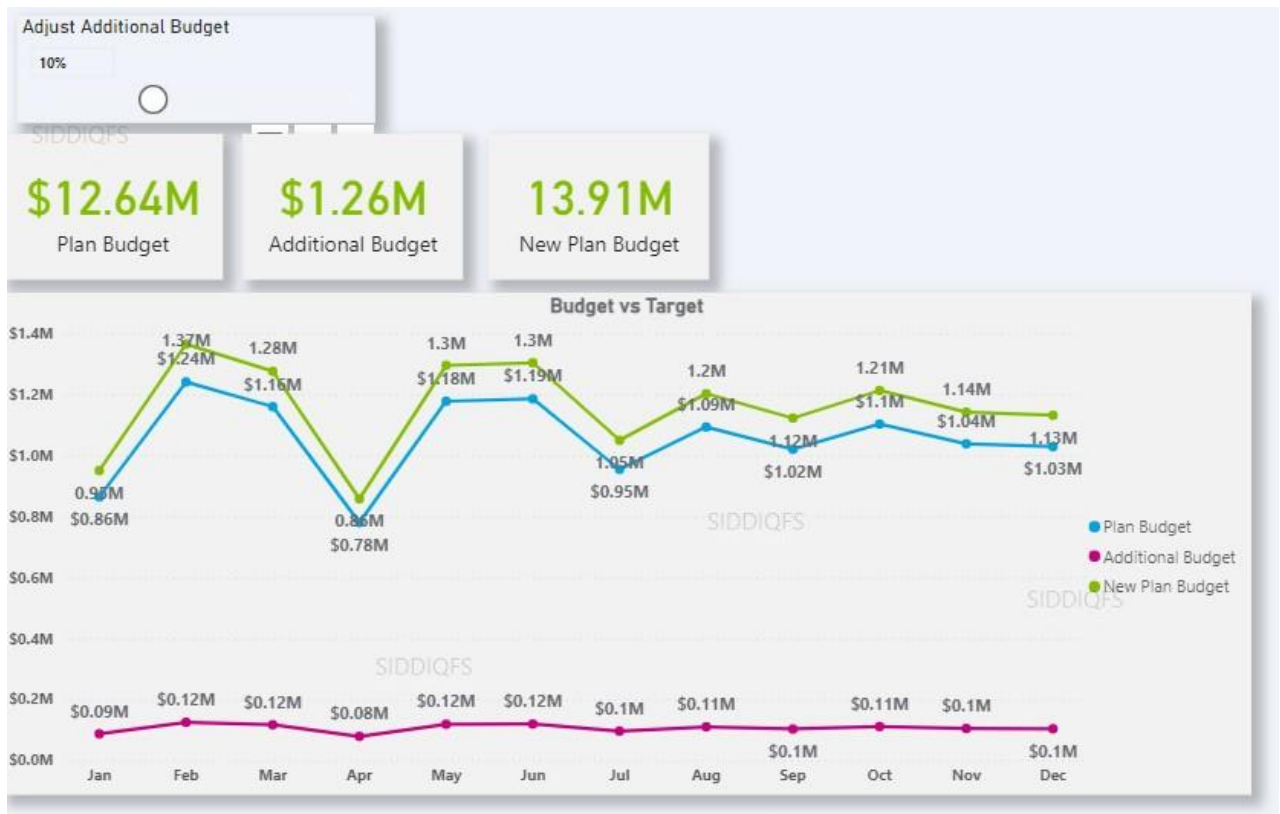


- e) Now, we need to create calculated measures where we can apply the what-if logic. In KPI's table, create following measures called **Additional Budget** & **New Plan Budget** as shown below:

Additional Budget = [Plan Budget] * 'Additional Budget %'[Additional Budget % Value]

New Plan Budget = [Plan Budget] + [Additional Budget]

- f) Insert Card & Line Chart & populate with Plan Budget, Additional Budget, New Plan Budget & Month Name.
- g) Change **Additional Budget %** from Slicer & analyze the impact.



h) Save your work.

4. Row-level Security (Optional):

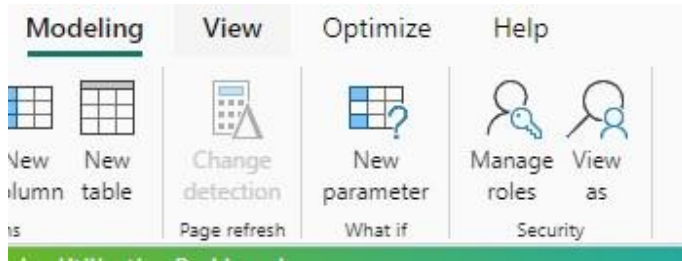
Row-level security (RLS) with Power BI can be used to restrict data access for given users. Filters restrict data access at the row level, and you can define filters within roles.

To learn more about row-level security, visit the link below:

[Row-level security \(RLS\) with Power BI - Power BI | Microsoft Docs](#)

Note: You can't perform below steps as it requires administrator rights on the server.

- a) From Modeling tab, click on **Manage roles**.



- b) We would like to create a role to restrict data for **Upstream** business line only and assign to required users.
- c) Click on **Create** and call it Upstream Only.

Manage roles

Roles

Upstream Only ...

Create

Delete

- d) Under the Tables, select Business Line table ➊ Add filter... ➋ Business Line

Manage roles

Roles

Upstream Only ...

Create

Delete

Tables

Business Line_Dim

Course Category BD

Course Category TD

Course Category_Dim

Date_Dim

Delivery Method_Dim

KPI's

Last Refresh Date

Location Mode_Dim

Training Cost_Fact

Training Data_Fact

Table filter DAX expression

GyBK7

Add filter...

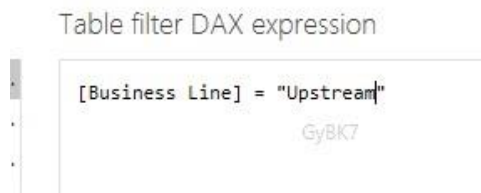
Copy table filter from...

Clear table filter

Hide all rows

[Business Line]

- e) In the generated DAX expression, type Upstream as value. Verify & click save.

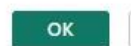


f) Click on **View as**. Select Upstream Only and click OK.



View as roles

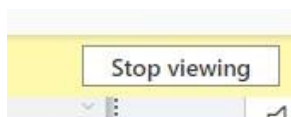
- ☐ None
- ☐ Other user
- ☒ Upstream Only



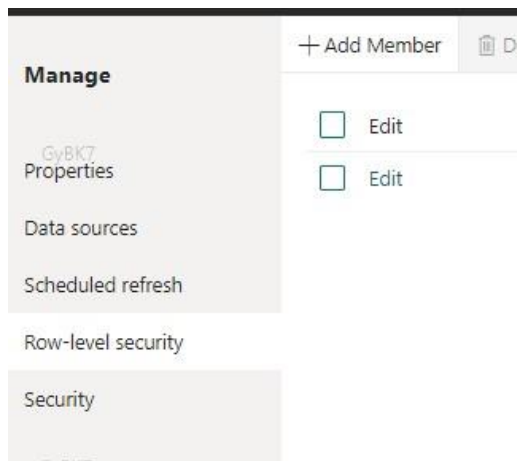
g) The whole dashboard is filtered for upstream data only.



h) Click on stop viewing to return back to normal view.



- i) In order to apply the above created role to specified users, we need to re-upload dashboard to the server & configure row level security.
- j) While in server, select your dashboard file **Manage** **Row-level security** **Add Member**.



k) Add user or a group & assign the required row level security.

Use this page to add members to one or more role:

Group or user:

Select one or more roles to assign to the group or user:

- ☐ Role ^
- ☐ IT General Training only
- ☐ Upstream & Downstream Data
- ☒ Upstream Only

OK **Cancel**

- l) When the user views the dashboard from the browser, the data will show as per the role assigned to the user.
- m) **Challenging Task:** Create a new role called **SAP only**. Restrict the data for the following course categories and assign it to specified users.
- General SAP Training
 - SAP Academy
 - SAP Certifications

Appendix A: Getting the Data from SAP Business Warehouse (BW)

Power BI can fetch the data from SAP Business Warehouse (BW) System.

The below steps explain how to extract the data from SAP BW (PRW System).

Note: You need to coordinate with your ISA to have the required role in **SAP PRW** System in Aramco.

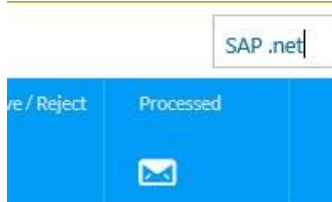


Note: It's not guaranteed that below steps will work due to of some **unresolved error**. You can explore the alternative option to fetch the data from SAP PRW as highlighted in **Appendix B**.

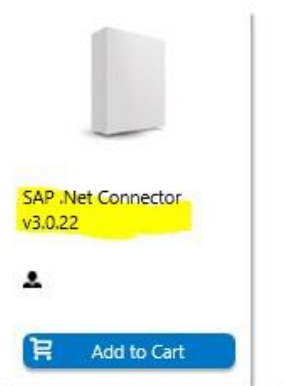
Steps:

1. install **SAP .Net Connector** from Aramco App Center.

<https://appcenter.aramco.com.sa>

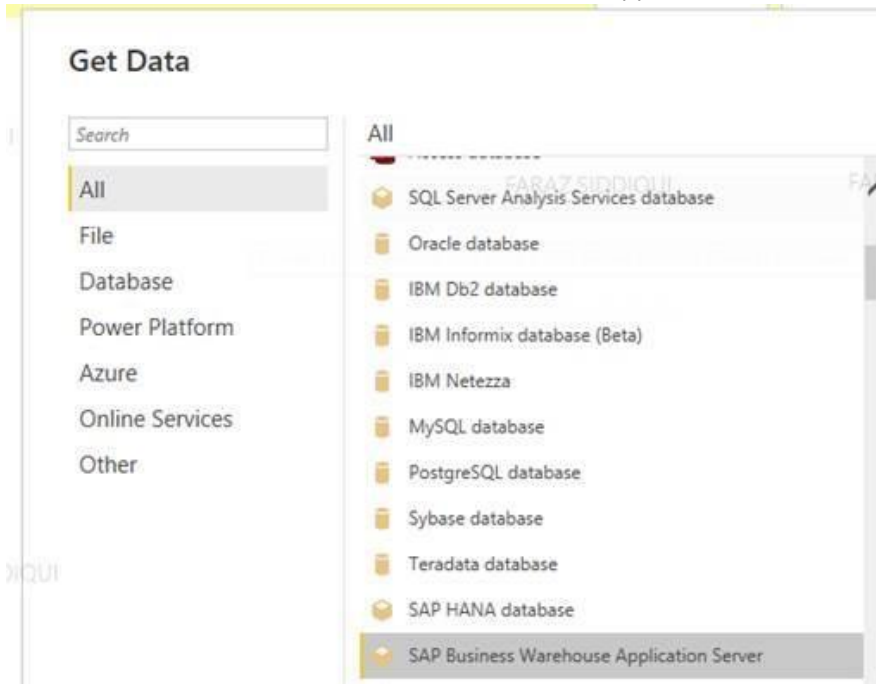


"SAP .net"
5 Search Results



2. Open Power BI Application.

3. From Get Data ⑦ SAP Business Warehouse Application Server



4. Provide the following details

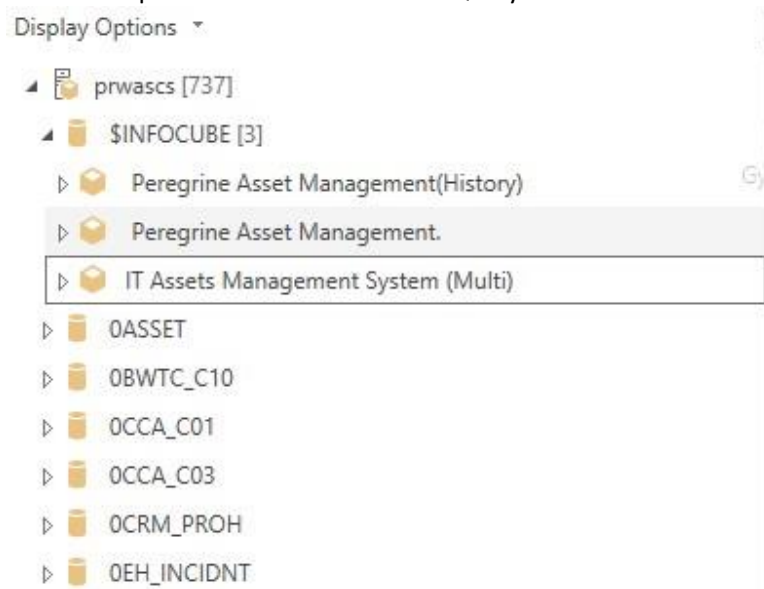
- a. Server: prwascs
- b. System number: 03
- c. Client ID: 242



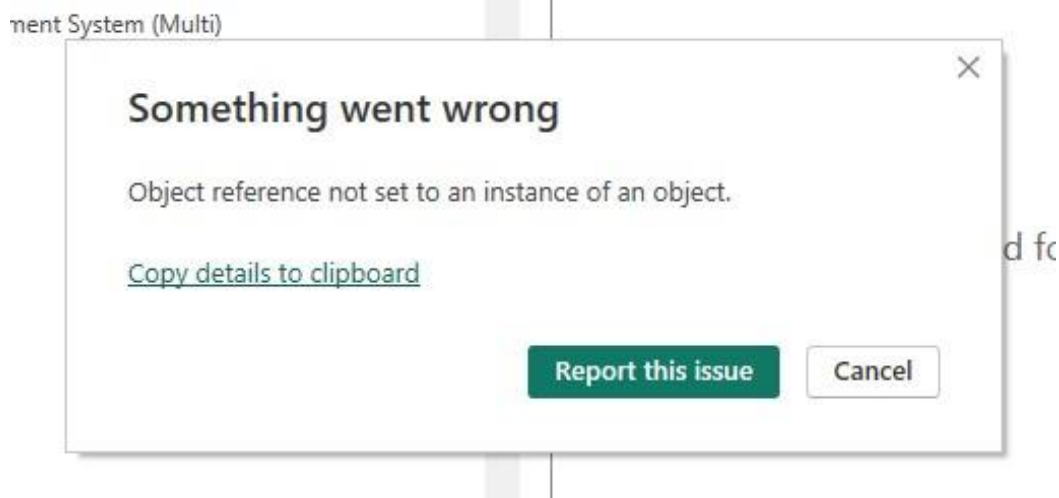
5. Select **Database** and type your network id and password.



6. You can able to view the Info Providers under Navigator Screen. Select the required Info Provider or BEx Query.



7. If you are getting the below error then its **unresolved issue** yet.



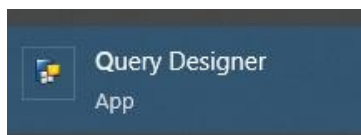
Appendix B: Getting the Data from SAP Business Warehouse (BW) through BEx Query & Web

You can fetch the data from SAP BW (PRW) through **BEx Query & Web** option in

Power BI. **What is a BEx Query?**

BEx queries allow users to define queries by selecting and combining **InfoObjects** – such as characteristics and key figures to explore the data within an InfoProvider.

Users can either consume existing queries or create their own BEx queries using **Query Designer tool**. For more details , see **Appendix C**.



Note: You need to coordinate with your ISA to have the required role in **SAP PRW** System in Aramco.

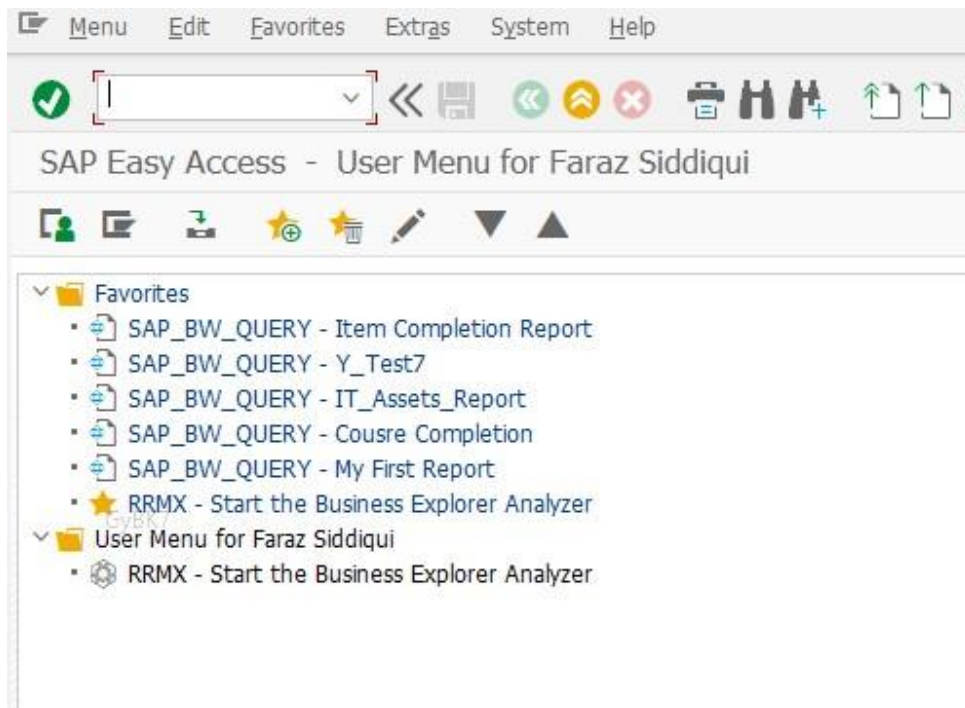


Steps:

1. We would like to consume available query. From myhome , launch PRW.



2. From PRW logon, double click to execute your query.



3. For example, we will run **IT_Assets_Report**. The report will be loaded in a browser.

5. Replace the **red part** with your query technical name and info provider technical copied from the above step.

[https://prw.aramco.com.sa:44303/sap\(bD1lbiZjPTI0MiZkPW1pbG==\)/bc/bsp/sap/zbw_reporting/execute_report.htm?query=InfoProvider Technical Name%2fQuery Technical Name](https://prw.aramco.com.sa:44303/sap(bD1lbiZjPTI0MiZkPW1pbG==)/bc/bsp/sap/zbw_reporting/execute_report.htm?query=InfoProvider Technical Name%2fQuery Technical Name)

6. **For example:**

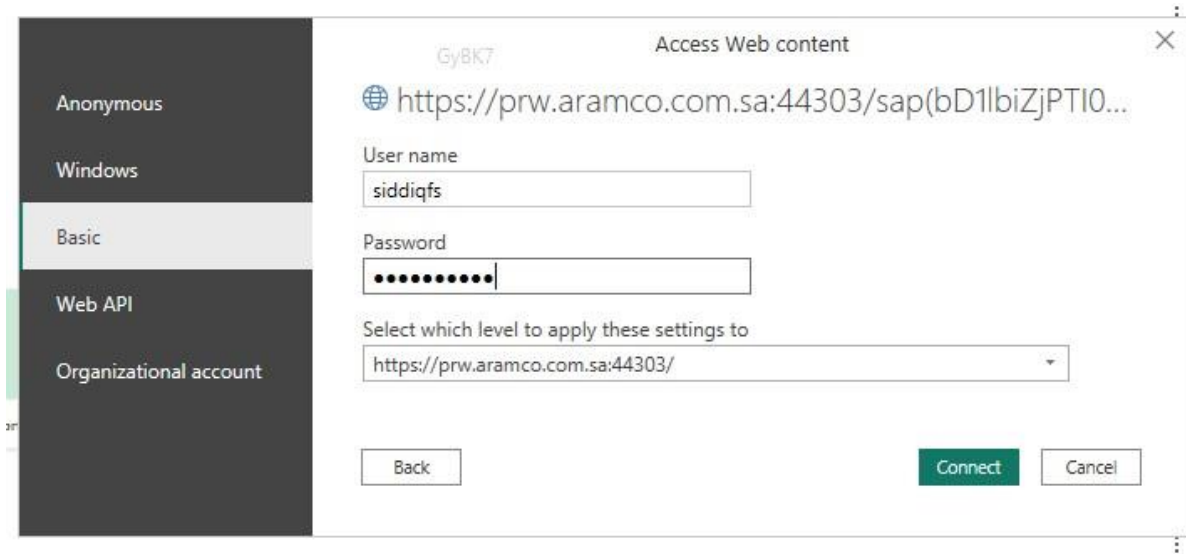
[https://prw.aramco.com.sa:44303/sap\(bD1lbiZjPTI0MiZkPW1pbG==\)/bc/bsp/sap/zbw_reporting/execute_report.htm?query=ZXAINV%2fY_IT_ASSESTS_REPORT](https://prw.aramco.com.sa:44303/sap(bD1lbiZjPTI0MiZkPW1pbG==)/bc/bsp/sap/zbw_reporting/execute_report.htm?query=ZXAINV%2fY_IT_ASSESTS_REPORT)

7. Open Power BI Desktop Application.
8. From Get Data, select **Web** and paste the query you created in above step. Click OK.



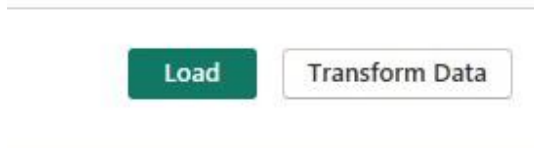
The screenshot shows the 'From Web' dialog box in Power BI. The 'Basic' tab is selected. The URL field contains the text 'bw_reporting/execute_report.htm?query=ZXAINV%2fY_IT_ASSESTS_REPORT'. The 'OK' button is highlighted in green.

9. In the credential screen, select **Basic** & type your network id & password (same password you set for PRW) & click **Connect**.

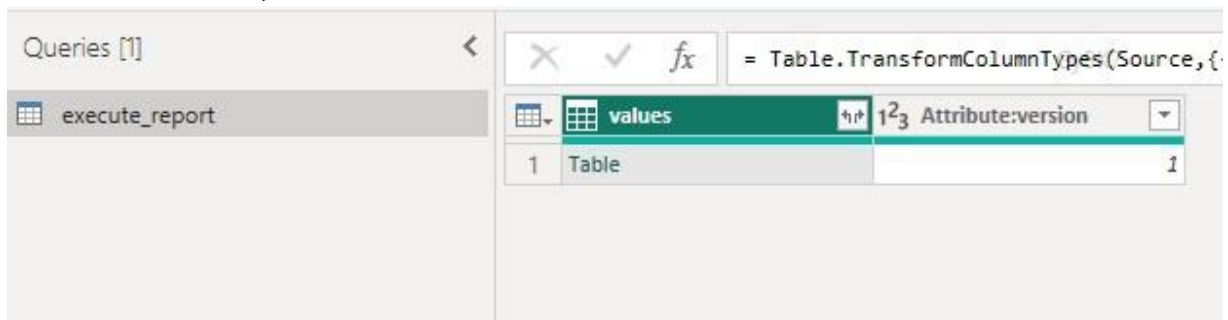


The screenshot shows the 'Access Web content' credential screen in Power BI. The 'Basic' tab is selected. The 'User name' field contains 'siddiqfs'. The 'Password' field is masked with dots. The 'Select which level to apply these settings to' dropdown is set to 'https://prw.aramco.com.sa:44303/'. The 'Connect' button is highlighted in green.

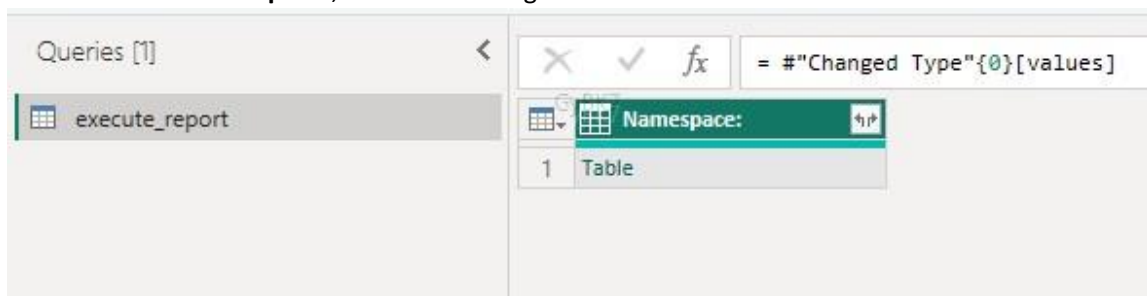
10. Click on **Transform Data**.



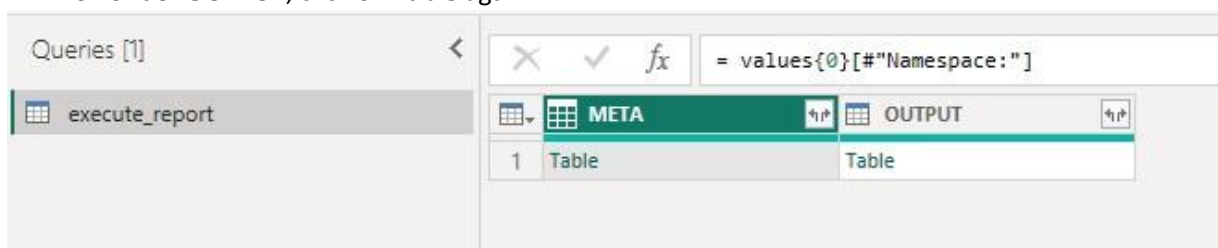
11. Under **values**, click on **Table**.



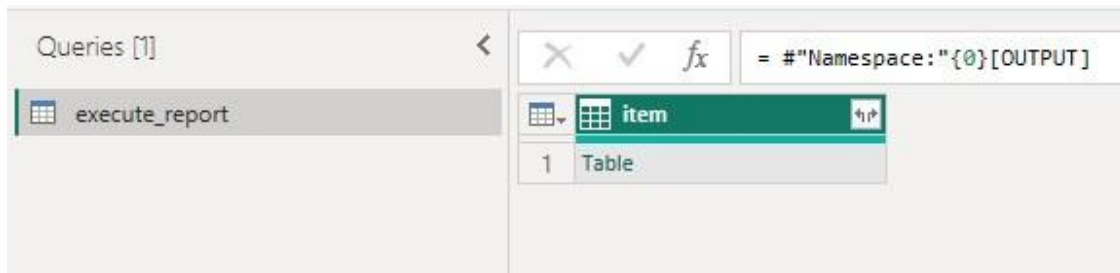
12. Under **Namespace:**, click on **Table** again.



13. Under **OUTPUT**, click on **Table** again.



14. Under **item**, Click on **Table** again.



15. You can now able to view the data.

Queries [1]

execute_report

Table.TransformColumnTypes(item,{{\"VALUE001\", Int64.Type}, {\"ZHRBUSLIN\", type text}, {

	1 ² VALUE001	A ^B _C ZHRBUSLIN	A ^B _C ZHRDEP	A ^B _C ZHRDIV
1	3	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
2	17	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
3	2	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
4	138	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
5	17	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
6	7	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
7	7	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
8	191	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
9	18	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
10	2	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
11	20	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
12	1	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div
13	1	30002950 Technical Services	30001956 MLD	30001963 Global Logistics Div

16. Double click on Column headers to give proper column names as per your BEX Query.

Table.RenameColumns(#\"Changed Type1\",

	1 ² Asset Count	A ^B _C Business Line	A ^B _C
1	3	30002950 Technical Services	30001956 MLD
2	17	30002950 Technical Services	30001956 MLD
3	2	30002950 Technical Services	30001956 MLD
4	138	30002950 Technical Services	30001956 MLD

17. Also give a proper name for your query under properties.

Query Settings

PROPERTIES

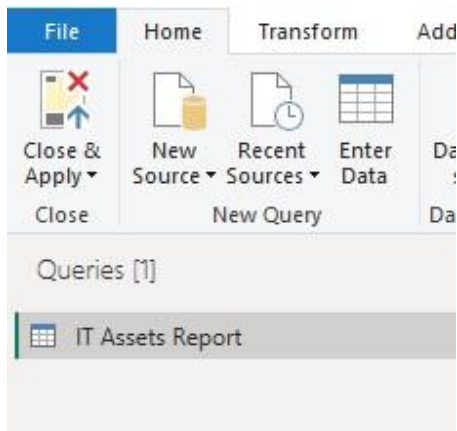
Name

IT Assets Report

All Properties

APPLIED STEPS

18. Click Close & Apply.



Appendix C: How to create your own BEx Query

You can create your own BEx queries using **Query Designer** Tool.

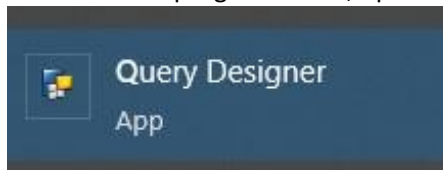
What is a BEx Query?

BEx queries allow users to define queries by selecting and combining InfoObjects – such as characteristics and key figures to explore the data within an InfoProvider.

Note: You need to coordinate with your ISA to have the required role in **SAP PRW** System in Aramco.

Steps:

1. From program menu, open **Query Designer** Tool.

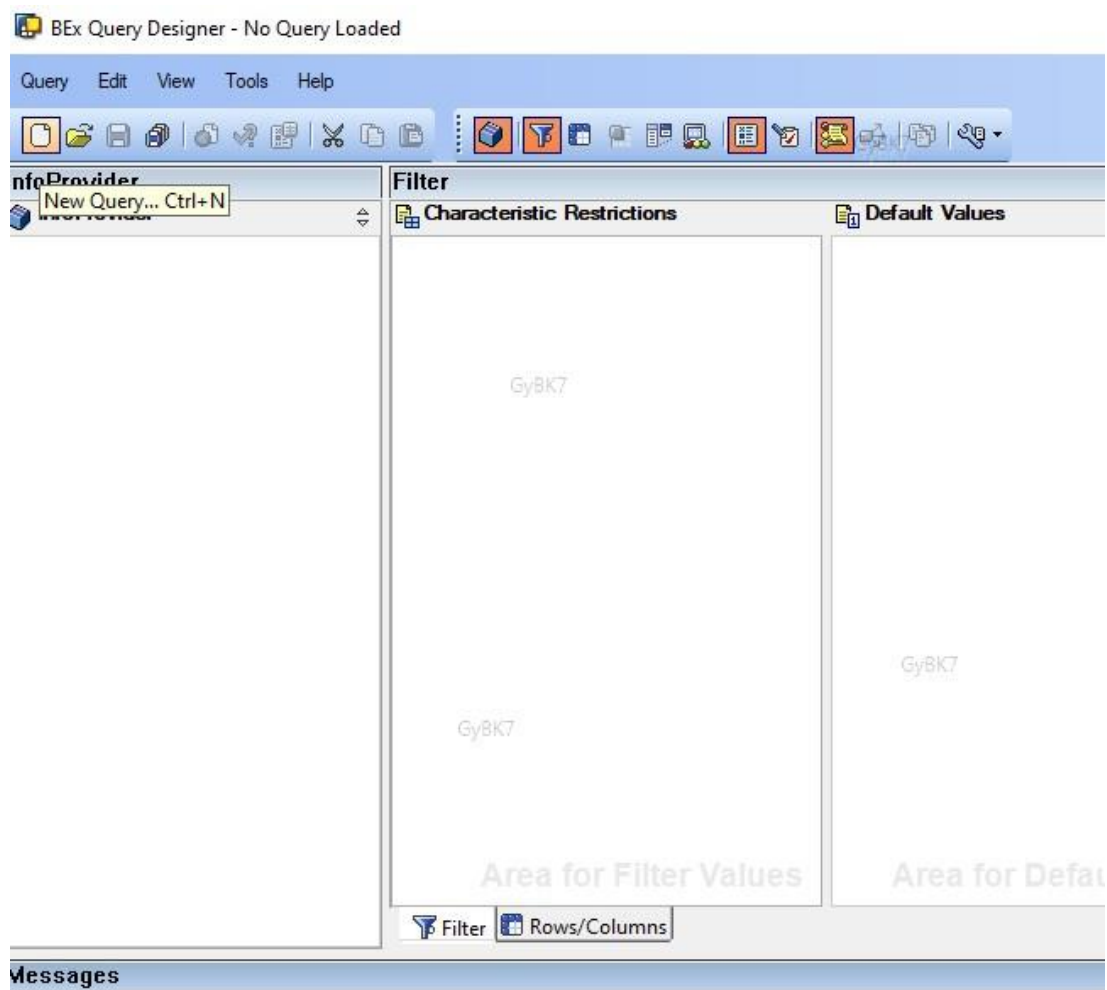


2. Type your network id & password. Make sure client is **242**. Click OK.

SAP Logon at BW - Direct ✕

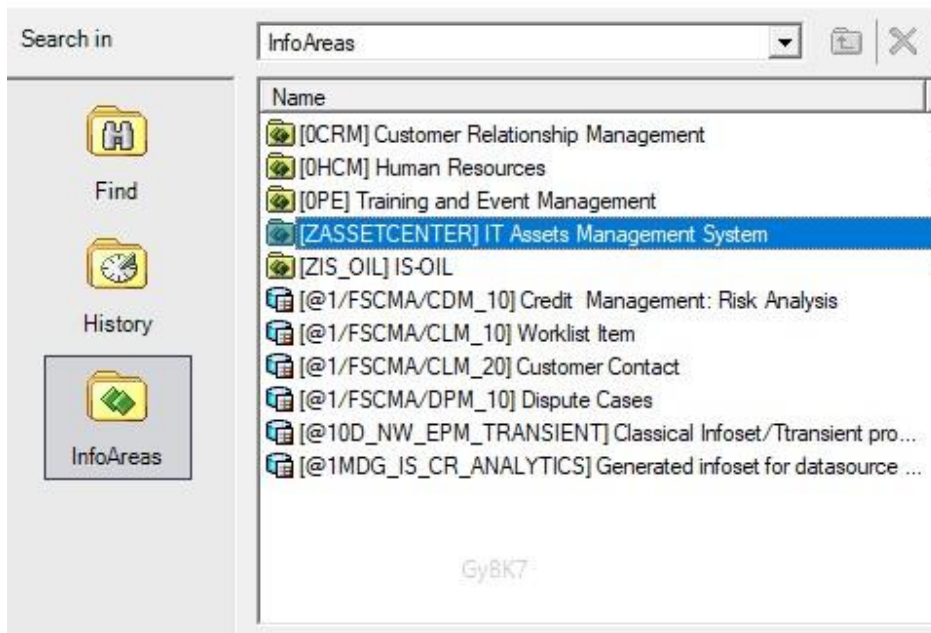
User Data	
Client	242
User	SIDDIQFS
Password	*****
Language	EN

3. Click on **New Query**.



4. Click on **InfoAreas** and select the required **InfoProvider**.

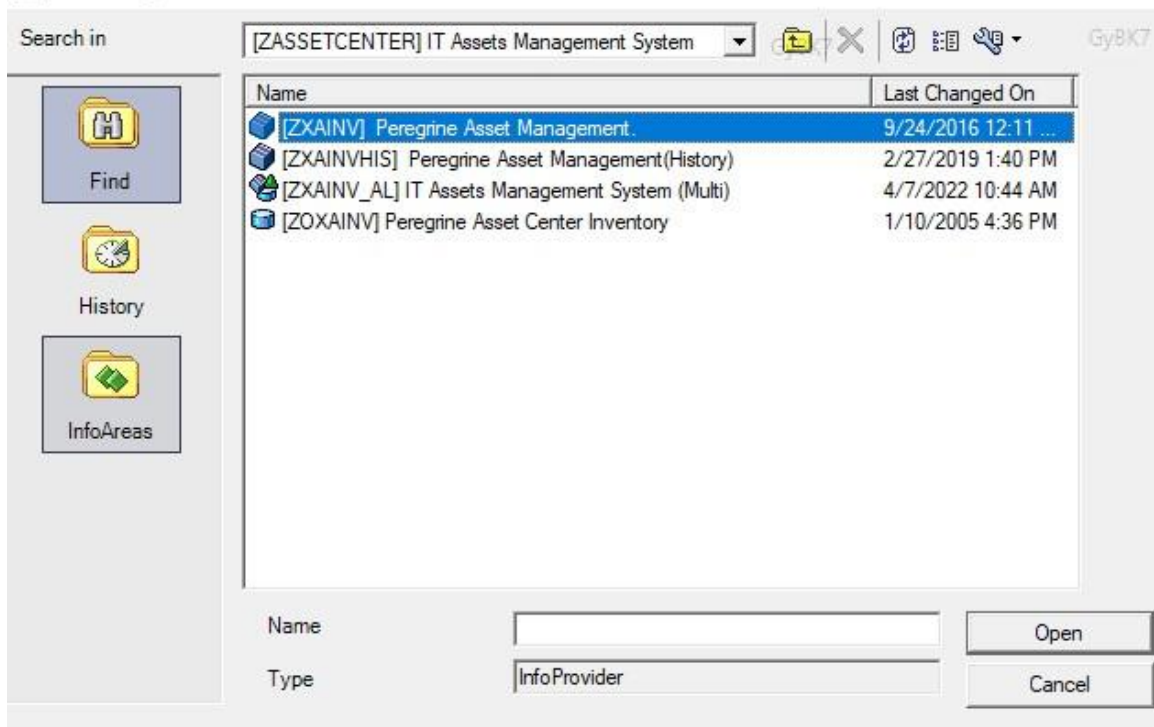
New Query: Select InfoProvider



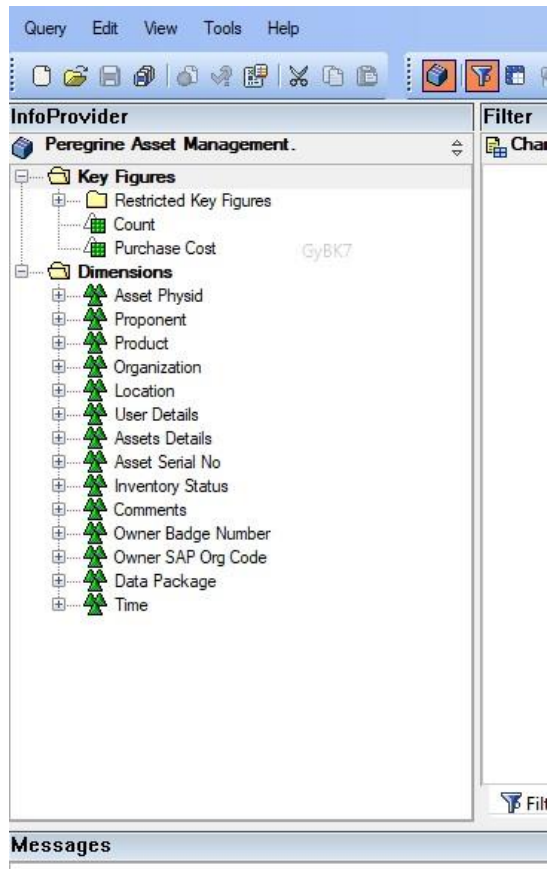
Note: you can view InfoProviders based on your assign Role. InfoProvider contains **InfoCubes** or **DSO's** which store the data exported from transactional systems like PRC, PRH etc.

5. Select the required InfoCube or DSO then click Open.

New Query: Select InfoProvider



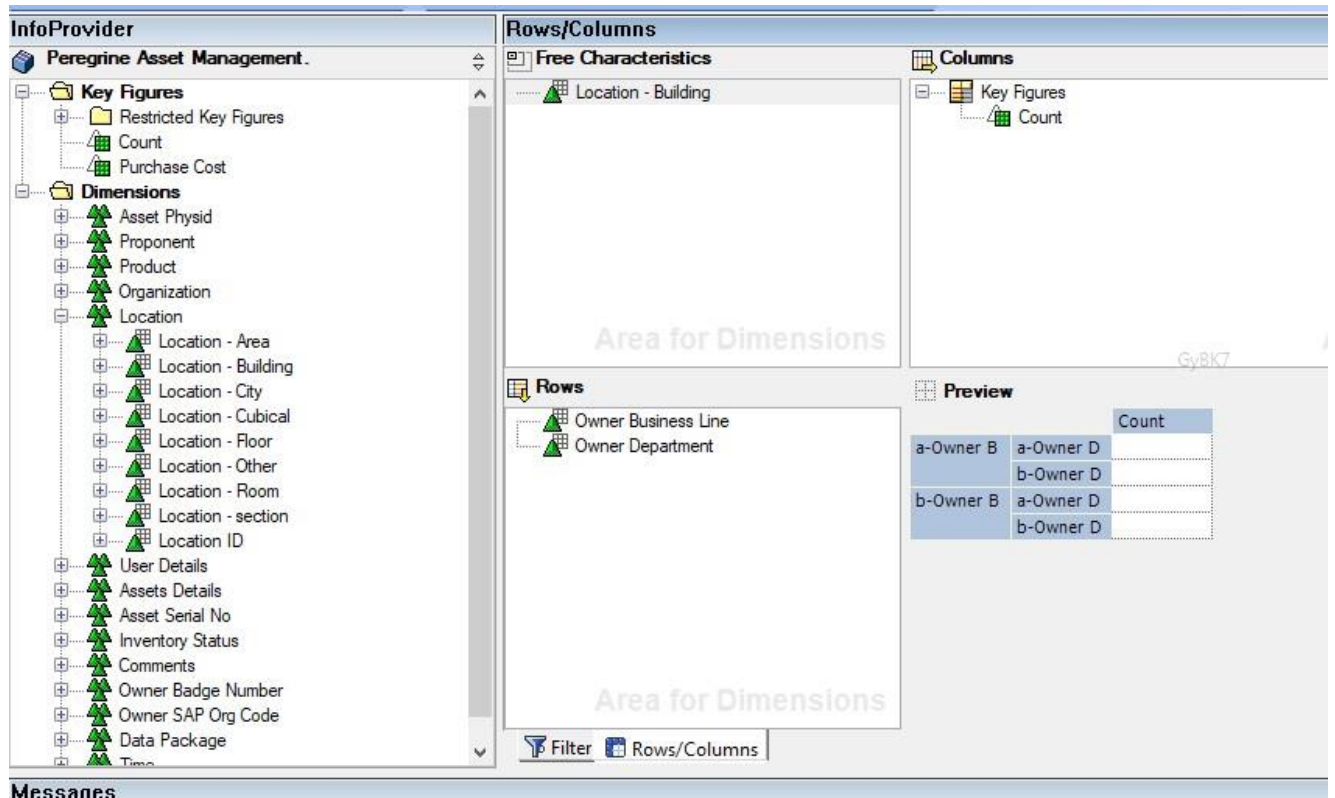
6. Data structure will be loaded on the left side.



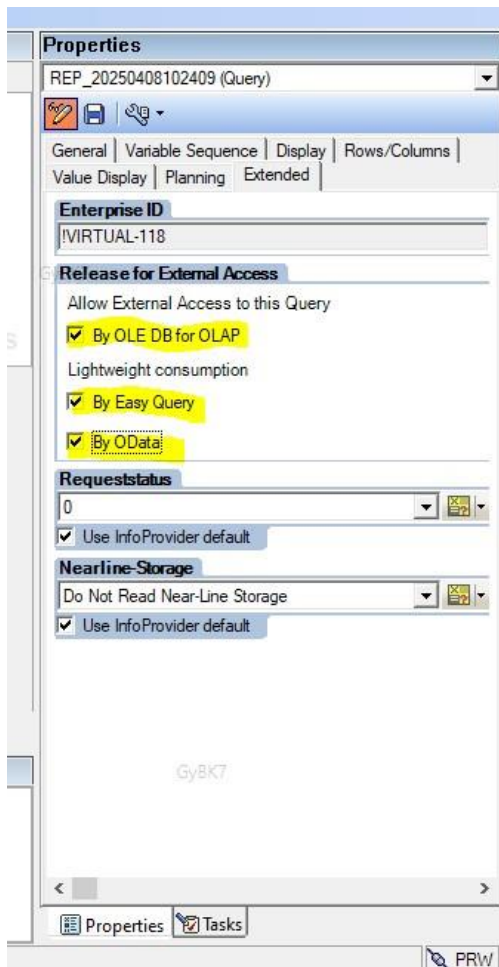
Note: InfoCube contains **Dimensions & Key Figures**. Dimensions are categorical data like Product, Location, Time etc. while Key Figures are numerical values like count, purchase cost etc.

7. Build your query by adding dimensions & key figures into **Rows & Columns** as per your requirement. For example, we can add Business Line and Department under **Rows**, count (*Total assets count*) as **Column** and Building as **free characteristics**. You can also add **filters** in your query if needed.

Note: **free characteristics** are dimension objects you can optionally add later in your report.

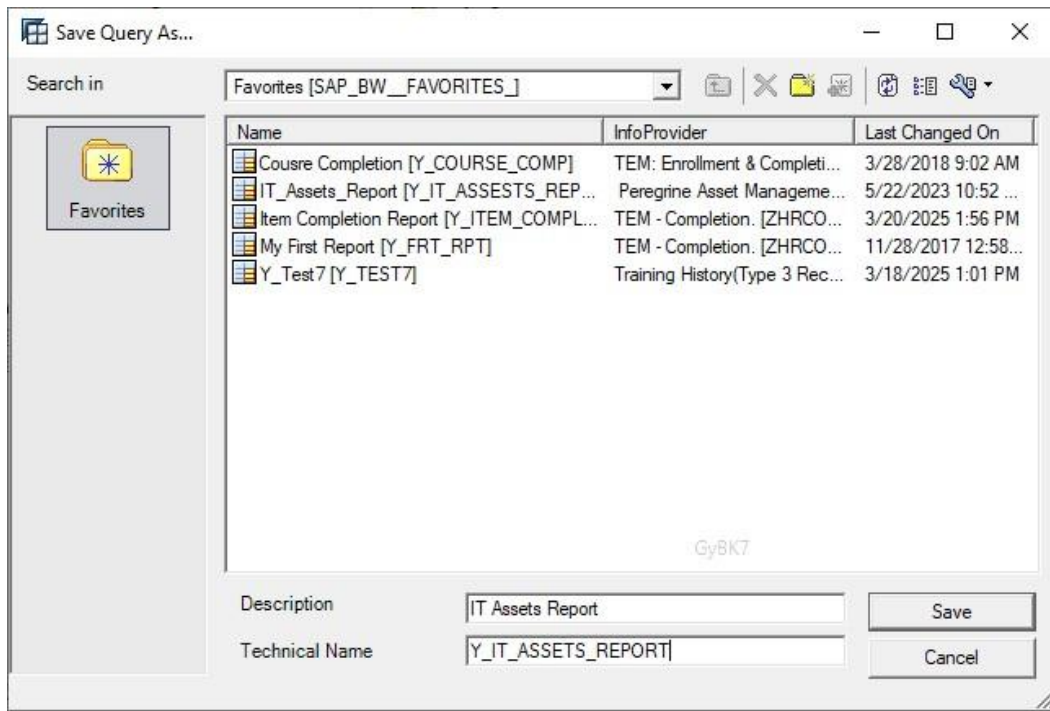


8. From right side, select **query properties** and make sure the highlighted options are **checked**.



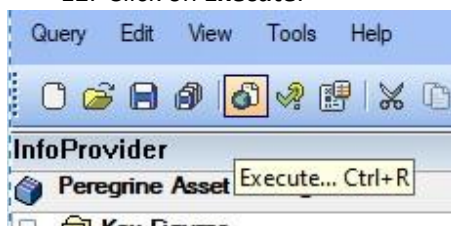
9. From main menu, click on **Save Query**.

10. Type the name of your query. Also type technical name. Technical name must start with **Y**.

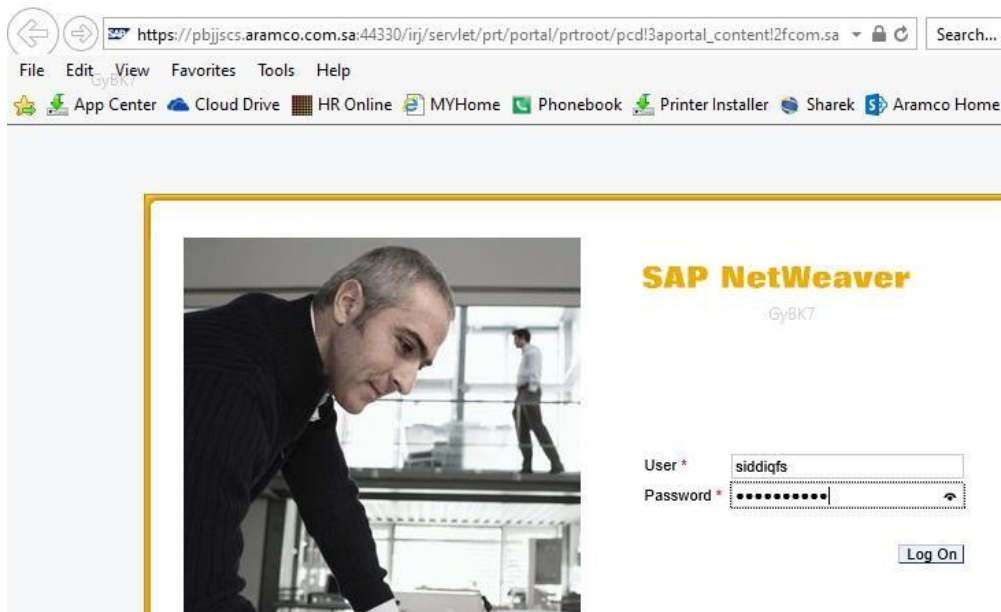


11. Click save. The query will save in your favorites.

12. Click on **Execute**.



13. The query needs to be run on a Brower. Type your network id/password and click **Logon**.



14. You can now able to view the data.

IT_Assets_Report2					Last Data Update
Open Save As... Display As Table Info Print Version Export to Microsoft Excel Notes Variables					
Columns	Owner Business Line		Owner Department		Count
Key Figures	30002950		Technical Services		21,214
Rows			30001956		MLD
Owner Business Line			30001987		MSD
Owner Department			30002006		PPD
Free characteristics			30002027		OPD
Location - Building			30002047		SCP&SD
			30003029		GFPD
			30003036		ECSD
			30003143		PMOD
			30003279		ID
			30003598		D&TPD
			30003650		EPD
			30003825		R&NGLPD
			30004186		PCD
			30004259		TSP&PMD
			30010926		P&CSD
			30017652		IPD
			30019851		C&PPD
			30026009		OMPPD
			30027069		OFMPPD
			30027516		OED

Note: you can modify your query any time in query designer. Select **Open Query** and add/remove objects as needed.

Appendix D: Getting the Data from SharePoint (ShareK)

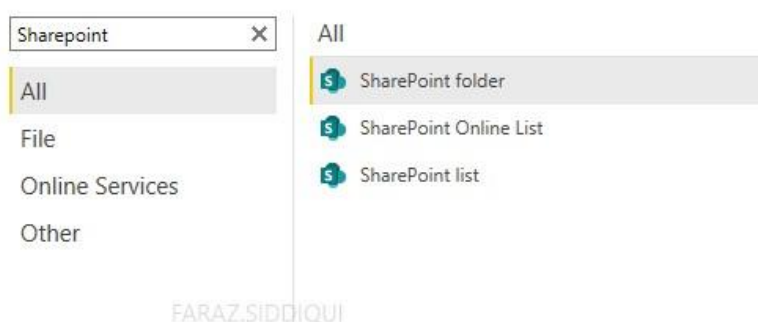
Power BI can fetch the data from SharePoint (*Also called **ShareK** in our environment*).

The below steps explain how to extract the data from SharePoint (ShareK).

Steps:

1. Open Power BI Application.
2. From Get Data, search SharePoint.

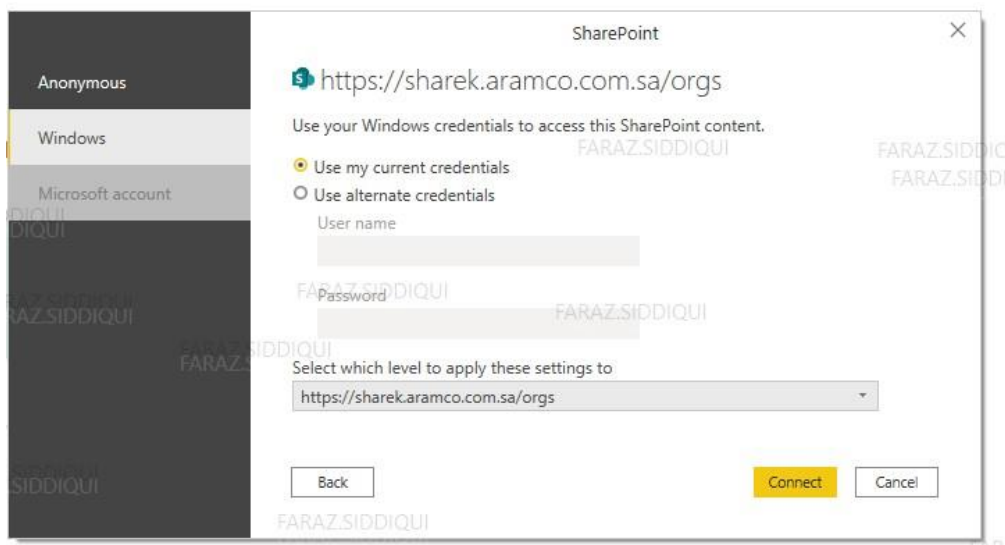
Get Data



3. You can read either through SharePoint folder or a list. For example, select SharePoint list & provide the URL of your Sharek site or a page.



4. It will ask your credentials.



5. Under the Navigator screen, Select the required data & click on load or transform.

Navigator

Display Options

<https://sharek.aramco.com.sa/orgs/300083...>

- ☐ Books_Selection
- ☐ CA_Website_Feedback_Form
- ☐ CDay_Test
- ☐ CH14 OOK Summary
- ☐ Composd Looks
- ☐ Corporate Academy Website Feedbac...
- ☐ COVID-19 CH14 By Region/Country
- ☐ COVID-19 Support Center Highlights
- ☒ Cyber Security Course registration
- ☐ Documents
- ☐ Domains
- ☐ gt
- ☐ Images
- ☐ In Kingdom Programs Summary
- ☐ Job Ladder Automation Awareness W...
- ☐ Kendo
- ☐ Learning Providers News
- ☐ Learning Providers Required Info
- ☐ license

Cyber Security Course registration

Preview downloaded on Tuesday, October 25, 2022

FileSystemObjectType	Id	ServerRedirectedEmbedUri	ServerRedirectedEmbedI
0	2	null	
0	6	null	

Load
Transform Data
Cancel

Appendix E: Getting the Data from Oracle/SQL Server

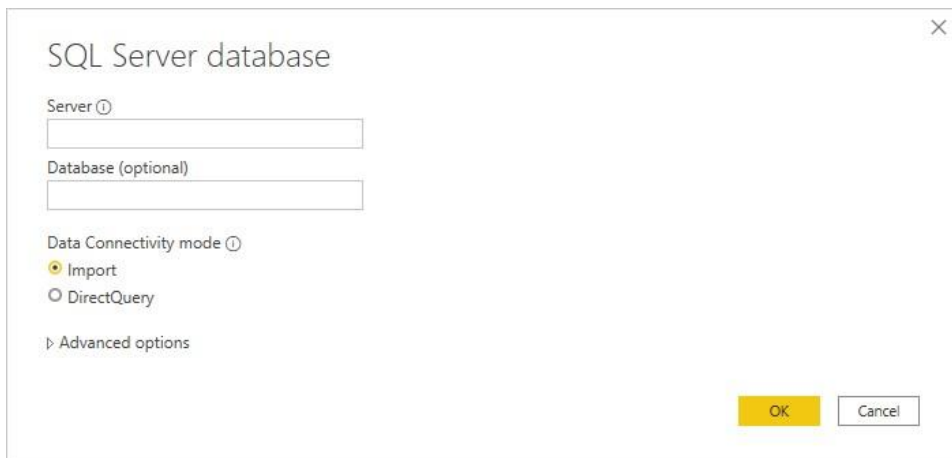
Power BI can fetch the data from Oracle, SQL Server & other relational databases.

The below steps explain how to extract the data from Oracle/SQL Server.

Note: In order to extract the data from Oracle database, you need to install first the **Oracle Client**. (*Oracle client is available in Aramco App Center*).

Steps (SQL Server):

1. Open Power BI Application.
2. From Get Data  SQL Server database.
3. In the **SQL Server database** dialog that appears, provide the name of the server and database (*optional*).



SQL Server database

Server 

Database (optional)

Data Connectivity mode 

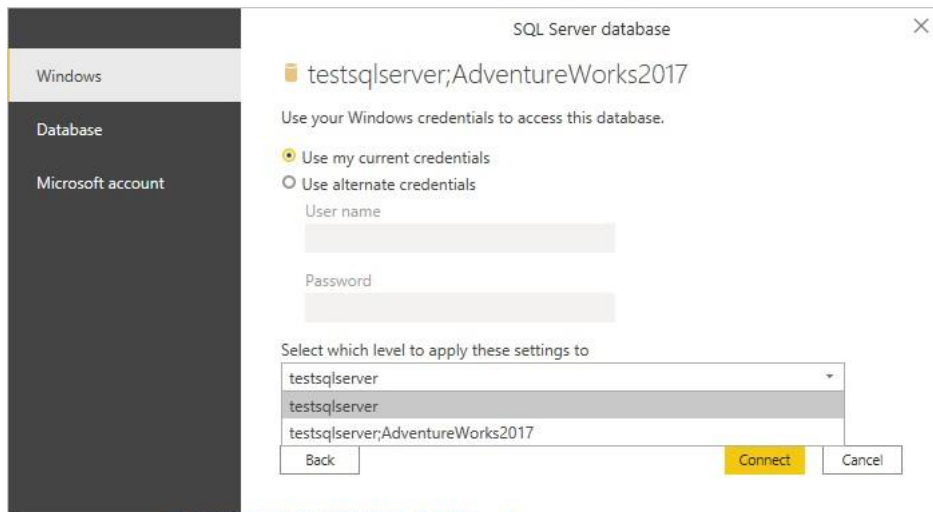
☒ Import

☐ DirectQuery

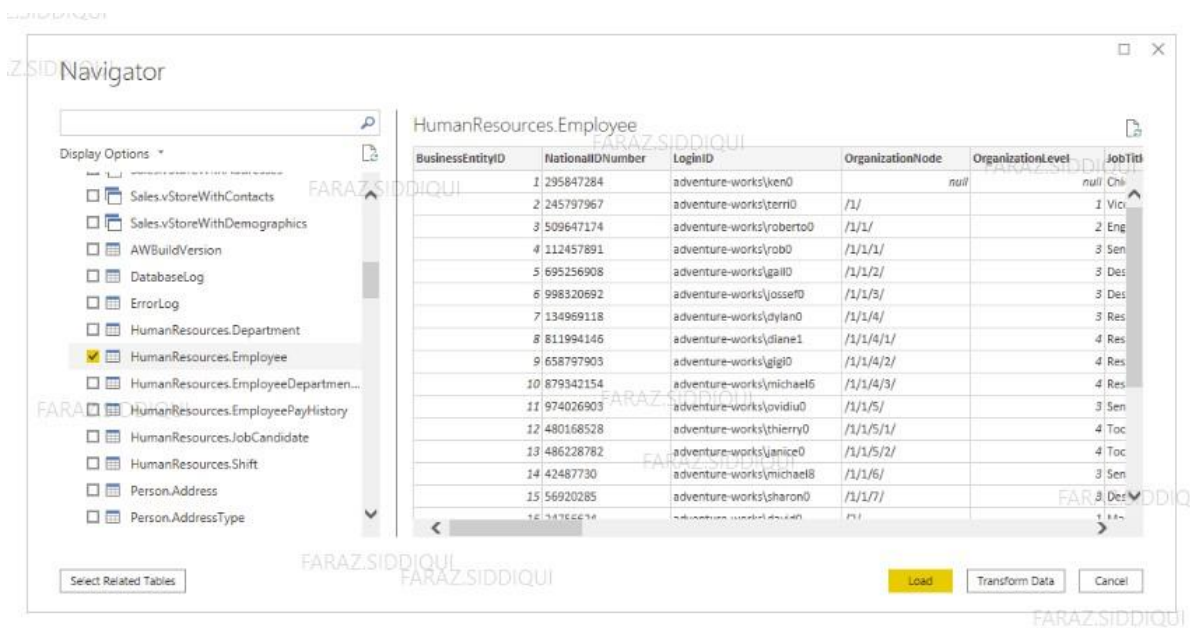
[Advanced options](#)

OK Cancel

4. Provide your credentials to access the database.



5. Under the Navigator screen, Select the required data & click on load or transform.



Steps (Oracle):

1. Open Power BI Application.
2. From Get Data  Oracle database.
3. In the **Oracle database** dialog that appears, provide the name of the server and SQL Query (optional).

Oracle database

Server

Data Connectivity mode ⓘ

☒ Import

☐ DirectQuery

Advanced options

Command timeout in minutes (optional)

SQL statement (optional)

☒ Include relationship columns

☒ Navigate using full hierarchy

OK Cancel

4. Provide your credentials to access the database.

Windows

Database

adwptr_low

User name

admin

Password

Back Connect Cancel

5. Under the Navigator screen, select the required data & click on load or transform.

Navigator

Display Options ▾

GGSYS

GSMADMIN_INTERNAL

GSMCATUSERDDIQUI

GSMUSER

LBACSYS

MDDATA

OML\$PROXY

ORACLE_OCM

ORDS_METADATA

ORDS_PUBLIC_USER

PTN [2]

☐ ADMIN_EMP

☒ SALES

REMOTE_SCHEDULER_AGENT

SH

SSB

SYSSUMF

SYSDG

SYSDG

SYSDG

SALES

ID	REGION	SALES	CLIENTS
1	APAC	11834000	1300
2	EMEA	51321000	7000
3	LATAM	13621000	3500
4	NORTH AMERICA	15003271	5257

Select Related Tables

Load

Transform Data

Cancel